# Distribution-Independent Tour-Length Estimation for the Traveling Salesman Problem Across Dimensions

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**Abstract**

We present the Geometrically Assisted Regression Tree (GART) 2.0, a distribution-independent estimator of Traveling Salesman Problem (TSP) tour length whose feature set is defined for arbitrary dimension. Existing estimators either assume a known sampling density or depend on planar geometry that becomes unstable as dimension grows. GART 2.0 is a LightGBM regressor that predicts the instance-specific ratio $\alpha = L_{\mathrm{TSP}}/L_{\mathrm{MST}}$ of tour length to minimum-spanning-tree (MST) length and returns $\hat{L}_{\mathrm{TSP}} = \hat{\alpha}\, L_{\mathrm{MST}}$; no distribution label, no density, and no convex hull enters the computation. Trained on synthetic instances alone, it transfers to TSPLIB95 and, through a multidimensional-scaling embedding, to non-Euclidean instances given only as a distance matrix. On all four benchmark aggregates GART 2.0 leads every baseline scored there, from a seven-baseline roster, and on the multidimensional benchmark its dispersion falls monotonically in the two arguments the feature set was built to carry—growing instance size and growing dimension—out to a hundred dimensions, a value withheld from training entirely. We then score a certified Held–Karp 1-tree lower bound, which consumes no training corpus and proves its own output, and use it to fix the boundary of the method rather than to argue around it. On the corpus aggregate the bound is the better instrument; that aggregate is mostly small instances, where the relaxation is nearly exact and nearly free, and read at fixed instance size the ordering turns over. Above a few hundred nodes GART 2.0 is on the cost/accuracy front at every dimension it was trained on, and in the plane it is strictly better than the bound on both axes from an ascent budget of 25 upward. The reason is structural rather than empirical: the bound pays a quadratic price in instance size for a fixed accuracy in every dimension, while the estimator's cost carries no accuracy term—so the estimator sharpens exactly where an exact solve becomes infeasible. Below that size, and off the training grid in dimension, the bound is preferable; both halves of that rule are needed to use either method well. The same bound doubles as an acceptance test for the labels every method is scored against: every label it reaches passes, with zero violations.

***K*eywords** Traveling Salesman Problem · Tour-length estimation · Distribution-independent · Arbitrary dimension

## 1 Introduction

The Traveling Salesman Problem (TSP) is a cornerstone of combinatorial optimization, used throughout logistics operations planning, network design, and resource allocation [Chien, 1992, Applegate et al., 2006]. The TSP is NP-hard [Karp, 1972, Papadimitriou, 1977], so exact solution times grow prohibitive on large instances and rapid tour-length estimates become essential for real-time decision-making. And in many cases we are interested in the cost of the tour rather than the tour itself, which is what makes estimation worth doing at all: tour-length estimates serve as inputs to other optimization problems where the TSP appears as a subproblem. A facility-location decision, for instance, does not need the route sequence, only the cost of the routes each candidate solution implies. The difficulty compounds when many candidates must be compared, because a full solve is then paid once per candidate rather than once.

### 1.1 Background and Motivation

The scale of those decisions is what makes estimation a strategic input rather than a convenience. Last-mile delivery accounts for approximately 53% of total shipping costs by one industry estimate [Finmile], and the UPS ORION system is reported to save roughly 100 million miles and $300–400 million annually through route optimization [Holland et al., 2017]. In such systems recalculation is frequent, because orders arrive continuously and the number of customers on a route is large, so the cost of a full solve is paid many times over. A tour-length estimator replaces that solve wherever the decision depends on the relative cost of candidate routes rather than on their exact values.

The same estimation problem recurs wherever a TSP appears as a subproblem, and it rarely arrives in the plane. Radiation hybrid mapping in bioinformatics orders genetic markers by a TSP over an abstract similarity metric [Agarwala et al., 2000], and mobile-sink routing in wireless sensor networks contains a TSP through the data-collection locations [Yuan and Peng, 2010]. Large-scale routing methods partition an instance into smaller sub-TSPs and compare candidate partitions [Mariescu-Istodor and Fränti, 2021]; Pan et al. [2023] instead learn a hierarchical policy whose upper level repeatedly selects a subset of nodes and whose lower level extends a partial route through them. A planner choosing among node sets, route splits, or cluster assignments needs the cheaper option to be predicted as cheaper, so an estimator serving these tasks must be accurate and rank-preserving at once.

None of those settings is planar, yet most classical tour-length estimators are built for low-dimensional Euclidean inputs and depend on geometric quantities—most often convex-hull area—that become unstable or intractable as dimension grows [Chazelle, 1993]. An estimator that serves the settings above must therefore be defined for arbitrary $d$ from a feature set that never builds a hull. That requirement also opens the non-Euclidean case: instances given as a distance matrix admit a Euclidean representation of bounded distortion once enough dimensions are allowed [Bourgain, 1985], and an estimator that stays accurate at those dimensions can consume the embedding. Dimension-scalability is the design goal of this paper, and Section 4 shows the estimator meeting it: on the multidimensional benchmark, dispersion falls monotonically as both the instance size and the ambient dimension grow.

### 1.2 Limitations of the Current Estimators

The need for a general TSP estimator is clear; no published estimator covers high-dimensional or non-Euclidean instances. The Beardwood–Halton–Hammersley (BHH) theorem [Beardwood et al., 1959] is stated for arbitrary dimension, but it is asymptotic in $n$ and its calibrating constants and density integral are not recoverable from an arbitrary finite point set; published constants exist for $d = 2$ [Johnson et al., 1996] and $d = 3$ [Percus and Martin, 1996], and above that only a large-$d$ approximation [Bertsimas and van Ryzin, 1990]. The estimators of Chien [1992] are planar and depend on particular depot and covering-region definitions. A parallel line of work is tuned to the 2D Euclidean case: Daganzo [1984b] gives a strip-based construction, Kwon et al. [1995] fit regression and neural models on rectangular-region predictors, and Çavdar and Sokol [2015] propose a distribution-free estimator built from per-axis spread statistics about the covering rectangle. Space-filling heuristics such as the planar construction of Bartholdi and Platzman [1982] instead build a tour outright in $O(n \log n)$. Every member of this family consumes either a convex-hull area or an axis-aligned covering box, and both degrade with dimension, so the embedding route out of the non-Euclidean case returns them to the regime in which their geometric inputs are unstable. Section 4 scores BHH and Çavdar and Sokol [2015] numerically, both on the full benchmarks and on the subset where their published assumptions hold (Table 16). We survey Chien [1992], Daganzo [1984b] and Kwon et al. [1995] without scoring them, because we could not obtain their primary sources and decline to publish numbers produced by constants we have not read in the original.

Recent neural approaches address the same gap from the learning side. Varol et al. [2024] report sub-1% deviation on standard distributions by combining a multilayer perceptron (MLP) with domain-knowledge features, but their pipeline requires solution-level features, which are computationally expensive and challenging to generalize across all instance sizes. Kou et al. [2022] use the standard deviation of random feasible tour lengths as a single-feature predictor, and evaluate it on both Euclidean and non-Euclidean instances. The predictor is distribution-independent but requires repeated tour sampling, which raises its per-instance cost. Kou et al. [2024] extend this randomization principle beyond the TSP, predicting optimal values for the vehicle-routing (VRP), MST, and matching problems from the standard deviation of random feasible solutions; like the original predictor it relies on repeated per-instance sampling, whose cost we exclude on the same grounds as solution-level features. The most direct methodological neighbor is Kou et al. [2023], which pairs a Sammon-map embedding with a distribution-independent estimator on non-Euclidean TSPLIB95 instances; our pipeline differs by using classical metric MDS for the embedding while computing MST-topology features on the original distance matrix rather than the embedded coordinates. What this line shares is a per-instance price—solution-level features or repeated tour sampling—that moves its cost back toward the solve it replaces.

A separate classical line does not estimate the tour at all: it bounds it, and it is the strongest comparator available to any tour-length estimator. Held and Karp [1970, 1971] relax the Hamiltonian-cycle constraint to a *1-tree*, a spanning tree on all but one node together with the two cheapest edges at that node, and any vector of node penalties yields a proven lower bound on the optimal tour, so an ascent over penalties stopped anywhere returns a certificate rather than a prediction. Three properties follow. The bound consumes no training corpus and no distributional assumption, so it has no split to contaminate and no domain to leave. It never invokes the triangle inequality, so it is defined wherever a symmetric distance matrix is, including the non-Euclidean instances Section 6 reaches only through an embedding. And its cost is an explicit knob, quadratic in $n$ per ascent iteration, so accuracy is bought continuously rather than fixed by a closed form. Section 5.1 gives the relaxation and the two step schedules we score as first-class rows on every benchmark (Table 1). Because the bound is certified, cheap in the plane only at small $n$, and quadratic in $n$ everywhere, it is the comparator that fixes where a learned estimator is worth running; Section 5 draws that line.

### 1.3 Contribution

We present the Geometrically Assisted Regression Tree (GART) 2.0, a LightGBM regressor [Ke et al., 2017] that predicts a per-instance scaling factor $\hat{\alpha}$ rather than applying a fixed constant. It is distribution-independent: no distribution label and no parametric density is supplied at inference, although accuracy varies by distribution type. That property is not by itself new—Çavdar and Sokol [2015] and Kou et al. [2022, 2023] share it—but those estimators remain tied to 2D coordinate geometry or to per-instance sampling. The contribution of GART 2.0 is to be distribution-independent *and* dimension-scalable at once: all 31 features are defined for arbitrary $d$ through MST topology, centroid dispersion and coordinate ranges, no feature requires a convex hull, and the same feature set applies unchanged as both dimension and instance size grow. Section 3.1 gives the feature set and Section 3.4 validates it against SHAP attributions on the held-out split.

We benchmark GART 2.0 against seven baselines: two classical estimators, three constant multiples of $L_{\mathrm{MST}}$ of which one is calibrated on the training split, the GART 1.0 predecessor [Feldman, 2025], and a space-filling-curve tour construction. Both classical estimators are scored on the full benchmarks and again on the i.i.d. uniform subset where their published assumptions hold, BHH there receiving the exact sampling region its theorem names. GART 2.0 has the lowest aggregate mean absolute percentage error (MAPE) and standard deviation of percent error (SDPE) among the baselines scored on each of the four benchmarks, ahead of a calibrated per-cell lookup table on the multidimensional set, of Çavdar–Sokol on planar i.i.d. uniform instances, and of the asymptotic MST ratio on TSPLIB EUC_2D. The behavior that matters for deployment is that these gains are not concentrated at one operating point. Dispersion falls monotonically as instances grow, and again as the ambient dimension grows, holding at a dimension withheld from training entirely. An estimator that sharpens in exactly the two directions that make an exact solve infeasible is the one worth putting into a routing or partitioning pipeline.

Section 5 then fixes the boundary of the method against a comparator no estimator can argue with: the certified Held–Karp 1-tree bound, defined on every instance in this paper and on some that GART 2.0 declines. On the corpus aggregate the bound is the better instrument. Stratified by size as well as by dimension (Table 5), the ordering turns over above a few hundred nodes, and the mechanism is the complexity separation of Section 5.5. Section 5.7 states the resulting rule in full; stating both halves of it is what makes the recommendation usable.

The bound is also an acceptance test for the labels themselves: $B > L$ refutes a stored label with no tour entering the test, and every label the bound reaches passes (Appendix E).

### 1.4 Paper Outline

Section 2 establishes the MST–TSP ratio $\alpha$ as an instance-dependent quantity that a fixed constant misestimates. Section 3 describes the features, the training corpus, and the fit. Section 4 benchmarks GART 2.0 against classical and constant-ratio baselines, on the full benchmarks and on held-out regimes. Section 5 prices it against the certified Held–Karp 1-tree bound and an exact solve, and states the operating rule. Section 6 extends it to non-Euclidean instances, and Section 7 states what the paper establishes and what bounds it.

## 2 Theoretical Foundations

This paper predicts the ratio $\alpha = L_{\mathrm{TSP}}/L_{\mathrm{MST}}$, and this section establishes that the ratio is a property of the instance rather than a constant. Let $\mathcal{G} = (V, E)$ be a complete undirected graph on the node set $V = \{1, \ldots, n\}$ with non-negative edge costs $c_{ij}$, typically the Euclidean distance between points in $\mathbb{R}^d$. Throughout, $n$ is the number of nodes in an instance and $d$ is the dimension of the space its coordinates occupy. The TSP asks for a minimum-cost Hamiltonian cycle. In the standard Dantzig–Fulkerson–Johnson formulation [Dantzig et al., 1954, Applegate et al., 2006] a binary $x_{ij}$ indicates whether edge $(i, j)$ is in the tour and $\sum_{i<j} c_{ij}x_{ij}$ is minimized subject to a degree-two constraint at every vertex and a subtour-elimination constraint on every proper subset of $V$; the problem is NP-hard [Karp, 1972, Papadimitriou, 1977]. What the formulation exposes, and what this paper exploits, is that any feasible tour is a 2-regular spanning subgraph, of which the minimum spanning tree (MST) is the cheapest connected relaxation.

The MST is solvable in polynomial time. Prim's algorithm constructs one in $O(n^2)$ on a complete graph with an adjacency-matrix representation [Prim, 1957]; Kruskal's must sort $\Theta(n^2)$ edges and so costs $O(n^2 \log n)$ there [Kruskal, 1956]. For nodes in $\mathbb{R}^d$ the pairwise distance computation scales as $O(n^2 d)$, linear in $d$, which is what makes the MST a computationally stable scaffold for high-dimensional inference. It also brackets the tour. Removing the heaviest edge from any tour leaves a spanning tree, so $L_{\mathrm{MST}} \leq L_{\mathrm{TSP}}$; the double-tree heuristic [Rosenkrantz et al., 1977] doubles the MST edges into an Eulerian multigraph and shortcuts repeated nodes to yield a feasible tour of length at most $2\,L_{\mathrm{MST}}$ on any instance obeying the triangle inequality. Together these give

$$L_{\mathrm{MST}} \leq L_{\mathrm{TSP}} \leq 2 \cdot L_{\mathrm{MST}}. \tag{1}$$

The upper constant cannot be reduced in general: for collinear point sets the optimal tour must traverse the line to its far end and return, so $L_{\mathrm{TSP}} = 2\,L_{\mathrm{MST}}$ exactly.

A separate and frequently conflated result concerns *algorithmic* approximation rather than this intrinsic ratio. The Christofides–Serdyukov algorithm [Christofides, 1976, Serdyukov, 1978] augments the MST with a minimum-weight perfect matching (MWPM) on the odd-degree nodes and returns a tour within a factor $\frac{3}{2}$ of the *optimum*; Karlin et al. [2021] improve on that factor with a different construction, a randomized $\frac{3}{2} - \epsilon$ approximation. Both are guarantees on an algorithm's output relative to the optimal tour, and neither tightens the intrinsic ratio $L_{\mathrm{TSP}}/L_{\mathrm{MST}}$, which can still equal 2. The MWPM costs $O(n^3)$ [Edmonds, 1965, Gabow, 1990, Cook and Rohe, 1999], far above the $O(n^2)$ MST construction, so the Christofides refinement is impractical as a fast estimation subroutine.

For symmetric metric instances we therefore treat $\alpha \in [1, 2]$ as an intrinsic, instance-dependent quantity to be predicted. For uniformly distributed points the TSP constant itself varies with dimension [Percus and Martin, 1996], and no published MST constant covers every $d$, so we establish the behavior of the ratio empirically: the standard deviation of $\alpha$ over our training split falls monotonically from 0.1837 at $d = 2$ to 0.0762 at $d = 50$. The relative difference in objective value, and not the computational complexity of either problem, decreases as $d$ grows, so a static estimator calibrated in the plane is biased in higher dimensions. GART 2.0 predicts $\alpha$ from the observed instance features instead.

## 3 Methodology: The Geometrically Assisted Regression Tree

GART 2.0 extends the estimator of Feldman [2025]. It predicts $\alpha = L_{\mathrm{TSP}}/L_{\mathrm{MST}}$ with a LightGBM ensemble and returns $\hat{L}_{\mathrm{TSP}} = \hat{\alpha}L_{\mathrm{MST}}$. We constrain the regression target to $\alpha \in [1, 2]$, the metric-TSP interval implied by Eq. (1). Two raw corpus values fall outside that interval, one training row at 0.975 and one validation row at 2.060, and target preparation clips them to the endpoints; an interior clip one millionth inside each endpoint keeps the logit target of Section 3.3 finite for those two rows.

### 3.1 Feature Selection

We extract 31 features from each instance: eleven geometric, scale and centroid descriptors, ten MST edge-weight statistics, nine MST topology and clustering proxies, and one constructive ratio. The count reflects the four aspects of instance structure the target responds to – spatial spread, edge-weight distribution, branching topology, and the gap between a constructed tour and the MST – the first three requiring several statistics apiece to cover the behaviors observed across distributions, the fourth a single ratio.

#### 3.1.1 Geometric and Centroid Descriptors

The geometric group supplies what the MST statistics deliberately factor out: scale and spread. MST edge weights and the degree sequence are invariant to rotation, translation and coordinate-system choice; the geometric descriptors are lightweight quantities such as bounding hypervolume and centroid distance that generalize to arbitrary $d$ without a convex hull, supplying the scale information that the volume term of the BHH formula carries [Beardwood et al., 1959] and adapting their centroid statistics from Çavdar and Sokol [2015]. The dimension $d$ enters as an explicit input rather than implicitly through coordinate geometry. Before any geometric descriptor is evaluated we rotate the coordinates into their principal-axis frame [Jolliffe, 2002], which costs $O(nd^2 + d^3)$ and is dominated by dense MST construction whenever $n$ is large relative to $d$; the rotation leaves the rotation-invariant features untouched and makes the axis-aligned bounding-box descriptors independent of the input orientation. Appendix O lists the geometric group.

#### 3.1.2 MST Edge and Topological Features

Nineteen features summarize the MST. Ten describe the edge-weight distribution: mean, standard deviation, skewness, kurtosis, maximum, and the 10/25/50/75/90 quantiles. Higher moments characterize heterogeneous edge-length distributions, including those produced by dense clusters separated by long connecting edges. Nine describe topology: dominance, gaps, leaves, degrees, large edges, and the raw and normalized MST diameter. The MST's total length is not among them, because it is the multiplicand $L_{\mathrm{MST}}$ and withholding it keeps the learned quantity a scale-free ratio. The thirty-first feature, `greedy_nn_over_mst`, is the length of an exact greedy nearest-neighbor tour divided by $L_{\mathrm{MST}}$. It is the only input GART 2.0 adds to the 30-feature block it inherits, and it is a constructed upper bound on the target: a greedy tour is a feasible tour, so its length is at least $L_{\mathrm{TSP}}$ and the ratio is at least $\alpha$. The feature also gates coverage. On the non-Euclidean path of Section 6, where the metric is given but coordinates are not, the estimator returns a status row rather than a prediction for any instance whose ratio falls outside $[1.035, 2.209]$. That interval is the range the ratio spans over the full 106,272-row corpus, training, validation and test together, which is what the implemented gate uses; the training split alone spans the narrower $[1.0465, 2.1295]$. We state the gate as the corpus range rather than as a training range because that is the constant the code applies. Definitions and computational costs are reported in Appendix C.

### 3.2 Dataset Generation

To train a general model, we constructed a synthetic dataset spanning four one-dimensional generators: Uniform, Normal (Gaussian), Clustered ($k$-means centers), and Correlated (autoregressive across dimensions). Each axis of each instance draws its generator independently, producing anisotropic configurations whose topology varies by axis. The generator is selected from sixteen letter codes that map onto these four functions, and the map is not balanced: eleven codes resolve to the uniform draw, two to normal, two to clustered, and one to correlated. Measured over all $2{,}444{,}256$ axes in the corpus, 68.7% are uniform, 12.5% clustered, 12.5% normal, and 6.2% correlated. The corpus is therefore predominantly uniform per axis and not balanced across the four classes, and the multidimensional results below should be read against that mix.

The full feature corpus comprises 106,272 solved instances with node counts $n \in [5, 1000]$ and dimensions $d \in \{2, 3, 4, 5, 6, 7, 8, 9, 10, 15, 20, 25, 30, 35, 40, 45, 50\}$ (17 training-visible dimension values), plus a held-out extrapolation bucket at $d = 100$. Instances are partitioned into training (69,768), validation (19,584), and test (16,920) via a stratified split on $(d, n,$ grid size$)$; all $d = 100$ instances are locked into the test partition so the model never sees that dimension during training or validation. The full training grid is given in Appendix B (Table 11); extrapolation to $d = 100$ is evaluated in Section 4.3.1.

For each instance, the generation pipeline ran Concorde [Applegate et al., 2006] and LKH-3 [Helsgaun, 2017] and retained the shorter tour. The canonical feature table stores the retained tour cost but not solver-specific costs or Concorde certificates; we therefore call these values reference tour costs and do not claim that every stored row is independently auditable as optimal. Solver-native integer quantization is not carried into the released labels: every reference cost is recomputed in float64 from the released coordinates. Instances with $n \leq 10$ are labeled by an exact Held–Karp dynamic program, a certificate; larger instances carry the float64 length of the retained tour, an upper bound on the optimum that is exact wherever the retained permutation is. Appendix E states the validation every label passes and the certification hierarchy.

### 3.3 Model Architecture and Training

The estimator for $\alpha$ is a *gradient-boosted ensemble* of regression trees. A single regression tree is easy to interpret but high-variance, so combining many trees yields a far more stable estimate. Random forests [Breiman, 2001] average trees grown independently on bootstrap samples; gradient-boosted decision trees (GBDT) [Friedman, 2001] instead grow the trees *sequentially*, each new tree fit to the error left over by the trees before it, so the ensemble keeps improving where it is currently weakest. We use LightGBM [Ke et al., 2017], a fast and memory-efficient GBDT implementation whose leaf-wise growth splits the leaf with the largest estimated loss reduction instead of expanding every depth level uniformly; its final prediction $\hat{\alpha}$ is the sum of all tree outputs.

The learner uses squared-error regression on the logit-transformed target $z = \log\big((\alpha - 1)/(2 - \alpha)\big)$, and predictions are returned through its inverse $\alpha = 1 + (1 + e^{-z})^{-1}$. The bound $\alpha \in [1, 2]$ is therefore structural: the inverse transform cannot leave the open interval $(1, 2)$, so the collinear extreme of Eq. (1) is approached rather than attained; a clip to $[1, 2]$ remains in the released wrapper as a guard against non-finite inputs, and it is never binding. On the 16,920 multidimensional test predictions $\hat{\alpha}$ fell in $[1.029, 1.922]$, so the transform is not saturating at either end. The hyperparameters are inherited and frozen, so Section 4 compares a tuned field against an untuned entrant; an Optuna search [Akiba et al., 2019] against in-distribution validation moved multidimensional MAPE by 0.011 percentage points while costing 0.16 points of TSPLIB MAPE, and we rejected it (Appendix P). Early stopping used cost-level MAPE on the validation split with a patience of 100 rounds; the model was fit at seed 42 on the training split only, with no train–validation refit, and the test split was scored once after that fit. The resulting ensemble contains 1,118 trees with 148 leaves per tree as the growth cap and 147.80 realized on average; its configuration is reported in Appendix P.

Two inputs carry non-increasing monotone constraints, $n$ and $d$, encoding that $\alpha$ falls toward 1 as an instance grows or the ambient dimension rises. The constraints are enforced inside the tree builder, so a sum of such trees cannot increase along a ceteris-paribus sweep of either variable; a probe over 1,000 held-out instances accordingly returns 100% non-increasing sweeps on both axes with zero violations, on log-spaced grids of 24 points, $d$ from 2 to 200 and $n$ from 5 to 4,000, deliberately overshooting the synthetic evaluation range. The probe overwrites one column of a real held-out instance and holds the other thirty at that instance's measured values, so a swept row corresponds to no realizable instance: it measures the shape of the learned function in one argument, and it says nothing about accuracy at those $n$ and $d$.

A constant multiple of $L_{\mathrm{MST}}$ is trivially non-increasing on both axes, so the informative control is the same fit with the constraints removed, which holds monotonicity on 8.4% of the dimension sweeps and 19.0% of the size sweeps. The constraint is therefore doing the work.

Inference cost is dominated by feature extraction, not tree traversal. The trained model performs $K = 1{,}118$ traversals with empirical mean root-to-leaf depth $\bar{D} \approx 11.83$ (maximum 40). Because the trees are unbalanced, traversal cost is better represented by $\bar{D}$ than by $\lceil \log_2 148 \rceil$; the ensemble therefore performs approximately $K\bar{D} = 1.3 \times 10^4$ comparisons per prediction, independent of $n$. With a dense distance matrix, the pipeline costs $O(n^2 d + nd^2 + d^3)$, combining pairwise distances and MST construction with the PCA rotation. In the plane the Euclidean MST is a subgraph of the Delaunay triangulation [Shamos and Hoey, 1975], so it can be built in $O(n \log n)$; sorting MST edges adds $O(n \log n)$, and the remaining topological statistics cost $O(n)$. The greedy nearest-neighbor tour behind `greedy_nn_over_mst` does not take that path: it scans a dense distance matrix in $O(n^2)$ for $n \leq 3000$ and switches to precomputed 16-neighbor candidate lists with an exact nearest-unvisited fallback above that threshold. Below $n = 3000$ the greedy tour, not the MST, is therefore the leading term of 2D extraction.

### 3.4 Feature Importance

We assess the 31-feature set with mean absolute SHAP (SHapley Additive exPlanations) magnitudes [Lundberg and Lee, 2017] against the released booster, on a 5,000-row sample of the 16,920-row held-out test split. The MST dominance ratio contributes 26.5% of the total magnitude and `greedy_nn_over_mst` 23.9%, so those two features carry 50.4% of it between them; node count and dimension jointly contribute 14.1%. The greedy ratio ranking second on its first appearance is the case for adding it; it is also the input that lets the estimator decline an instance, since the coverage gate of Section 6 is a range test on this feature and no other. Appendix D reports the complete ranking.

## 4 Benchmarking

We report four error metrics because no one of them ranks estimators correctly on its own. MAPE reports average magnitude. SDPE is the primary precision metric and the first column of each per-benchmark results table, but it is blind to bias: an estimator that is uniformly 20% high scores an SDPE of zero. We therefore pair it with MAPE, with the median absolute percentage error, the |Median| column of the tables, as an outlier-robust companion, and with the mean signed percent error (MSPE) so that a systematic offset stays visible. For the rows that predict a multiple of $L_{\mathrm{MST}}$ we also report $R^2_\alpha = 1 - \sum_i (\hat{\alpha}_i - \alpha_i)^2 / \sum_i (\alpha_i - \bar{\alpha})^2$, with $\bar{\alpha}$ the mean of $\alpha$ over the reported bucket: a negative value means the estimator predicts $\alpha$ worse than that bucket mean does. The distinction is not hypothetical here. BHH given the exact sampling region carries a $-8.65\%$ offset on uniform instances with only 7.76% SDPE, and the custom Hilbert sort has 30.95% MAPE against 14.19% SDPE; reading either row on SDPE alone would misdescribe it. SDPE is defined as:

$$\mathrm{SDPE} = 100 \cdot \sqrt{\frac{1}{N-1} \sum_{i=1}^{N} (e_i - \bar{e})^2}, \qquad \bar{e} = \frac{1}{N} \sum_{i=1}^{N} e_i.$$

Here $e_i = (\hat{L}_i - L_i)/L_i$ is the signed relative error and $N$ is the number of scored instances; the leading factor of 100 expresses SDPE in percentage points. In the per-benchmark results tables of the appendices, bold marks GART 2.0, the model proposed here; it does not mark the best value in a column, and several baselines beat it in individual cells. The ladder tables of Section 5 follow a different bold convention, stated at the start of Section 5.2. We accompany each SDPE with a 95% bootstrap confidence interval over 1,000 resamples, seeded at 42, and suppress the interval where a group holds fewer than three instances. Those intervals are computed separately for each estimator and therefore ignore the pairing, so overlapping intervals establish nothing on their own; Appendix J supplies paired tests of the difference.

### 4.1 Benchmark Roster

We compare GART 2.0 against seven baselines in four families: classical region-based estimators, constant-ratio references, the GART 1.0 predecessor, and a constructive heuristic. One rule governs the roster: every classical row receives the quantity its source defines, read from a primary document we hold, and we state for every row which inputs it receives. Where only a secondary rendering of an estimator exists we survey it and decline to score it; that rule excludes Daganzo, Chien, and Kwon–Golden–Wasil, whose coefficients circulate only as mutually inconsistent transcriptions, so those three are surveyed in Section 1.2 and scored nowhere in this paper (Appendix F).

**Classical region-based estimators.** Each estimator now receives the quantity its source defines and is gated to the domain its source covers. The BHH form is $\hat{L} = \beta_d n^{(d-1)/d}\mathcal{V}^{1/d}$, where $\mathcal{V}$ is the measure of the sampling region [Beardwood et al., 1959] and $\beta_2 = 0.7124$ [Johnson et al., 1996]; for $d = 3$ we use $\beta_3 = 0.6979$ [Percus and Martin, 1996] and for $d \geq 4$ the conjectured large-$d$ asymptotic $\beta_d \approx \sqrt{d/(2\pi e)}$ [Bertsimas and van Ryzin, 1990]. Çavdar and Sokol [2015] gives $\hat{L} = 2.791\sqrt{n\,\mathrm{cstdev}_x\mathrm{cstdev}_y} + 0.2669\sqrt{n\,\mathrm{stdev}_x\mathrm{stdev}_y A/(\bar{c}_x\bar{c}_y)}$, where $A$ is the area of the rectangle covering the nodes, $\mathrm{stdev}_j$ is the standard deviation of the raw coordinates on axis $j$, and $\bar{c}_j$ and $\mathrm{cstdev}_j$ are the mean and standard deviation of the absolute distances from the nodes to that rectangle's midpoint line on axis $j$. The Çavdar–Sokol estimate is divided by the finite-$n$ adjustment factor its source fits, held fixed outside the node range that fit covers. Both forms are transcribed from primary documents we read: the BHH constants from the sources cited above, and the Çavdar–Sokol estimator together with both of its constant sets and its adjustment factor from Çavdar [2014]. Appendix F records what each estimator receives, and why three further classical estimators are surveyed but not scored.

**Constant-ratio references.** Three rows bound what is achievable without learning, and each answers a different question. The $\alpha = 1$ floor returns $L_{\mathrm{MST}}$ itself: it is the certified lower bound of Eq. (1), the only constant defined at every $d$, and the reference $R^2_\alpha$ is measured against, so it isolates what the learned multiplier adds to the scaffold it multiplies. The asymptotic ratio uses the published constants $\beta_{\mathrm{TSP}}/\beta_{\mathrm{MST}} \approx 1.1253$ [Johnson et al., 1996, Cortina-Borja and Robinson, 2000] and is defined at $d = 2$ only, the one constant here taken from the literature instead of fitted on our data. The calibrated row uses $\hat{\rho}(d, n)$, the mean of $\alpha$ within each dimension–size cell of the training split, which is the ceiling of this family: the best constant obtainable from the training split conditioned on everything cheap that is known about an instance. Two further constants stay in the released tidy tables and are not printed as baselines: an author-chosen schedule whose $d \geq 3$ values sit 8 to 15 percentage points below the training-split mean of $\alpha$, and the per-dimension mean $\hat{\rho}(d)$, which $\hat{\rho}(d, n)$ dominates everywhere and which the $\alpha = 1$ floor itself beats on TSPLIB EUC_2D. The $\alpha = 1$ and asymptotic rows are one estimator at two values of $c$ in $\hat{L} = cL_{\mathrm{MST}}$, so their dispersion is related exactly by $\mathrm{SDPE}(c) = c\,\mathrm{SDPE}(1)$; we keep both because that identity does not extend to MAPE.

**Prior model.** GART 1.0 [Feldman, 2025] is our prior model and the one learned estimator in the baseline count. It is quadratic in construction and planar only, which is what this paper set out to remove: it appears only on the 2D synthetic and TSPLIB EUC_2D benchmarks (Tables 22, 23). Its artifact-driven extractor uses a dense pairwise-distance matrix, so its effective time and memory cost is $O(n^2)$ rather than $O(n \log n)$, and it treats the lexicographically first unique coordinate as a depot-like reference for two features although no depot is defined in these datasets. It differs from GART 2.0 in data, features, and training, so the comparison isolates no single causal change.

**Constructive heuristic.** The Hilbert row builds a tour from a bare $d$-dimensional Hilbert ordering after independent per-axis min–max normalization to a 16-bit grid. It constructs a tour rather than estimating one, and the planar guarantee of Bartholdi and Platzman [1982] does not transfer to this multidimensional variant.

**Certified lower bound.** One further row is not an estimator at all. The Held–Karp 1-tree bound [Held and Karp, 1970, 1971] returns a proven lower bound on the optimal tour rather than a prediction of it, at a cost fixed by an explicit ascent budget $k$. Two rows carry it, differing only in the step rule of the subgradient ascent: the Volgenant–Jonker schedule [Volgenant and Jonker, 1982], which consults no upper bound, and the Polyak step [Polyak, 1969], which sizes each move from the gap to a constructively built tour. Both are scored on all four benchmarks – the 2D diverse set, the multidimensional set, TSPLIB EUC_2D, and the non-EUC_2D TSPLIB instances – both raw and scaled by a single constant fitted per budget on the training split, and Section 5 reports them in full. We report the bound outside the seven-baseline count because it answers a different question than an estimator does. It is nevertheless the comparator that decides this paper.

**Region measure and matched domains.** A *matched domain* is the subset of a benchmark on which a classical estimator's own published assumptions hold, so that scoring it there measures the method rather than the mismatch. Every synthetic generator draws coordinates inside $[0, G]^d$ for a coordinate-grid side length $G$, so the sampling region measure is $G^d$ exactly. BHH is the only estimator here whose source names that measure, and on the synthetic benchmarks we report it as two rows, one given $G^d$ and one given the convex hull of the realized sample. Both are reported because the region measure is the source-faithful input only where the generator is uniform on that region: a near-collinear point set has one-dimensional effective support, and handing an area-based estimator $G^2$ inflates its prediction by an amount that describes the data and not the estimator. TSPLIB defines no sampling region at all, so there BHH receives the convex hull as a stated plug-in. Çavdar–Sokol takes no region input: its $A$ is the area of the rectangle covering the nodes, a statistic of the sample, so it has a single row on every benchmark and supplying it $G^2$ would be our construction rather than its authors'. Section 4.4 reports both estimators on the subset where their published assumptions hold. Neither assignment is a free parameter: the region measure is $G^d$ by construction of the generator, and the matched domain is pinned by the source itself at i.i.d. uniform sampling, so a reader can reconstruct the subset without reference to our results.

Table 1: Benchmark roster. "Domain" is the range each source fits or derives; Section 4.4 rescores the classical estimators inside it.

| **Estimator** | **Formula** | **Domain** | **Cost** | **Source** |
|---|---|---|---|---|
| *Classical region-based estimators* | | | | |
| BHH | $\beta_d\, n^{(d-1)/d}\mathcal{V}^{1/d}$, $\mathcal{V}$ the sampling-region measure | i.i.d. uniform, $n \to \infty$ | $O(nd)$ | Beardwood et al. [1959], Johnson et al. [1996], Bertsimas and van Ryzin [1990] |
| Çavdar–Sokol | $2.791\sqrt{n\,\mathrm{cstdev}_x\mathrm{cstdev}_y} + 0.2669\sqrt{n\,\mathrm{stdev}_x\mathrm{stdev}_y A/(\bar{c}_x\bar{c}_y)}$ | Minimum-area rectangle; $n \geq 100$ correction | $O(nd)$ | Çavdar [2014] |
| *Constant-ratio references* | | | | |

*Table 1 continued from the previous page*

| **Estimator** | **Formula** | **Domain** | **Cost** | **Source** |
|---|---|---|---|---|
| $L_{\mathrm{MST}}$ ($\alpha = 1$) | $L_{\mathrm{MST}}$ | All | $O(n^2 d)$ | Lower bound of Eq. (1) |
| Asymptotic MST ratio | $(\beta_{\mathrm{TSP}}/\beta_{\mathrm{MST}})\, L_{\mathrm{MST}} \approx 1.1253\, L_{\mathrm{MST}}$ | $d = 2$, uniform, $n \to \infty$ | $O(n^2 d)$ | Johnson et al. [1996], Cortina-Borja and Robinson [2000] |
| Calibrated MST ratio $\hat{\rho}(d, n)$ | $\hat{\rho}(d, n)\, L_{\mathrm{MST}}$, per dimension–size cell | Trained cells | $O(n^2 d)$ | This work |
| *Prior model and constructive heuristic* | | | | |
| GART 1.0 | 2D-only feature set, GBDT regression on $\alpha$ | $d = 2$ | $O(n^2)$ | Feldman [2025] |
| Custom Hilbert sort | Bare $d$-dimensional ordering, per-axis normalized, Hilbert order $h = 16$ | Constructs a tour | $O(n \log n)$ | After Bartholdi and Platzman [1982] |
| *Certified lower bound* | | | | |
| Held–Karp 1-tree, Volgenant–Jonker step, budget $k$ | the incumbent $\hat{w}_k = \max_{j \le k} w(\pi_j)$ over $k$ ascent steps; step schedule consults no upper bound | Any symmetric distances, any $d$; certificate | $\Theta(n^2 d + kn^2)$ | Held and Karp [1970, 1971], Volgenant and Jonker [1982], Helsgaun [2000] |
| Held–Karp 1-tree, Polyak step, budget $k$ | The same bound; step $\gamma(\mathrm{UB} - w)/\lVert g \rVert^2$ against a constructive tour | Any symmetric distances, any $d$; certificate | $\Theta(n^2 d + kn^2)$ | Held and Karp [1970], Polyak [1969] |
| Calibrated 1-tree | $c_k$ times either bound, $c_k$ fitted per budget on the training split | As trained | $\Theta(n^2 d + kn^2)$ | This work |

### 4.2 Datasets

We evaluate on a multidimensional synthetic set, a diverse 2D synthetic set, and TSPLIB95, whose EUC_2D instances form the third benchmark and whose screened non-EUC_2D instances form the fourth, constituted in Section 6. The synthetic sets are held-out splits from our generation pipeline (Appendix B). The evaluation assumes symmetric metric distances; Section 6 lists the ten EXPLICIT TSPLIB matrices whose structural triangle-inequality violations put them outside every metric-dependent aggregate here.

#### 4.2.1 Multidimensional Synthetic

The benchmark contains 16,920 held-out instances with $n \in [5, 1000]$ and $d$ drawn from the 17 training-visible values plus $d = 100$, sampled by stratified dimension–size–grid buckets from the same four per-axis generators as the training set: uniform, normal, clustered ($k$-means [Lloyd, 1982]), and autoregressive-correlated. Stratification enforces zero row overlap with the training split, but the generator families and axis mix of Section 3.2 are shared, so this benchmark measures held-out sampling and not held-out geometry. Section 4.5 withholds whole dimensions and whole size ranges from training to separate the two. The full $(d, n)$ count grid is given in Appendix B.3 (Table 10).

#### 4.2.2 2D Diverse Synthetic

The benchmark contains 2,580 instances with $d = 2$, $n \in [5, 1000]$, and coordinate-grid side length $G \in [1000, 10000]$. It was generated by the separate 2D benchmark pipeline; an identifier-level audit found zero overlap with the 106,272-row training/validation/test corpus. Five distribution classes (Table 2) each probe a different assumption made by classical 2D estimators. Two of the thirteen sub-generators test geometry the training corpus does not contain: Line Noise, which is near-collinear, and `grid`, a jittered square lattice. Near-collinear and piecewise-linear point sets are a standard degenerate TSP test case [Smith-Miles et al., 2010, Bossek et al., 2019]. Appendix B.2 gives the construction of both classes, and Table 17 reports `grid` as its own row instead of inside its class aggregate.

Table 2: 2D benchmark dataset composition (2,580 instances).

| **Class** | **Generators** | **Count** | **Stress target** |
|---|---|---|---|
| Isotropic [Çavdar and Sokol, 2015, Golden and Stewart, 1985] | Uniform, Normal, Triangular, Truncated Exp. | 840 | Control: $A_{\text{hull}} \approx A_{\text{box}}$. |
| Biased / Skewed [Çavdar and Sokol, 2015] | Squeezed, Uni-Tri, Tri-Squeezed, Correlated | 840 | Aspect-ratio distortion: $\sigma_x \neq \sigma_y$. |
| Geometric Struct. [Reinelt, 1991, Bossek et al., 2019] | Grid†, Boundary (hollow), X-Central | 630 | Lattice regularity, hollow voids. |
| Clustered [Golden and Stewart, 1985, Bossek et al., 2019] | $k$-Means ($k \in [5, 20]$) | 60 | Bimodal edge-weight distribution. |
| Line Noise† | Boundary-clipped $y = mx + b + \epsilon$; realized as a 2–4 segment union | 210 | Locally 1D structure; hull/box divergence. |
| **Total** | | **2,580** | |

† Not represented in training; included to measure generalization to unseen geometries.
`Grid` is one of the three Geometric Struct. generators; Table 17 reports it as its own row.

#### 4.2.3 TSPLIB95 EUC_2D

The benchmark contains 78 EUC_2D instances with $n$ from 51 (`eil51`) to 18,512 (`d18512`) [Reinelt, 1991]. Every numerically evaluated 2D baseline is defined on all 78, the paper's common instance set. The 23 retained non-EUC_2D instances are evaluated separately in Section 6. For the largest instances, a Delaunay-based MST construction [Delaunay, 1934] avoids materializing the dense distance matrix.

### 4.3 Results

GART 2.0 holds the lowest aggregate MAPE and SDPE of the roster on each benchmark below, and the margins concentrate where instances are large. The appendix tables disaggregate every aggregate by size, dimension and generator class, using the metrics of the section preamble; each estimator in a bucket uses the stated common instance set.

For synthetic tables, the separate reference-tour row reports the median generation time stored in the canonical benchmark results. The logs contain one timed estimator invocation per instance, with no recorded warm-up repetitions or within-instance dispersion; feature and inference time exclude file I/O and reference-tour generation. All timings are therefore descriptive single-machine measurements, and none supports an inferential comparison. We omit TSPLIB reference-generation timing because the available logs have incomplete instance coverage and mix hardware sources. Two timing protocols appear in this paper and they are never mixed in one comparison: the Time columns of the multidimensional and 2D detail tables carry one cold harness invocation per instance, including interpreter and just-in-time warm-up, while Table 23 and every ladder of Section 5 carry a solo protocol, one estimator per process with warm-up outside the clock, run in a quiet window except where Section 5.4 discloses otherwise; the first reads an order of magnitude above the second for the same work. On the solo protocol, *corpus median* names one statistic wherever this paper uses it: the cost of the typical instance, the median over the corpus of each instance's own median over its repeats.

#### 4.3.1 Multidimensional Benchmark

The multidimensional benchmark is 16,920 held-out instances, of which the 16,846 with a recoverable reference cost are scored (Appendix E). We report it in two orthogonal slices of the same instances, by dimension (Table 20) and by size (Table 21), with the error distributions in Figure 1.

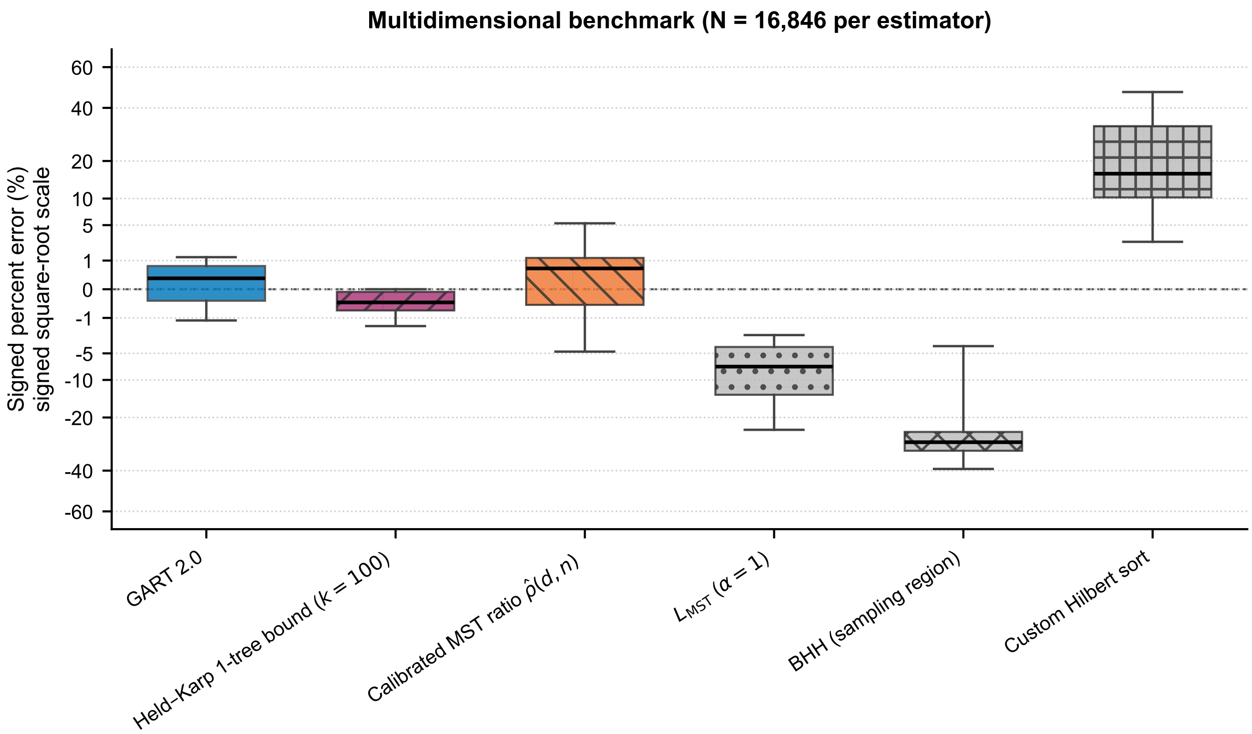


Figure 1: Signed percent error on the multidimensional benchmark ($N = 16{,}846$ per estimator); the certified 1-tree bound is drawn at ascent budget $k = 100$.

GART 2.0 obtains 0.62% MAPE and 0.99% SDPE overall. The strongest baseline in the roster of Section 4.1 that uses no learned model is the calibrated ratio $\hat{\rho}(d, n)$ at 1.82%/2.94%, so the learned $\alpha$ improves on a lookup table of 102 constants fitted on the same training split by a factor of 2.9 on MAPE and 3.0 on SDPE. The $\alpha = 1$ floor reaches 9.71%/7.26%, the custom Hilbert sort 21.21%/14.40%, and BHH given the exact sampling region 28.01%/13.57%. Figure 1 draws the certified Held–Karp 1-tree bound beside them, and the shape of its box is a difference in kind rather than in degree: it lies wholly at or below zero because the bound cannot exceed the optimum, so the sign of its error is known before the instance is seen. Several estimator boxes also fall on one side of the line, but only as a fitted tendency observed after the fact; none of them can promise a side on the next instance. Section 5.3 scores that bound against GART 2.0 on cost and accuracy together.

SDPE decreases monotonically across all six size buckets, from 1.56% at $n \leq 10$ to 0.46% at $n \in [201, 500]$ and 0.45% at $n \in [501, 1000]$. Across dimension groups it falls from 2.37% at $d = 2$ to 0.48% at $d \in [30, 50]$, and falls again, to 0.44%, at the unseen $d = 100$, while MAPE rises there from 0.29% to 0.59%. The standard deviation of $\alpha$ over the training split falls monotonically with dimension, from 0.1837 at $d = 2$ to 0.0762 at $d = 50$, so the target concentrates as $d$ grows and the prediction problem becomes easier in exactly the direction the extrapolation runs. The $d = 100$ result is therefore favorable-direction extrapolation, and Section 4.5 supplies the unfavorable-direction tests. Figure 2 draws the whole accuracy surface at once, per dimension and size band, and the held-out row is the one visible break in it.

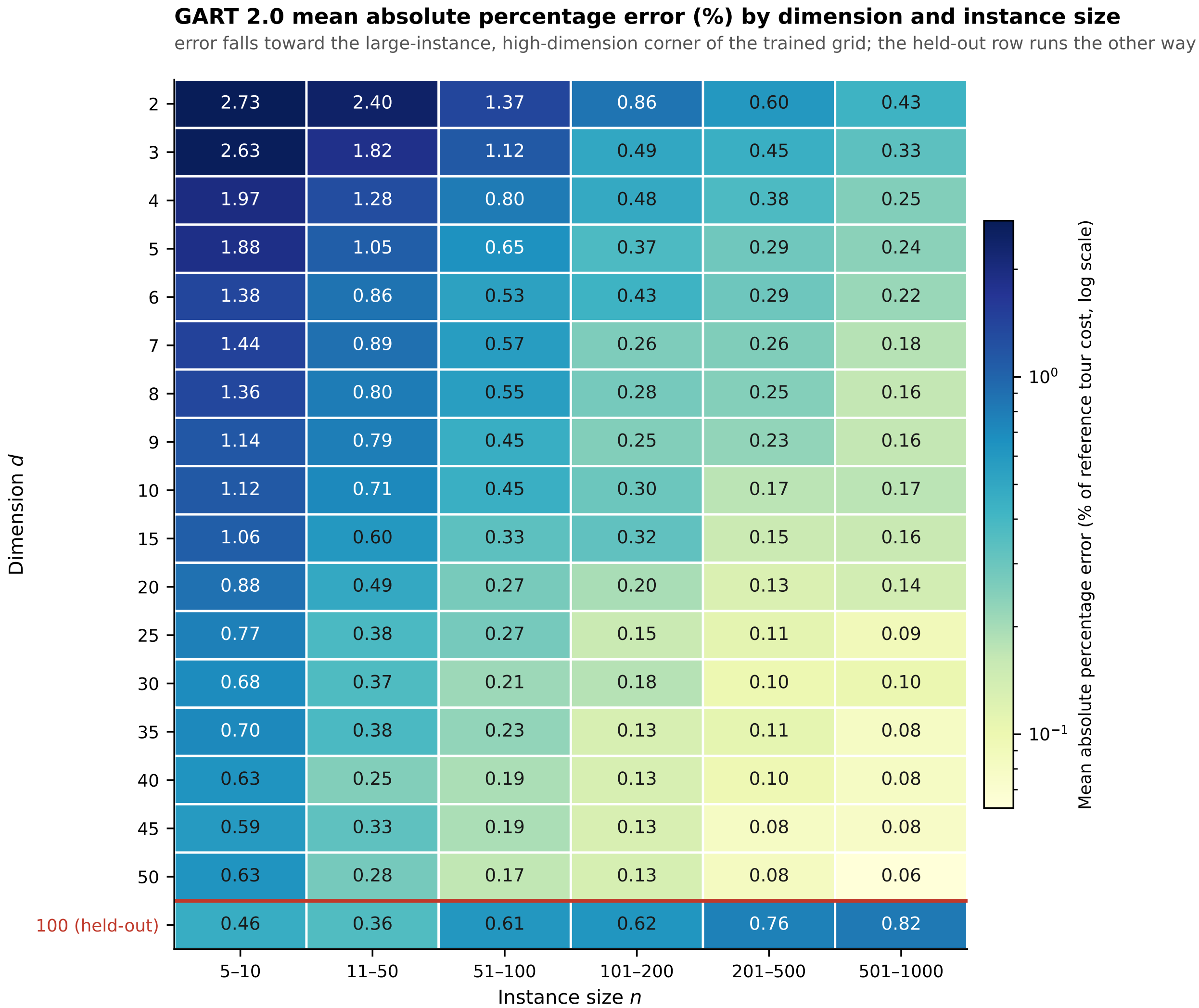


Figure 2: GART 2.0 MAPE by dimension and instance-size band, log color scale. The red rule separates the trained dimensions from the held-out $d = 100$ row, the one row whose error grows with instance size.

BHH's 28.01% error deserves its own explanation, because it is not a defect of the reimplementation. The dominant term is the constant: for $d \geq 4$ the scored form substitutes a conjectured large-$d$ asymptotic for $\beta_d$ (Appendix F), and the row's signed error is negative at every $d \geq 3$, the direction an understated constant produces. Two premises of the theorem fail here as well: only 68.7% of axes are uniform, so the region measure $G^d$ exceeds the effective support of the rest — visible as the positive signed error at $d = 2$ — and $n \leq 1000$ with $d \geq 4$ is far from the asymptotic regime the theorem describes, where the sample must fill the region. Supplying the exact region rather than the convex hull nonetheless cuts its MAPE from 39.07% to the 28.01% in Table 20, which isolates how much of the convex-hull form's error is an input substitution; the convex-hull row itself is in the released tidy tables rather than here, because at $d > 3$ that hull is replaced by the coordinate bounding box (Appendix F) and the row would mix two region definitions across dimensions.

#### 4.3.2 2D Diverse Benchmark

The 2D benchmark asks whether accuracy survives geometry the multidimensional grid never produces. Its 2,580 instances fall in five distribution classes, each probing a different geometric regime; Figure 3 shows the common-coverage error distributions and Appendix M disaggregates by size.

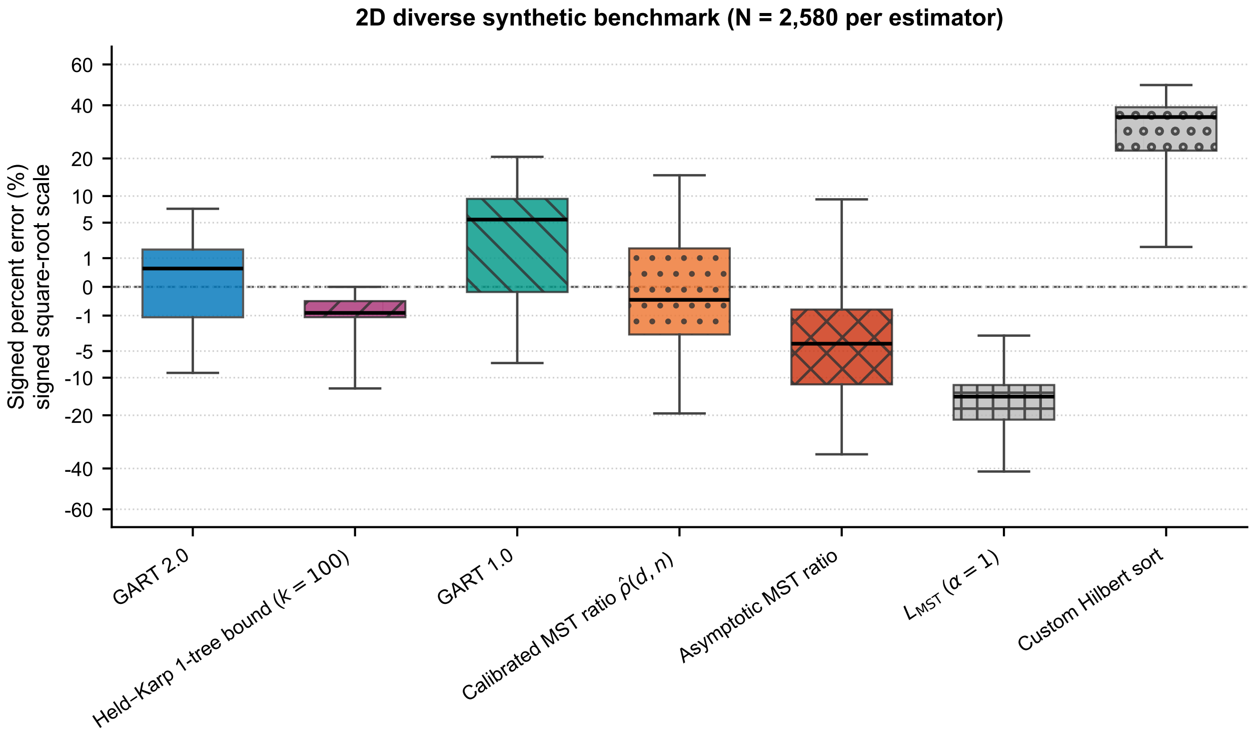


Figure 3: Signed percent error on the diverse 2D benchmark ($N = 2{,}580$ per estimator); axes and bound budget as in Figure 1.

GART 2.0 obtains 2.89% MAPE and 4.68% SDPE overall, against 5.76%/9.38% for the calibrated ratio $\hat{\rho}(d, n)$, 7.30%/7.90% for GART 1.0, 9.20%/11.55% for the asymptotic ratio, and 17.95%/10.27% for the $\alpha = 1$ floor. Its SDPE stays below 5.9% in every size bucket and falls to 2.75% in the $[501, 1000]$ bucket, and its aggregate is less than half the asymptotic ratio's. The custom Hilbert sort has 30.95% MAPE against 14.19% SDPE, a gap that identifies its error as a near-constant positive offset and not scatter, which is what a space-filling-curve tour should produce. The certified bound plotted beside them in Figure 3 is the one row there that is not an estimator, and it reads that way: its interquartile range is the narrowest in the figure and it never crosses zero, which is what a one-sided guarantee looks like when it is drawn next to two-sided estimates.

Table 17 disaggregates the five stress classes, which the aggregate hides. GART 2.0 ranges from 1.47% MAPE on the isotropic class to 10.82% on Line Noise, and every estimator degrades on Line Noise in the same direction: the calibrated ratio moves from 3.31% to 20.45%, the asymptotic ratio from 7.12% to 25.88%, and the $\alpha = 1$ floor from 17.15% to 34.13%. Near-collinear point sets drive $\alpha$ toward its upper bound of 2, and no estimator anchored on a typical ratio tracks that. Line Noise is not the only unrepresented geometry, and the other one fails in the opposite direction. On the 210 `grid` instances GART 2.0 reads 7.07% MAPE against 1.96% on the 420 Geometric Struct. instances that are represented, and its MSPE equals that MAPE exactly, so every one of those predictions is high. Realized $\alpha$ averages 1.05 on `grid` against 1.56 on Line Noise, so the two unrepresented generators sit at opposite ends of the target range and the estimator errs toward the middle on both. Both therefore confound unseen geometry with an extreme target value. One consequence is worth stating exactly: on `grid` the $\alpha = 1$ floor is more accurate than the released model, at 4.54%, the only row anywhere in this paper on which the floor beats it.

#### 4.3.3 TSPLIB95 EUC_2D Benchmark

This is the transfer test: real geography and circuit layouts, no sampling region, and the narrowest margin in the paper.

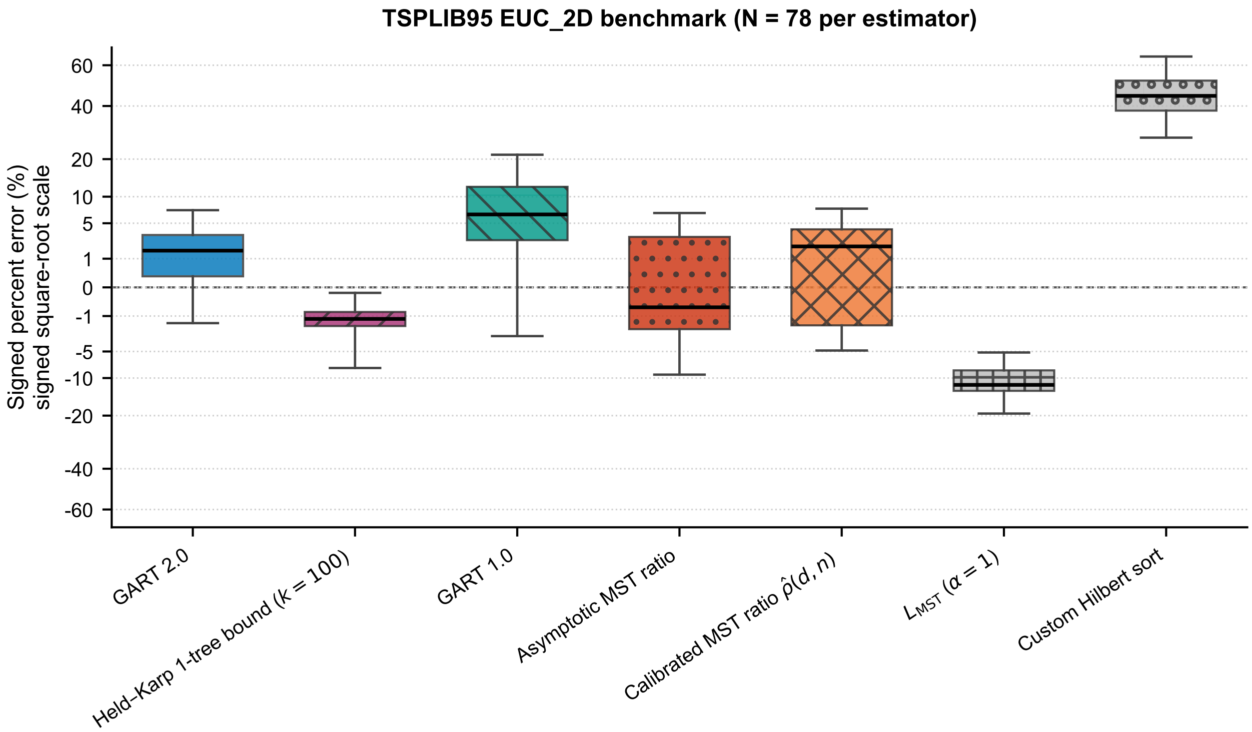


Figure 4: Signed percent error on the TSPLIB95 EUC_2D benchmark ($N = 78$ per estimator); axes and bound budget as in Figure 1.

On the full EUC_2D benchmark GART 2.0 obtains 2.53% MAPE and 2.92% SDPE, against 3.74%/4.48% for the calibrated ratio $\hat{\rho}(d, n)$, 3.53%/4.74% for the asymptotic ratio, and 8.43%/7.45% for GART 1.0. Table 23 reports all 78 instances, spanning $n = 51$ to 18,512, in three prespecified size buckets (23/16/39 instances), and Figure 4 shows the error distributions; the coarse $n > 400$ bucket should not be read as a homogeneous asymptotic regime. This is the narrowest margin in the paper, and it is narrow for a structural reason: $\alpha$ on these 78 instances has mean 1.1306 and standard deviation 0.0558, so a single well-chosen constant is already close. The MAPE-minimizing constant on these instances is 1.1275 and reaches 3.52%, so no constant multiplier does better on MAPE, and GART 2.0's 2.53% improves on that bound by 0.99 percentage points. The dispersion gap is the larger one, 2.92% SDPE against 4.74%, but SDPE alone cannot carry that comparison: for $\hat{L} = cL_{\mathrm{MST}}$ the signed error is $c/\alpha - 1$, so SDPE is proportional to $c$ and shrinking $c$ drives it to zero at ruinous MAPE. The comparison holds because GART 2.0 attains the lower SDPE at the lower MAPE. Figure 4 puts the certified bound on the same axis, and the two interquartile ranges fall on opposite sides of it: GART 2.0's lies wholly above the zero line, the bound's wholly below. On the stratum where GART 2.0 is strongest the two are therefore comparable in magnitude and opposite in direction, and only one of them fixes its direction in advance.

The advantage is not uniform across size. In the coarse $n > 400$ bucket the GART 2.0 and asymptotic-ratio SDPE intervals overlap, on $[3.20, 3.62]$. In the post hoc $n > 10{,}000$ slice of five instances GART 2.0 is lower on both metrics, 0.57% MAPE and 0.72% SDPE against the asymptotic ratio's 2.74% and 1.46%. The largest instance, `d18512`, is more than $18\times$ the training-size cap. For i.i.d. points of bounded density, normalized TSP and MST lengths each converge to dimension-dependent constants [Beardwood et al., 1959, Steele, 1988], which motivates a fixed-ratio baseline but does not establish that the heterogeneous $n > 400$ bucket is asymptotic; the advantage at extreme sizes is an observation on five instances and not proof of a convergence mechanism.

Both classical estimators are far behind on this benchmark, and the reason is the same one that motivates the paper. Çavdar–Sokol obtains 23.87% MAPE and BHH 25.14%. TSPLIB instances are drawn from real geography and circuit layouts rather than from a uniform density on a known region, so an area-based estimator is applied outside the conditions it was derived under: BHH receives a convex hull in place of a sampling region that does not exist, and Çavdar–Sokol is evaluated on graphs that do not carry a node at the corners of their covering rectangle, which every one of its training graphs did. Both overpredict, by $+13.66\%$ and $+20.07\%$ respectively. These rows measure the cost of that mismatch, not the quality of the cited methods, which is why Section 4.4 evaluates both where their assumptions hold.

### 4.4 Matched-Domain Comparison

This section evaluates each classical estimator where its published assumptions hold. The 210 instances of the `random` generator, the Uniform member of the Isotropic class of Table 2, are drawn i.i.d. uniformly on $[0, G]^2$, which is exactly the sampling model BHH assumes, and the region measure $A = G^2$ is known exactly. Çavdar–Sokol carries the same single row here that it carries above, because it consumes no region: between the two panels its instance set changes and nothing else, which is what makes the pair readable as a domain effect. Table 16 reports both panels.

Both estimators' errors fall substantially on the matched domain. BHH falls from 23.80% MAPE on the full 2D benchmark to 8.91% here. Two things change at once between its two panels, the region input and the instance set, so that figure is the combined effect and not the isolated value of the region input. What is left is a systematic underprediction, $-8.65\%$ signed against 7.76% SDPE, which is what an asymptotic constant evaluated at $n$ as low as 5 should do. Çavdar–Sokol falls from 18.24% to 8.16%, and here only the instance set changed, so that figure is a clean domain effect. It has the lower MAPE of the two classical estimators on every panel of Table 16, and the lower median absolute error here, but not the lower dispersion, and it is also the least even: at 14.28% SDPE its dispersion is the widest in the panel, and its median absolute error of 2.84% sits at just over a third of its mean, so it is close on most uniform instances and badly wrong on a few.

GART 2.0 beats both classical estimators on the matched domain, on both metrics, by a smaller and more credible margin than the full-benchmark tables suggest. On the 210 uniform instances it obtains 1.31% MAPE against Çavdar–Sokol's 8.16%, BHH's 8.91% and the $\alpha = 1$ floor's 15.60%. A factor of 6.2 over Çavdar–Sokol on i.i.d. uniform draws states what this model adds over the classical literature in the plane; the wider gaps in the full-benchmark table describe domain mismatch instead.

### 4.5 Generalization Beyond the Training Grid

The multidimensional test split is held out by row, not by geometry: it shares the four generators and the 68.7% uniform axis mix of the training split, and it is stratified on the same $(d, n,$ grid size$)$ cells. Its 0.62% MAPE therefore measures held-out sampling, and a reader is entitled to ask how much of that is interpolation within a seen stratum. We answer with three refits that hold the hyperparameters and seed fixed and vary only the training data, so no run-to-run noise confounds the comparison. A baseline refit reproduces the released model bit-for-bit at 1,118 trees, reading 0.6177% MAPE over the 16,846 scored rows; the refit script enforces this, requiring an identical serialized booster and a maximum absolute prediction difference of zero over all 106,272 corpus rows before it will report anything else. The pipeline is therefore deterministic under the fixed seed on this machine, and every difference below is attributable to the training data and not to refit noise.

Withholding both $d = 15$ and $d = 25$ from training, 11.8% of the training rows, raises test MAPE at $d = 15$ from 0.50% to 0.58% and at $d = 25$ from 0.35% to 0.38%, while the neighboring trained dimensions $d \in \{10, 20, 30\}$ do not degrade at all, moving from 0.43% to 0.42%. The loss is specific to the withheld strata rather than to the reduced training volume, and it takes the form of a bias shift: mean signed error at $d = 15$ moves from $-0.008\%$ to $+0.134\%$. Training only on $n \leq 200$ and testing on $n \in (200, 1000]$ raises MAPE from 0.41% to 0.59%, again almost entirely as a uniform offset, since SDPE moves only from 0.47% to 0.56% while mean signed error rises from $+0.28\%$ to $+0.51\%$, a factor of 1.8.

Neither held-out regime degrades by more than two tenths of a MAPE point against its matched in-grid baseline. The estimator does exploit having seen a cell of the grid; off the grid it acquires a systematic offset and keeps its structure. We demonstrate no correction for that offset. The Line Noise class of the 2D benchmark supplies the complementary test on geometry rather than on scale, and Section 4.3.2 shows the cost there is larger. Section 4.6 decomposes that cost.

### 4.6 Failure on Near-Collinear Geometry

Section 4.3.2 reports 10.82% MAPE on the near-collinear Line Noise class against 1.47% on the isotropic class, and the error is almost entirely one-sided: MSPE is $-10.82$, so the model under-predicts. Two explanations compete: the training split may never show the model this regime, or the feature set may be unable to represent it. We separate them below, and both hold.

The support gap is exact. The training split contains zero instances with $\alpha > 1.5$ at $n \geq 200$, and every training instance with $\alpha > 1.7$ has $n \leq 10$. Its $\alpha$ distribution has a 99th percentile of 1.544, and 54% of the 210 benchmark Line Noise instances lie above that percentile. No tree can map large $n$ at high $\alpha$ onto a high $\alpha$ when the fit never contains the combination. The symptom matches: regressing predicted $\alpha$ on true $\alpha$ over the 90 Line Noise instances at $n \geq 200$ gives a slope of 0.36 against a correlation of 0.95, so the model ranks those instances almost perfectly and declines to move its $\alpha$ level.

A $2 \times 2$ ablation separates the two causes. Holding hyperparameters and seed fixed, we vary the feature set, extending the 30-feature baseline with five degeneracy, local-intrinsic-dimension and MST-topology descriptors, and we vary the training support, adding 874 newly generated and solved instances at large $n$ and high $\alpha$ to the training split only. Measured by the Line Noise slope at $n \geq 200$, features alone move the baseline from 0.294 to 0.569, support alone moves it to 0.332, and the two together reach 0.798. Neither intervention approaches sufficiency alone and the two are not additive, so the deficit is joint. The ladder is anchored on the 30-feature model, whose slope on this slice is 0.294; the released 31-feature model reaches 0.362 on the same slice, so these figures compare the two interventions against each other rather than against the released estimator.

We attempted the repair twice against criteria fixed before either candidate was scored, and rejected both. Adding all 874 instances to the training split clears every one of nine gates, moving 2D MAPE from 2.89% to 2.04%, Line Noise MAPE from 10.82% to 5.83% and the slope from 0.366, that experiment's own baseline reading, to 0.838, but two defects disqualify it. The slope is not a property of the model: eight permutations of the row order of the identical training rows, at one seed, span $[0.637, 0.803]$, and the reported 0.838 lies above that entire band. And the gain is not a cross-family result, because the augmentation's $d = 2$ lattice generator and the benchmark's `grid` generator place their sites, spacing, cell centers and $\pm 5\%$ jitter identically, so half of the 2D gain is measured on the candidate's own training geometry. A de-contaminated second candidate, re-run over seven seeds fixed in advance, removes that overlap and clears ten of eleven gates at a median slope of 0.735, but forfeits the paper's one distribution-independent dispersion claim to do so: its Holm-adjusted advantage over the asymptotic ratio on TSPLIB reaches only $p = 0.065$ at the median against 0.0048 for the released model. We did not make that trade. A per-family multiplicative constant, an oracle a deployed estimator does not have, recovers 97.9% of the first candidate's Line Noise gain by itself, so most of what either candidate buys is a per-family offset rather than a repaired response to geometry. The degenerate-geometry failure is therefore diagnosed and not repaired, the released model is unchanged, and the full criteria, seed bands and per-candidate statistics are released with the results.

### 4.7 Rank Agreement and Calibration

Everything above judges GART 2.0 on pointwise accuracy. Many of the downstream uses motivated in Section 1 never consume a single cost in isolation: they compare instances and act on the comparison, asking which candidate node set is cheaper or which route split to keep. For those uses the estimator must *order* instances the way the stored references do even when its absolute numbers are off, so we measure order agreement directly. Spearman's $\rho$ correlates the cost *ranks* and Kendall's $\tau$ is the concordant minus discordant fraction of instance pairs; both equal 1 for perfectly matching order and 0 for an unrelated one, while the close-pair statistic below is a plain concordant fraction and sits at $0.5$ under an unrelated ordering. Global rank correlation is close to 1 for every estimator here, so it separates almost nothing. Table 18 reports both metrics with $L_{\mathrm{MST}}$ itself as the control. GART 2.0 obtains Spearman $\rho$/Kendall $\tau$ of 0.9994/0.983 on 2D, 1.0000/0.998 on the multidimensional set, and 0.9993/0.987 on TSPLIB EUC_2D; the $\alpha = 1$ control obtains 0.9961/0.962, 0.9996/0.986, and 0.9986/0.979. Cross-instance cost varies over orders of magnitude and $L_{\mathrm{MST}}$ already tracks that variation, so a high $\rho$ demonstrates nothing about the learned ratio. Every constant multiplier is a monotone transform of $L_{\mathrm{MST}}$ and therefore scores identically to it, which is why the asymptotic-ratio and $\alpha = 1$ rows coincide exactly on 2D and TSPLIB.

Close-pair ordering is the discriminating measure. For pairs satisfying $|L_A - L_B|/\max(L_A, L_B) < 5\%$, GART 2.0 orders 74.5% of 74,826 2D pairs, 92.4% of 1,647,719 multidimensional pairs, and 72.2% of 54 TSPLIB pairs correctly, against 71.0%, 69.1%, and 61.1% for the $\alpha = 1$ control. The multidimensional gain of 23.4 percentage points is the substantive one; the 2D gain of 3.4 points is not, and on TSPLIB 54 pairs cannot support an 11-point claim. At a 10% threshold GART 2.0 obtains 83.6%, 96.1%, and 81.0%. Every pair set is enumerated exhaustively rather than sampled (Appendix I), and the percentages remain descriptive because the pairs share instances and we compute no block-bootstrap interval over them. GART 1.0 trails GART 2.0 on the TSPLIB close-pair statistic at the tighter threshold, at 70.4%. Cost accuracy and pair ordering are distinct properties, and 54 pairs resolve none of these gaps.

Two calibration caveats bound the cost-level results. The $n > 400$ TSPLIB bucket has $R^2_\alpha = -0.071$, verified with both a library implementation and the sum-of-squares definition computed directly; a negative value means the model predicts $\alpha$ worse than the bucket mean does. That bucket has low target variance and an upward mean prediction offset while its tour-cost MAPE remains 2.90%, so the cost-level result does not remove the ratio-calibration failure. The aggregate margins are also not uniformly significant. We test each comparison on the instances where both estimators produce a finite prediction, using the paired difference in absolute percent error with a 1,000-resample paired bootstrap and a Wilcoxon signed-rank test (Appendix J). On TSPLIB EUC_2D the difference between GART 2.0 and the asymptotic MST ratio is $-1.00$ percentage points with a 95% interval of $[-1.77, -0.22]$ and $p = 0.0049$: on that benchmark GART 2.0's MAPE advantage is small but resolvable, and its larger gain is dispersion, 2.92% SDPE against 4.74%.

### 4.8 Computational Cost

Cost is dominated by feature extraction, not by the ensemble. In the TSPLIB timing log GART 2.0 spends 92.90% of its wall time on feature extraction and MST construction against 5.03% on LightGBM inference, so the tree traversal is not the target for any speed work and the ensemble size is not what the cost claim turns on. The Time column of Table 23 puts every estimator on one protocol: one estimator per process, a single thread, the median of 11 repeats. On that protocol GART 2.0 costs 6.12 ms over the full EUC_2D set, matching GART 1.0, roughly twice the 2.57–3.08 ms of the rows that build an MST or a Hilbert sort, and roughly an order of magnitude over the 0.645–0.880 ms of the two classical closed forms, which compute a convex hull and then evaluate an algebraic expression. That ordering is the price of the feature set and it is stable across the size range; at $n > 400$ GART 2.0 is the most expensive row in the table at 20.54 ms.

Those multiples are the wrong comparison for the deployment case, because the alternative in that case is a solve. On the multidimensional benchmark the median prediction and the median reference-tour generation cost the same, 121 ms against 122 ms, since most of those instances are small enough to solve outright and an estimator buys nothing. The argument holds only where the solve is expensive, and there it holds decisively: at $n \in [501, 1000]$ the reference tour takes 3,799 ms against GART 2.0's 460 ms, and on TSPLIB `d18512`, at $n =$ 18,512 and more than $18\times$ the largest training instance, GART 2.0 predicts in 239 ms. Every row in this accounting is an estimator, and none of them certifies anything. Section 5 runs the accounting again against a comparator that does.

## 5 Comparison Against a Certified Lower Bound

Every baseline of Section 4.1 is an estimator: it returns a number that carries no guarantee and is scored only by how close that number lands. This section admits a comparator of a different kind, the Held–Karp 1-tree lower bound, and uses it to establish where a learned estimator is the right instrument and where it is not. The bound is the hardest available test, because it consumes no training corpus, certifies its own output, and is defined on every instance in this paper. We score it on all four benchmarks, on cost and accuracy together. On the aggregate of every one of them the bound is both cheaper and more accurate than GART 2.0, on TSPLIB EUC_2D once scaled by a single training-split constant, so the estimator is off the cost/accuracy Pareto front of all four and no aggregate in this paper argues otherwise. What the aggregates conceal is the one property that separates the two families, which is how each scales in $n$: the ascent is quadratic and the planar estimator is not. That difference, and not any error figure, fixes the operating regime we recommend at the end of this section.

### 5.1 The Held–Karp 1-tree Bound

Two properties separate this row from every other row in this paper, and both follow from the same relaxation. Fix a node $s$. A *1-tree* is a spanning tree on $V \setminus \{s\}$ together with any two edges incident to $s$; it has $n$ edges and total degree $2n$, and every Hamiltonian tour is a 1-tree, so the minimum 1-tree — a minimum spanning tree on $V \setminus \{s\}$ plus the two cheapest edges at $s$ — is a relaxation of the tour problem. Introduce node potentials $\pi \in \mathbb{R}^n$ and transformed costs $c^{\pi}_{ij} = c_{ij} + \pi_i + \pi_j$. Under $c^{\pi}$ every tour's cost shifts by exactly $2\sum_i \pi_i$, because every node has degree two, so the optimal tour under $c^{\pi}$ costs $L_{\mathrm{TSP}} + 2\sum_i \pi_i$ and for *any* real $\pi$

$$w(\pi) \;=\; L_{\mathrm{1tree}}(\pi) \;-\; 2\sum_{i \in V} \pi_i \;\le\; L_{\mathrm{TSP}}, \tag{2}$$

which is the Held–Karp bound [Held and Karp, 1970, 1971]. The map $w$ is concave and piecewise linear, a subgradient at $\pi$ is $g_i = \deg_i(\pi) - 2$, and $\max_\pi w(\pi)$ is approached by subgradient ascent. Subgradient ascent is not monotone in $w$, so the scored quantity at budget $k$ is the incumbent $\hat{w}_k = \max_{j \le k} w(\pi_j)$, and $k$ is an explicit knob trading cost against tightness; on any one ascent the incumbent is non-decreasing in $k$ by construction. Validity needs neither $\pi \ge 0$ nor $c^{\pi} \ge 0$: the argument above is sign-free, and the spanning tree is built on the perturbed costs directly.

The first property is the certificate. Eq. (2) holds at every $\pi$, so $\hat{w}_k$ is a *certificate* whatever the ascent did: the printed number is a proven lower bound on that instance's optimal tour, and a reader can verify it from the 1-tree alone. No other row in this paper certifies anything about its own output, GART 2.0 included. The second is that the bound consumes no training corpus, so no question of coverage, distribution shift, or leakage arises about it at all.

The certificate is paid for with one-sidedness. Because $\hat{w}_k$ never exceeds the optimum, its error against a reference tour is a duality gap plus whatever that reference is itself above the optimum, and not the two-sided spread of an estimator. Turning the bound into an estimator requires one fitted scalar, and we report that row too: the *calibrated* bound $c_k\,\hat{w}_k$, where $c_k$ is a single scalar fitted per budget on the planar training split and never on any scored instance. That row is not a certificate, since $c_k > 1$ pushes some predictions above the optimum, and it is not training-free. It is the like-for-like comparator against a trained estimator, and it is the stronger of the two rows on every corpus that carries it; its constant is planar-fitted, so the multidimensional ladder reports the raw bound alone.

Both ascents run on all four benchmarks, and each headline table reports the arm that leads on its own corpus. The planar tables use the Volgenant–Jonker step schedule with Helsgaun's direction smoothing [Volgenant and Jonker, 1982, Helsgaun, 2000]. The multidimensional tables use a Polyak step [Polyak, 1969] against a constructive upper bound, a nearest-neighbor tour improved by 2-opt and built from coordinates alone, with the stored label never read during the ascent. Neither ascent is proved to attain $\max_\pi w(\pi)$, so every accuracy figure below is a floor on what this family reaches and not a ceiling. The two arms disagree: on the multidimensional split the Volgenant–Jonker arm returns the higher bound on 4.30% of instances, and taking the per-instance maximum of both, at the cost of running both, improves the converged multidimensional error from 0.0518% to 0.0486%. Validity was checked against exhaustively enumerated optima on small instances and against a second exact solver, with no violation of Eq. (2) anywhere.

### 5.2 Results on TSPLIB95 EUC_2D

Cost is measured on the protocol of Table 23, one estimator per process inside a single quiet window, and both methods are compared on the 77 of the 78 EUC_2D instances the bound's dense kernel covers; the one excluded instance is reported at the end of this subsection. Every ladder in this section shares its columns; the one exception is the multidimensional ladder, which carries a single MAPE column, the raw bound alone. The ms column is the median per-instance time, and the accuracy beside it is a MAPE and therefore a mean. $\times$ compares the bound's time with GART 2.0's: the ratio of the printed medians here and on the 2D and non-Euclidean ladders, and the median of back-to-back per-instance ratios on the multidimensional ladder. Raw is the certified incumbent $\hat{w}_k$; Cal. is $c_k \hat{w}_k$ with $c_k$ fitted per budget on the planar training split; bold marks every cell both cheaper and more accurate than GART 2.0; a *rung* is one budget row. Table 3 gives the ladder.

Read on the corpus median, the raw certified bound first matches GART 2.0's MAPE at an ascent budget of 50, where it costs 1.46 times GART 2.0 for a margin of 0.0041 percentage points, and where the paired win rate is exactly 50.0%. The calibrated row settles what the raw row only balances. At a budget of 25 it costs 0.90 times GART 2.0 and reaches 2.00% MAPE against 2.56%: strict domination on both axes, on the same instances and the same protocol. That multiple is a ratio of typical-instance costs; summed over the corpus instead, the same pair reads 24.94, because the bound is cheap on the small instances that set the median and dear on the large ones that set the total. The ladders report the first statistic, the one a caller holding one instance faces. GART 2.0 is therefore not on the cost/accuracy Pareto front of this corpus median once the family is admitted.

Splitting by size shows what the median hides, and the split rather than the median is this section's useful output. On the 23 smallest instances the calibrated bound at that same budget costs 0.16 times GART 2.0 at 1.69% MAPE against 2.01%, and even the raw uncalibrated certificate reaches 22.2% lower error at 0.46 times the cost. In the middle bucket nothing dominates, and the raw bound needs the top of the ladder to match at all. In the largest bucket the raw bound matches at 6.65 times the cost and again nothing dominates. The reversal is monotone in $n$, which is why the output of this comparison is a rule keyed to instance size.

The cost column carries two disclosures. The bound is far more load-sensitive than GART 2.0, its median rising by a factor of 1.25 between a quiet and a noisy window at the crossing budget of 50 against 1.05 for GART 2.0, so every published repeat is from one quiet window and the noisy repeats are retained only as the control; mixing the two would have overstated the bound's cost and flattered GART 2.0. And the excluded instance is real cost rather than a hidden exclusion: on `d18512`, above the bound's dense kernel switch, the bound costs 167 times GART 2.0 at the crossing budget and 2,078 times at the top of the ladder, 493 seconds against 237 milliseconds (the 239 ms of Section 4.8 is the same work under the harness protocol).

Table 3: TSPLIB95 EUC_2D cost/accuracy ladder over the 77 matched instances; column conventions are defined at the start of this subsection.

| | $n \in [51, 150]$, $N = 23$ | | | | $n \in [151, 400]$, $N = 16$ | | | | $n > 400$, $N = 38$ | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Row | ms | $\times$ | Raw | Cal. | ms | $\times$ | Raw | Cal. | ms | $\times$ | Raw | Cal. |
| GART 2.0 | 3.73 | 1.00 | 2.01 | | 4.55 | 1.00 | 2.38 | | 18.29 | 1.00 | 2.97 | |
| $k = 0$ | 0.19 | 0.05 | 11.83 | 3.12 | 0.45 | 0.10 | 11.31 | 3.59 | 24.49 | 1.34 | 9.13 | 4.01 |
| $k = 10$ | 0.37 | 0.10 | 8.47 | 2.04 | 0.91 | 0.20 | 9.85 | 3.76 | 43.93 | 2.40 | 7.63 | 2.53 |
| $k = 25$ | 0.59 | 0.16 | 2.91 | **1.69** | 1.59 | 0.35 | 4.84 | 3.09 | 72.52 | 3.97 | 3.48 | 1.72 |
| $k = 50$ | 0.96 | 0.26 | 2.09 | **1.68** | 2.79 | 0.61 | 3.82 | 3.03 | 121.59 | 6.65 | 2.31 | 1.56 |
| $k = 100$ | 1.72 | 0.46 | **1.56** | **1.20** | 5.16 | 1.13 | 2.97 | 2.29 | 222.22 | 12.15 | 1.96 | 1.33 |
| $k = 200$ | 3.22 | 0.86 | **1.57** | **1.22** | 10.10 | 2.22 | 2.96 | 2.28 | 427.85 | 23.40 | 1.88 | 1.26 |
| $k = 500$ | 7.88 | 2.11 | 0.97 | 0.74 | 25.65 | 5.64 | 1.93 | 1.42 | 1120.37 | 61.27 | 1.62 | 1.12 |

### 5.3 Results on the Multidimensional Benchmark

On the multidimensional benchmark the result is not a trade-off. Table 4 reports the ladder by dimension group. At an ascent budget of 200 the bound reaches 0.111% MAPE against GART 2.0's 0.618%, a factor of 5.56, at 0.71 times the cost, and it wins the paired comparison on 82.9% of instances. It first overtakes GART 2.0 at a budget of 100, where it costs 0.40 times as much. Each ratio is the median, over the timed instances, of the instance's own bound-to-GART 2.0 cost ratio measured back-to-back in one serial single-thread process; reweighting the dimension-stratified timing sample to the corpus composition puts the cost ratio at a budget of 200 at 0.54.

The domination is not an artifact of one dimension group. It holds at $d \in [4, 10]$, at $d \in [15, 50]$ and at $d = 100$, and it is widest exactly where the estimator was never trained: at $d = 100$ a budget of 500 costs 0.35 times GART 2.0 and is 140.8 times more accurate. The one group GART 2.0 holds is $d \in \{2, 3\}$, where matching its accuracy costs the bound 1.27 times as much.

It is, however, an artifact of instance size, and the aggregate above conceals that entirely. Of the 16,846 scored instances 10,569, or 62.74%, carry at most 100 nodes, a regime in which the relaxation is very nearly exact and very nearly free; a corpus mean over that composition is largely a statement about small instances rather than about the estimator. Stratifying by size as well as by dimension separates the two. We draw 20 instances per (dimension, size band) cell, matched on size across dimensions so that the cost ratio reports a dimension effect and not a size effect, and price the ladder on them under the protocol of Table 3; accuracy in each cell is measured on every instance both methods score, between 108 and 2,209 of them. Table 5 reports the result. GART 2.0 is undominated in 10 of the 18 cells, and the eight in which the bound is both cheaper and more accurate are the six at $n \in [20, 100]$ together with the two at $d = 100$. At $n \geq 200$ the estimator is on the cost/accuracy front at every dimension it was trained on, and it is dominated only at $d = 100$, the dimension withheld from training, where its own accuracy degrades to 0.824% against 0.062% at $d = 50$.

At deployment scale the margin is not narrow. In the $n \in [600, 1000]$ band GART 2.0 reaches 0.429% MAPE at 10.0 ms at $d = 2$, where the tightest rung we priced, $k = 500$, is both dearer and looser at 0.784% and 246.1 ms; 0.332% at 25.7 ms at $d = 3$, against 0.450% at 143.9 ms for $k = 200$; and 0.251% at 58.2 ms at $d = 4$, against 0.291% at 86.2 ms for the same budget. At $d = 10$ and $d = 50$ no rung dominates it in either direction and the cell is a genuine trade-off. The mechanism is the one Section 5.5 states: the bound is $\Theta(kn^2)$ in every dimension, so the cost of matching a fixed accuracy grows quadratically while the estimator's cost carries no accuracy term, and outright reversal — GART 2.0 dominating — arrives only in the plane and at $d = 3$, from roughly 200 nodes upward.

Three checks support the finding. The ascent never reads the stored label, so the comparison is not circular: its upper bound is a nearest-neighbor tour improved by 2-opt from coordinates alone, and supplying the released optimum in its place converges faster early and to the same value. On the 3,069 scored instances whose label the generation run recorded as a Concorde-proven optimum — a record of that run, not a certificate the released artifacts retain

(Section 3.2) — the target is an optimum rather than a heuristic tour, and GART 2.0 scores 1.03% against the bound's 0.046% at a budget of 200. And the relaxation closes exactly, returning the optimum rather than a bound, on 37.2% of the split. That last figure is also the mechanism: pairwise distances concentrate as $d$ grows, so the integrality gap this relaxation is usually described by is a planar fact rather than a general one.

Roughly one instance in seven exhausts the largest budget we ran, so at the top of the size range the reported accuracy is a floor and not a converged value. Two limits bound the size-stratified reading in particular. The cost figure in each cell is a median over 20 timed instances against an accuracy figure over the whole cell, so the cost column is the weaker of the two and a cell whose verdict turns on a few milliseconds should be read as a trade-off rather than as a ranking; budgets that close their gap terminate early, which is why a larger budget can occasionally read cheaper by a fraction of a millisecond. And the estimator's own asymptotics change with dimension: `compute_mst` dispatches to the Delaunay construction only for $d \leq 3$ and to a dense kernel from $d = 4$ upward, so above three dimensions both families are quadratic in $n$ and the separation is a constant rather than an exponent. That constant favors the estimator at $n \geq 200$ and the bound below it, which is what Table 5 measures; it is not a complexity argument and we do not extrapolate it past $n = 1000$, the largest size this corpus carries.

Table 4: Multidimensional cost/accuracy ladder, all 16,846 scored instances, Polyak ascent; conventions as in Section 5.2. Only the raw bound is reported, the calibration constant being planar-fitted.

| | $d \in \{2, 3\}$, $N = 1{,}295$ | | | $d \in [4, 10]$, $N = 4{,}534$ | | | $d \in [15, 50]$, $N = 5{,}170$ | | | $d = 100$, $N = 5{,}847$ | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Row | ms | × | MAPE | ms | × | MAPE | ms | × | MAPE | ms | × | MAPE |
| GART 2.0 | 5.87 | 1.00 | 1.453 | 3.30 | 1.00 | 0.728 | 3.59 | 1.00 | 0.340 | 3.26 | 1.00 | 0.593 |
| $k = 0$ | 2.43 | 0.24 | 12.190 | 0.13 | 0.04 | 7.446 | 0.37 | 0.07 | 3.834 | 0.23 | 0.07 | 2.133 |
| $k = 10$ | 3.51 | 0.31 | 7.389 | 0.24 | 0.08 | 4.500 | 0.36 | 0.12 | 2.164 | 0.26 | 0.08 | 1.026 |
| $k = 25$ | 5.18 | 0.47 | 4.746 | 0.38 | 0.13 | 3.240 | 0.55 | 0.17 | 1.461 | 0.37 | 0.12 | 0.691 |
| $k = 50$ | 7.37 | 0.69 | 3.177 | 0.62 | 0.21 | 1.972 | 0.92 | 0.27 | 0.870 | 0.49 | 0.15 | **0.412** |
| $k = 100$ | 12.90 | 1.27 | 1.388 | 1.21 | 0.39 | **0.692** | 1.95 | 0.56 | **0.301** | 0.76 | 0.23 | **0.147** |
| $k = 200$ | 22.97 | 2.18 | 0.557 | 2.15 | 0.69 | **0.161** | 3.30 | 0.95 | **0.053** | 1.19 | 0.37 | **0.025** |
| $k = 500$ | 50.80 | 4.92 | 0.384 | 4.51 | 1.38 | 0.067 | 6.05 | 1.75 | 0.011 | 1.12 | 0.35 | **0.004** |

### 5.4 Results on the 2D Diverse and Non-Euclidean Benchmarks

The remaining two ladders are measured on the protocol of Section 5.2: one estimator per process, one checkpointed ascent per instance.

Table 6 gives the 2D diverse ladder. GART 2.0 costs 5.00 ms on the typical instance at 2.889% MAPE. The raw certified bound passes it under the Volgenant–Jonker step at a budget of 50, at 0.23 times the cost for 2.702% MAPE and a paired win rate of 54.8%; the calibrated row passes it a rung earlier, at 0.14 times the cost for 2.564%. The reversal is monotone in $n$ and steeper than the one Section 5.2 reports, because this corpus reaches down to five nodes. On the 360 instances of its smallest size bucket, as Table 22 defines those buckets, the raw bound at a budget of 25 reads 0.439% MAPE against GART 2.0's 3.212% at 0.07 times the cost, and wins 95.0% of the paired comparisons. On the 610 instances of its largest, nothing on the ladder dominates: the raw bound is more accurate than GART 2.0 only at the top budget, and there it costs 26.08 times as much. This is the rule of Section 5.2 read over a wider range of $n$, not a second rule.

Table 7 gives the non-Euclidean ladder, where the estimator needs the MDS embedding of Section 6 and the bound reads the distance matrix directly. Over the 29 instances both methods score, GART 2.0 costs 4.88 ms, of which the embedding is 9.21%, at 3.814% MAPE. The Polyak step at a budget of 500 reaches 0.509% MAPE at 0.69 times that cost and wins all 29 paired comparisons; at a budget of 200 it reads 0.762% for 0.34 times the cost, again on every

Table 5: The multidimensional frontier stratified by size and dimension; "Dominates" means strictly better on cost and accuracy at the listed budgets. The bound is the Polyak arm.

| $d$ | $n$ band | $N$ | GART 2.0 MAPE (%) | GART 2.0 (ms) | Verdict against the ladder |
|---|---|---|---|---|---|
| | [20, 100] | 243 | 1.829 | 2.9 | bound dominates ($k = 200,\ 500$) |
| 2 | [200, 500] | 108 | 0.663 | 5.7 | **GART 2.0 dominates** ($k = 25,\ 50,\ 100,\ 200,\ 500$) |
| | [600, 1,000] | 135 | 0.429 | 10.0 | **GART 2.0 dominates** ($k = 25,\ 50,\ 100,\ 200,\ 500$) |
| | [20, 100] | 243 | 1.431 | 3.4 | bound dominates ($k = 100,\ 200,\ 500$) |
| 3 | [200, 500] | 108 | 0.459 | 12.2 | **GART 2.0 dominates** ($k = 100,\ 200$) |
| | [600, 1,000] | 135 | 0.332 | 25.7 | **GART 2.0 dominates** ($k = 25,\ 50,\ 100,\ 200$) |
| | [20, 100] | 243 | 1.015 | 2.4 | bound dominates ($k = 200$) |
| 4 | [200, 500] | 108 | 0.403 | 14.2 | trade-off at every rung |
| | [600, 1,000] | 135 | 0.251 | 58.2 | **GART 2.0 dominates** ($k = 200$) |
| | [20, 100] | 243 | 0.566 | 2.4 | bound dominates ($k = 200$) |
| 10 | [200, 500] | 108 | 0.204 | 14.7 | trade-off at every rung |
| | [600, 1,000] | 135 | 0.173 | 100.2 | trade-off at every rung |
| | [20, 100] | 243 | 0.214 | 2.8 | bound dominates ($k = 200$) |
| 50 | [200, 500] | 108 | 0.095 | 22.7 | trade-off at every rung |
| | [600, 1,000] | 134 | 0.062 | 114.3 | trade-off at every rung |
| | [20, 100] | 2,209 | 0.501 | 3.4 | bound dominates ($k = 100,\ 200$) |
| 100 | [200, 500] | 977 | 0.723 | 26.8 | bound dominates ($k = 50,\ 100$) |
| | [600, 1,000] | 1,185 | 0.824 | 120.4 | bound dominates ($k = 50,\ 100$) |

instance. The cheapest domination by a certified row anywhere in this study is on this corpus: the Volgenant–Jonker step at a budget of 25 costs 0.065 times GART 2.0 and reads 2.953% against 3.814%. The accuracy factor Section 6 quotes is therefore bought below the estimator's own cost rather than above it.

The two step rules do not order the same way on the two corpora. Volgenant–Jonker leads Polyak at every 2D budget that runs an ascent except 200, where the two are separated by half a hundredth of a percentage point, 3.606% against 7.515% at a budget of 25, and Polyak leads it from a budget of 200 on the non-Euclidean corpus, 0.762% against 1.228%. The assignment of Section 5.1 is therefore not arbitrary, but the crossover is a property of the corpus rather than of the budget, and neither schedule dominates the other across these four benchmarks.

Four checks run over every cell of both tables and none fails. Repeating each timed call returns zero spread across all 18,060 instance–budget pairs of the 2D corpus and all 217 of the non-Euclidean one. The timed call returns the same bound the accuracy sweep scores, bit-identical on the Volgenant–Jonker arm and within one unit in the last place of a double on the Polyak arm. Every bound printed is at or below the reference cost of its own instance, and every accuracy ladder is monotone in $k$, over all 2,580 series on the 2D corpus and the 31 the bound covers on the non-Euclidean one. On the instances small enough to solve outright the bound is checked against an exact dynamic program as well, again with no violation.

Two disclosures bound the cost columns, and both were measured rather than assumed. These sessions did not run on a quiet machine, and the two methods are not equally sensitive to that: a paired control against the published quiet-window measurements puts GART 2.0 at 1.48 times its published cost and the bound at 1.10 to 1.14 times, so every cost multiple above is between 0.74 and 0.77 of its quiet-box value. That direction runs against the bound, and correcting for it costs it exactly one domination, the raw Volgenant–Jonker row at a budget of 200 on the 2D corpus, whose 0.79 becomes 1.03. And the checkpointed protocol does not flatter the bound: running the released single-budget call once per rung, sharing nothing between rungs, costs 0.77 to 0.95 times the checkpointed ladder on the 2D corpus and 0.90 to 1.14 times on the non-Euclidean one.

Table 6: 2D diverse cost/accuracy ladder, all 2,580 instances, both step rules; conventions as in Section 5.2.

| | Volgenant–Jonker step | | | | Polyak step | | | |
|---|---|---|---|---|---|---|---|---|
| Row | ms | $\times$ | Raw | Cal. | ms | $\times$ | Raw | Cal. |
| GART 2.0 | 5.00 ms at $1.00\times$; 2.889% MAPE | | | | | | | |
| $k = 0$ | 0.23 | 0.046 | 13.938 | 5.654 | 0.31 | 0.062 | 13.938 | 5.654 |
| $k = 10$ | 0.44 | 0.087 | 8.352 | 4.257 | 0.49 | 0.098 | 10.572 | 6.872 |
| $k = 25$ | 0.70 | 0.14 | 3.606 | **2.564** | 0.73 | 0.15 | 7.515 | 6.393 |
| $k = 50$ | 1.15 | 0.23 | **2.702** | **2.330** | 1.12 | 0.22 | 5.526 | 4.978 |
| $k = 100$ | 2.05 | 0.41 | **2.048** | **1.763** | 1.88 | 0.38 | 3.194 | **2.788** |
| $k = 200$ | 3.95 | 0.79 | **1.883** | **1.618** | 3.36 | 0.67 | **1.878** | **1.650** |
| $k = 500$ | 9.42 | 1.88 | 1.274 | 1.102 | 7.15 | 1.43 | 1.447 | 1.331 |

Table 7: Non-EUC_2D TSPLIB95 ladder over the 29 instances both methods score, nine of which violate the triangle inequality; conventions as in Section 5.2, except that the planar $c_k$ makes the Cal. columns indicative only.

| | Volgenant–Jonker step | | | | Polyak step | | | |
|---|---|---|---|---|---|---|---|---|
| Row | ms | $\times$ | Raw | Cal. | ms | $\times$ | Raw | Cal. |
| GART 2.0 | 4.88 ms at $1.00\times$; 3.814% MAPE | | | | | | | |
| $k = 0$ | 0.03 | 0.007 | 15.606 | 6.181 | 0.05 | 0.011 | 15.606 | 6.181 |
| $k = 10$ | 0.17 | 0.035 | 8.150 | **3.189** | 0.16 | 0.033 | 10.087 | 5.481 |
| $k = 25$ | 0.32 | 0.065 | **2.953** | **1.784** | 0.28 | 0.057 | 6.501 | 5.053 |
| $k = 50$ | 0.56 | 0.11 | **1.783** | **1.436** | 0.49 | 0.100 | 3.817 | **3.129** |
| $k = 100$ | 1.10 | 0.23 | **1.269** | **1.037** | 0.92 | 0.19 | **1.990** | **1.589** |
| $k = 200$ | 2.15 | 0.44 | **1.228** | **1.018** | 1.66 | 0.34 | **0.762** | **0.581** |
| $k = 500$ | 5.27 | 1.08 | 1.139 | 0.995 | 3.38 | 0.69 | **0.509** | **0.492** |

### 5.5 Complexity of the Estimator Families

Table 8 states the dominant term of every family in this paper as the code executes it, not as the underlying algorithm is usually quoted, together with whether the row certifies its own output and whether it needs a training corpus. Three points differ from the textbook reading. The classical closed forms are dominated by a row-deduplication pass rather than by their arithmetic. The convex-hull blow-up this paper cites as a motivation never occurs at runtime, because the implementation caps hull construction at three dimensions and falls back to a bounding box above it. And the 1-tree cannot use the Delaunay shortcut the MST family relies on in the plane: the node potentials break the Euclidean metric, so after the first iteration the perturbed minimum spanning tree is not a subgraph of the triangulation. That is measured rather than assumed. At 144 potential vectors drawn from real ascent trajectories the Delaunay-restricted 1-tree is heavier than the exact one at 141 of them, so the shortcut is unavailable at 141 of the 144, and that is why the bound sits a complexity class above the MST family and not a constant factor above it.

That class difference is the separation that survives everything else in this section, and Figure 5 is what it looks like measured. In the plane GART 2.0 is $\Theta(n \log n)$ and the bound is $\Theta(n^2 d + kn^2)$, and the measured log–log cost exponents in $n$ agree: 0.975 for GART 2.0 above a thousand nodes against 2.08–2.13 for the bound over the same window. The consequence is a ratio that grows rather than a fixed multiple. GART 2.0 costs a flat 2.0 to 2.3 times the MST-ratio family across four orders of magnitude in $n$, which is the signature of a shared complexity class; against the 1-tree at a fixed budget the ratio moves from 7.7 at a thousand nodes to 211 at sixteen thousand. The bound's advantage at small $n$ is a statement about constants and the estimator's advantage at large $n$ is a statement about growth, and only the second is stable.

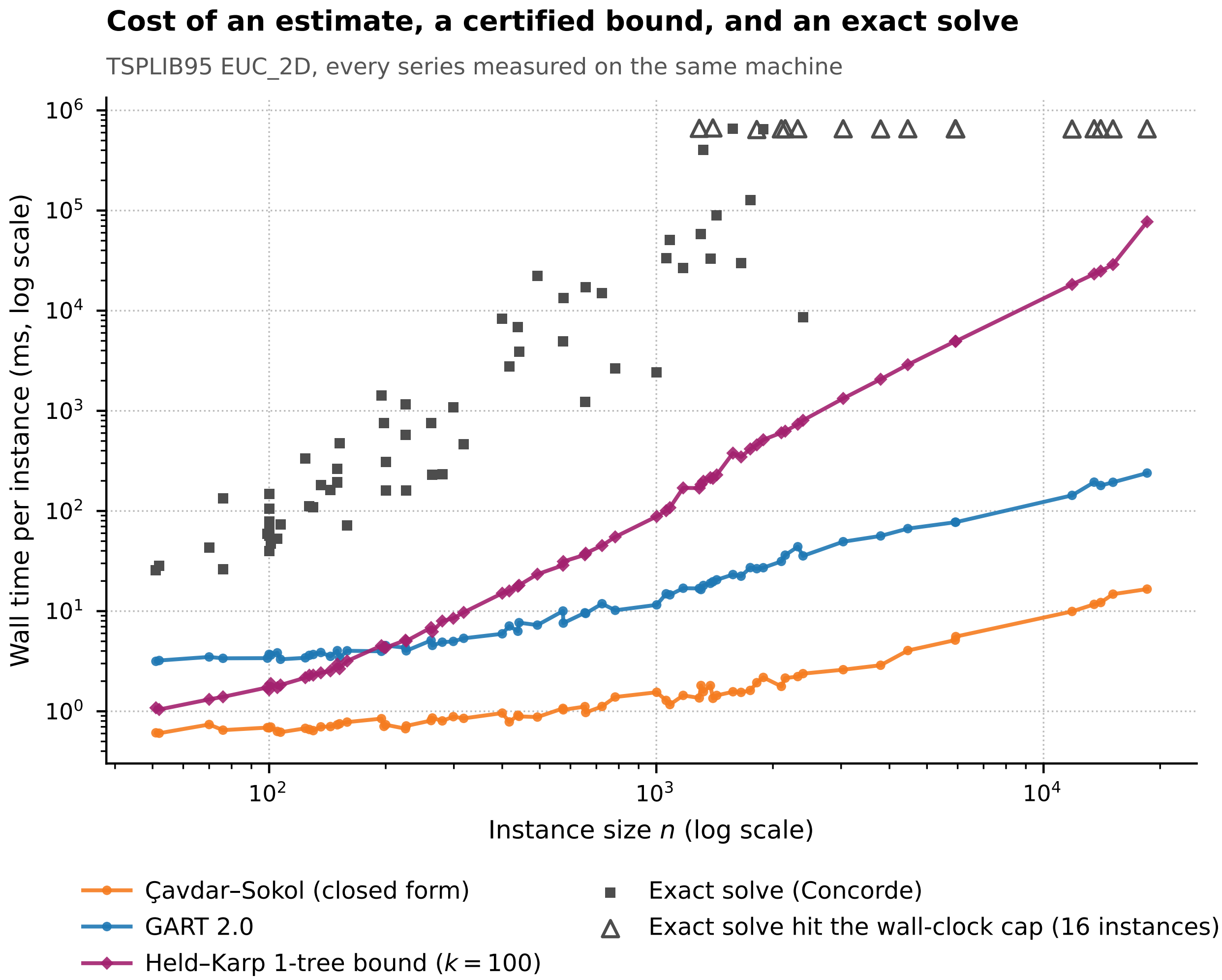


Figure 5: Per-instance cost against instance size on TSPLIB95 EUC_2D, log–log, every series on one machine. The exact solver's capped runs make its curve a lower bound.

The separation is planar, and we do not extend it. On the multidimensional benchmark GART 2.0's own measured cost exponent in $n$ is 2.02 at $d \in [4, 10]$ and 1.82 at $d \in [15, 50]$, against 1.785–1.889 for the bound, because the MST construction falls back to a dense all-pairs kernel from $d = 4$ upward. Only at $d \in \{2, 3\}$ is that exponent 0.56. On that benchmark both families are quadratic, the separation is a constant, and the constant points the wrong way.

### 5.6 Comparison with an Exact Solver

This paper's cost argument has always had an unbounded anchor above it, and that anchor is a floor. We measured Concorde on this machine over the whole EUC_2D benchmark, on the same box as every other number on the cost axis, so no cross-hardware correction is needed. It proves optimality on most of the corpus in well under a second and takes minutes on the rest; the largest instances reach the wall-clock cap without proving anything, which makes the exact solver's cost a lower bound rather than a measurement. Figure 5 draws it beside the estimator, the closed form and the certified bound, and the four families separate by roughly five orders of magnitude at the largest instance. The load-bearing statement is the complexity rather than the ratio: branch-and-cut over an exponential family of subtour constraints admits no polynomial bound, and even the Held–Karp dynamic program [Held and Karp, 1962], the exact method this paper's own label pipeline uses at small $n$, is $\Theta(n^2 2^n)$, so every row of Table 8 is asymptotically cheaper

Table 8: Dominant per-instance cost of each family as the released code executes it; $k$ is the ascent budget, $h$ the Hilbert order.

| **Family** | **Dominant term as implemented** | **Guarantee** | **Training** |
|---|---|---|---|
| Classical closed forms | $\Theta(d\,n\log n)$; a row-deduplication pass, not the arithmetic | None | Constants fitted in the source |
| Constant multiples of $L_{\mathrm{MST}}$ | $\Theta(n\log n)$ at $d=2$ (Delaunay); $\Theta(n^2 d + n^2\log n)$ otherwise | Lower bound only at $\alpha=1$ | The two calibrated rows only |
| Space-filling curve (Hilbert) | $\Theta(ndh)$ interpreted, plus an $O(n\log n)$ sort | Upper bound: it builds a tour | None |
| GART 2.0 (learned head on this feature block) | Feature cost as the MST family above, except the greedy tour's dense $\Theta(n^2)$ scan below its candidate-list threshold; head $O(1)$ in $n$ | None | Yes |
| GART 1.0 | $\Theta(n^2 d)$ time and $\Theta(n^2)$ memory; a dense matrix for two scalars | None | Yes |
| Held–Karp 1-tree, budget $k$ | $\Theta(n^2 d + kn^2)$; no Delaunay path exists after the first iteration | Lower bound | None |
| Calibrated 1-tree | As above | None | One scalar per budget |
| Exact solver | $\Theta(n^2 2^n)$ for the Held–Karp dynamic program; branch-and-cut admits no polynomial bound | Exact | None |

than an exact solve and being cheaper than one distinguishes nothing among them. The comparison that discriminates is the one against the 1-tree, which is polynomial, certified, and in the same cost decade as the estimators.

### 5.7 Operating Regime

GART 2.0 is cheaper than a 1-tree bound above roughly 400 nodes in the plane and dearer below roughly 150. Its accuracy sits between the bound's and the closed forms': against the MST-ratio family it is separated by 1.00 percentage points on the full 78-instance EUC_2D set, against 1.35 points from the converged bound on the same set.

On the aggregate of every benchmark here the bound is the better instrument, and we report that unchanged. It does not follow that the estimator is superseded, because the multidimensional aggregate is a corpus 62.74% composed of instances of at most 100 nodes, and that is the regime the aggregate describes. Read at fixed size instead (Table 5), the ordering turns over: at $n \geq 200$ GART 2.0 is on the cost/accuracy front at every dimension it was trained on, and in the two larger size bands it is strictly better than the bound on both axes from an ascent budget of 25 upward in the plane and over the mid-ladder budgets at $d = 3$. The aggregate and the stratified reading are both correct, and they answer different questions; only the second is the question a practitioner asks.

The rule that follows has three arguments rather than one. Below roughly 200 nodes, prefer the certified bound at any dimension: it is cheaper, it is more accurate, it needs no training corpus, and it tells the caller how wrong it can be. Above that size prefer the estimator at any dimension it was trained on, most decisively in the plane and at $d = 3$, where the ordering not only reverses but keeps reversing, because each ascent iteration costs the bound $\Theta(n^2)$ at a fixed accuracy in every dimension while the estimator's planar cost stays subquadratic. At $d = 100$, outside the training grid, prefer the bound at every size; that is the one cell in Table 5 where large $n$ does not rescue the estimator, and the reason is its own degradation off the grid rather than the bound's strength. Given only a distance matrix, prefer the bound at any size: it reads the matrix directly, and the estimator pays an embedding to reach it at all (Section 6). And where inputs are known to be i.i.d. uniform on a known region and an error in the high single digits of percent is acceptable, the classical closed forms remain the cheapest instrument in this paper by an order of magnitude, with no MST to build. Large instances re-evaluated many times under changing input are the last-mile and partitioning setting of Section 1; they are the regime in which an exact solve is out of reach and a quadratic bound is already expensive, and they are the regime in which GART 2.0's accuracy is at its best rather than its worst. A practitioner choosing between the methods should read Table 5 at their own instance size, and Figure 2 at their own dimension, before reading any aggregate.

## 6 Application: Non-Euclidean Estimation via MDS Embedding

The preceding benchmarks use Euclidean point clouds, and the certified bound of Section 5 needs no more than a symmetric distance matrix. This section asks what the estimator can do on distance-matrix instances, where it needs an embedding and the bound does not. TSPLIB95 contains 33 non-EUC_2D instances: 4 CEIL_2D, 2 ATT, 10 GEO, and 17 EXPLICIT. An exhaustive triangle-inequality scan identifies ten EXPLICIT matrices with structural violations (worst direct-edge ratio at least 1.2), which are outside every metric-dependent aggregate in this paper: `brg180`, `brazil58`, `gr120`, `gr48`, `gr24`, `gr21`, `bays29`, `hk48`, `dantzig42`, and `gr17`. That leaves 23 instances, seven of them EXPLICIT, and three of those (`swiss42`, `pa561`, and `fri26`) carry mild rounding-level violations, so conclusions are restricted to this screened set rather than to all non-Euclidean TSPs. We evaluate a hybrid pipeline that computes MST features from the original matrix and geometric features from a multidimensional-scaling (MDS) embedding. The paired reference is a constant multiple of the original-distance MST, a fixed $\alpha = 1.136$ close to the MAPE-minimizing constant $1.134$ over the full 111-instance TSPLIB set. The MAPE-minimizing constant over these 23 instances is $1.1718$ and reaches $7.55\%$, which is the floor any constant multiplier faces on this metric.

### 6.1 MDS Preprocessing

CEIL_2D cases use their native coordinates. For ATT, GEO, and EXPLICIT cases, classical MDS applied to the distance matrix produces an approximate positive-eigenvalue Euclidean representation [Torgerson, 1952, Gower, 1966, Borg and Groenen, 2005, Cox and Cox, 2000]: we double-center $D^{(2)}$ and take leading positive-eigenvalue components of the resulting Gram matrix [Gower, 1966]. We select the smallest embedding dimension $\kappa$ retaining at least 99.9% of positive eigenvalue mass, subject to $\kappa \leq 100$, and pass $\kappa$ as the model's `dimension` feature. The symbol is distinct from the ascent budget $k$ of the certified lower bound. The cap prevents five of 19 MDS cases from reaching 99.9%; hence this criterion does not imply distance preservation. Geometric features use the approximate embedding, whereas MST features are computed from the original matrix $D$ and are therefore unaffected by embedding distortion. Appendix Q reports the recovered distance diagnostics.

### 6.2 Results on Non-Euclidean TSPLIB95

On the instances it accepts, the hybrid pipeline clears the constant-multiplier floor decisively; Table 9 reports the paired results by distance type.

Table 9: Paired evaluation on the 23 screened non-EUC_2D instances. GART 2.0 declines `si1032` at the coverage gate, so the Total row is not like-for-like.

| Distance type | Count | Estimator | $N$ | SDPE (%) [95% CI] | MAPE (%) | \|Median\| (%) | MSPE (%) |
|---|---|---|---|---|---|---|---|
| ATT | 2 | GART 2.0 | 2 | 1.42 [—, —] | 3.66 | 3.66 | -3.66 |
| | | Fixed $\alpha = 1.136$ | 2 | 4.12 [—, —] | 3.38 | 3.38 | -3.38 |
| CEIL_2D | 4 | GART 2.0 | 4 | 5.16 [0.11, 6.35] | 5.77 | 6.05 | 4.04 |
| | | Fixed $\alpha = 1.136$ | 4 | 6.44 [1.08, 7.86] | 7.54 | 7.77 | 5.95 |
| GEO | 10 | GART 2.0 | 10 | 2.60 [1.48, 3.23] | 2.51 | 2.90 | -1.63 |
| | | Fixed $\alpha = 1.136$ | 10 | 9.49 [2.13, 10.93] | 9.62 | 4.10 | -9.62 |
| EXPLICIT screened | 7 | GART 2.0 | 6 | 3.11 [1.44, 3.87] | 3.01 | 2.35 | -2.15 |
| | | Fixed $\alpha = 1.136$ | 7 | 9.55 [5.04, 10.88] | 8.12 | 10.16 | 1.75 |
| **Total** | **23** | **GART 2.0** | **22** | **3.89 [2.52, 4.79]** | **3.34** | **3.21** | **-0.93** |
| | | Fixed $\alpha = 1.136$ | 23 | 10.48 [6.94, 12.93] | 8.26 | 6.29 | -2.91 |

Table 9 reports the paired results. On the 22 of the 23 screened instances it accepts, GART 2.0 has aggregate SDPE 3.89% and MAPE 3.34%, against 10.48% and 8.26% for the fixed $\alpha = 1.136$ multiplier scored on all 23, and 7.55% for the MAPE-minimizing constant on this set. The margin over the fixed multiplier therefore measures what the feature set adds to a constant. The Total row is not a like-for-like comparison: GART 2.0 declines `si1032` rather than extrapolate below the greedy-to-MST ratio it was trained on, and the $N$ column marks where the instance sets differ. The dispersion gap is the substantive result: $\alpha$ ranges from 1.006 to 1.511 across the set, and a single multiplier cannot track that spread without trading one metric against the other. The advantage is not uniform. On the two ATT instances the fixed constant has the lower MAPE, 3.38% against 3.66%, and on CEIL_2D the two estimators lie within 1.3 percentage points of SDPE of one another. Groups of two and four instances cannot separate estimators, and we report them only so the aggregate is not read as a uniform result. We do not estimate the relationship between per-instance embedding distortion and prediction error, so this remains a feasibility study for the hybrid pipeline rather than evidence of embedding robustness.

The certified bound needs no embedding here, and that is the whole of the comparison. The Held–Karp relaxation never invokes the triangle inequality, so it is defined on any symmetric matrix and reads these instances directly. On the 29 instances both methods score over all 33 non-EUC_2D files — a set wider than the metric screen above, since the certificate tolerates non-metric input — GART 2.0 obtains 3.81% MAPE against 1.27% for the raw certified bound at an ascent budget of 100 under the Volgenant–Jonker step and 0.51% at a budget of 500 under the Polyak step, a factor of 7.5. The two steps are not interchangeable on this corpus, and Section 5.4 prices both. The bound also scores `brg180` and `si1032`, the two instances GART 2.0 declines, one for a structural metric violation and one for falling below the coverage gate. The MDS pipeline is therefore a way to reach these instances with a Euclidean estimator, and not the best way to estimate their tour length.

## 7 Conclusion

We present GART 2.0, a TSP tour-length estimator that predicts the MST-to-tour scaling factor $\alpha$ from features defined for arbitrary dimension, combining MST edge and topology statistics with centroid and coordinate-range descriptors and computing no convex hull. Across synthetic planar and multidimensional benchmarks and both the Euclidean and the screened non-Euclidean parts of TSPLIB95, it has the lowest aggregate MAPE and SDPE among the baselines scored on each of the four benchmark aggregates, ahead of a calibrated constant on the multidimensional set, of Çavdar–Sokol on planar i.i.d. uniform instances, and of the asymptotic ratio on TSPLIB EUC_2D, where the paired difference resolves and the dispersion gap is the wider of the two. On the multidimensional benchmark, dispersion falls monotonically across the size range and across the dimension range, including a dimension held out of training entirely, and where cost-level differences compress, the close-pair test of Section 4.7 still orders near-ties better than every baseline on all three benchmarks. The same trained model reads non-Euclidean distance matrices through an MDS embedding, well ahead of a fixed multiplier on the instances it accepts.

We then fix the boundary of the method against a certified Held–Karp 1-tree bound, which consumes no training corpus, proves its own output, and needs nothing but a symmetric distance matrix. On the corpus aggregate the bound is the better instrument, more accurate on the multidimensional benchmark at lower cost, and more accurate again on the non-Euclidean instances it reads directly and GART 2.0 reaches only through an embedding. That is the general shape of the result: dimension-scalability is what lets one model reach these instances at all, and the advantage over a relaxation that is already defined at every dimension must come from scale, not from dimension. But that aggregate is mostly small instances, and it is a statement about them: read at fixed size it reverses. Above a few hundred nodes GART 2.0 is on the cost/accuracy front at every dimension it was trained on, and in the two larger size bands it is strictly better than the bound on both axes, from an ascent budget of 25 upward in the plane and over the mid-ladder budgets at $d = 3$. What the estimator owns is scale, because the bound pays a quadratic price in instance size for a fixed accuracy in every dimension while the estimator's cost carries no accuracy term; in the plane the measured cost exponents put the two families in different complexity classes, and an instance many times larger than anything in training is predicted in a fraction of a second. Large instances scored repeatedly are where tour-length estimation earns its place, and below that size, off the training grid in dimension, or given only a distance matrix, the certified bound should be preferred. Both halves of that rule are needed to use either method well.

Every claim above rests on labels, and the labels are validated rather than trusted. Every stored tour is recomputed on its own released coordinates, the small residue that fails is quarantined rather than scored, and every label the certification screen of Appendix E reaches passes it, zero violations of $B \leq L$ across every bound check the released artifacts supply. The model weights, the scored benchmark outputs, and the scripts that regenerate every table and figure are released (Appendix A), so the numbers in this paper can be reproduced from the published artifacts.

Three limitations constrain the estimator. Accuracy depends on geometry far more than on scale: error on the near-collinear class is several times what it is on the isotropic class, and Section 4.6 diagnoses that failure as a training-coverage gap and a feature gap at once, with two pre-registered repair attempts rejected and the released model unchanged. Withholding a whole dimension or the large-$n$ range from training costs a fraction of a MAPE point and appears as a systematic offset rather than as scatter, so the estimator degrades predictably rather than erratically off its training grid, and we demonstrate no correction for that offset. And the labels are not uniformly certificates: some are certified by a 1-tree bound that closes on the tour and some by exact dynamic programming, while the remainder carry an upper bound whose residual on the scored test split is narrow but not zero.

Each limitation marks a direction for further work, and one question is larger than the others. Both instruments in this paper are scored against reference tours in isolation; the open question is whether the precision advantage, or the certificate, changes the decisions a routing or partitioning algorithm actually makes when either is embedded inside it.

## References


Richa Agarwala, David L Applegate, Donna Maglott, Gregory D Schuler, and Alejandro A Schäffer. A fast and scalable radiation hybrid map construction and integration strategy. *Genome Research*, 10(3):350–364, 2000.

Takuya Akiba, Shotaro Sano, Toshihiko Yanase, Takeru Ohta, and Masanori Koyama. Optuna: A next-generation hyperparameter optimization framework. In *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, pages 2623–2631. ACM, 2019. doi: 10.1145/3292500.3330701.

David L. Applegate, Robert E. Bixby, Vašek Chvátal, and William J. Cook. *The Traveling Salesman Problem: A Computational Study*. Princeton University Press, 2006.

John J. Bartholdi, III and Loren K. Platzman. An O(N log N) planar travelling salesman heuristic based on spacefilling curves. *Operations Research Letters*, 1(4):121–125, 1982. doi: 10.1016/0167-6377(82)90012-8.

Jillian Beardwood, John H. Halton, and John Michael Hammersley. The shortest path through many points. *Mathematical Proceedings of the Cambridge Philosophical Society*, 55(4):299–327, 1959. doi: 10.1017/S0305004100034095.

Dimitris J. Bertsimas and Garrett van Ryzin. An asymptotic determination of the minimum spanning tree and minimum matching constants in geometrical probability. *Operations Research Letters*, 9(4):223–231, 1990. doi: 10.1016/0167-6377(90)90066-E.

Ingwer Borg and Patrick J.F. Groenen. *Modern Multidimensional Scaling: Theory and Applications*. Springer, 2nd edition, 2005. doi: 10.1007/0-387-28981-X.

Jakob Bossek, Pascal Kerschke, Aneta Neumann, Markus Wagner, Frank Neumann, and Heike Trautmann. Evolving diverse TSP instances by means of novel and creative mutation operators. In *Proc. 15th ACM/SIGEVO Conference on Foundations of Genetic Algorithms (FOGA XV)*, pages 58–71, 2019. doi: 10.1145/3299904.3340307.

Jean Bourgain. On Lipschitz embedding of finite metric spaces in Hilbert space. *Israel Journal of Mathematics*, 52 (1–2):46–52, 1985. doi: 10.1007/BF02776078.

Leo Breiman. Random forests. *Machine Learning*, 45(1):5–32, 2001.

Bahar Çavdar. *A computation-implementation parallelization approach to time-sensitive applications*. PhD thesis, Georgia Institute of Technology, Atlanta, GA, 2014. URL https://repository.gatech.edu/handle/1853/52322. H. Milton Stewart School of Industrial and Systems Engineering; advisor Joel S. Sokol. Chapter 4 is the tour-length estimation model later published as Çavdar and Sokol (2015).

Bahar Çavdar and Joel Sokol. A distribution-free TSP tour length estimation model for random graphs. *European Journal of Operational Research*, 243(2):588–598, 2015. doi: 10.1016/j.ejor.2014.12.020.

Bernard Chazelle. An optimal convex hull algorithm in any fixed dimension. *Discrete & Computational Geometry*, 10(4):377–409, 1993. doi: 10.1007/BF02573985.

T. William Chien. Operational estimators for the length of a traveling salesman tour. *Computers & Operations Research*, 19(6):469–478, 1992. doi: 10.1016/0305-0548(92)90002-M.

Youngmin Choi. *Adjustment Factors and Applications for Analytic Approximations of Tour Lengths*. PhD thesis, University of Maryland, College Park, 2021. DRUM handle 1903/27698.

Nicos Christofides. Worst-case analysis of a new heuristic for the travelling salesman problem. Technical Report 388, Carnegie-Mellon University, 1976. Reprinted in Operations Research Forum 3:20, 2022, doi:10.1007/s43069-021-00101-z.

William J. Cook and André Rohe. Computing minimum-weight perfect matchings. *INFORMS Journal on Computing*, 11(2):138–148, 1999.

Mario Cortina-Borja and Tony Robinson. Estimating the asymptotic constants of the total length of Euclidean minimum spanning trees with power-weighted edges. *Statistics & Probability Letters*, 47(2):125–128, 2000. doi: 10.1016/S0167-7152(99)00147-9.

Trevor F. Cox and Michael A.A. Cox. *Multidimensional Scaling*. Chapman and Hall/CRC, 2nd edition, 2000. ISBN 9781584880943.

Carlos F. Daganzo. The distance traveled to visit N points with a maximum of C stops per vehicle: An analytic model and an application. *Transportation Science*, 18(4):331–350, 1984a. doi: 10.1287/trsc.18.4.331.

Carlos F. Daganzo. The length of tours in zones of different shapes. *Transportation Research Part B: Methodological*, 18(2):135–145, 1984b. doi: 10.1016/0191-2615(84)90027-4.

George B. Dantzig, D. Ray Fulkerson, and Selmer M. Johnson. Solution of a large-scale traveling-salesman problem. *Operations Research*, 2(4):393–410, 1954. doi: 10.1287/opre.2.4.393.

Boris Delaunay. Sur la sphère vide. *Bulletin de l'Académie des Sciences de l'URSS, Classe des Sciences Mathématiques et Naturelles*, 6:793–800, 1934.

Jack Edmonds. Paths, trees, and flowers. *Canadian Journal of Mathematics*, 17:449–467, 1965. doi: 10.4153/CJM-1965-045-4.

Stephen Feldman. Tour length estimation guided vehicle routing. Master's thesis, Rensselaer Polytechnic Institute, 2025. URL https://search.proquest.com/openview/d4a8aaeb70d19bb62af72359b5caf7f7/1?pq-origsite=gscholar&cbl=18750&diss=y. Introduces the original GART 1.0 tour-length estimator. Available via ProQuest Dissertations Publishing.

Miguel A. Figliozzi. Planning approximations to the average length of vehicle routing problems with time window constraints. *Transportation Research Part B: Methodological*, 43(4):438–447, 2009. doi: 10.1016/j.trb.2008.08.004.

Finmile. Revolutionizing last-mile delivery for unparalleled efficiency and customer satisfaction. Finmile industry resource page, 2025. URL https://finmile.co/resources/finmile-revolutionizing-last-mile-delivery. Accessed 2026-08-26.

Herbert Freeman and Ruth Shapira. Determining the minimum-area encasing rectangle for an arbitrary closed curve. *Communications of the ACM*, 18(7):409–413, 1975. doi: 10.1145/360881.360919.

Jerome H. Friedman. Greedy function approximation: A gradient boosting machine. *Annals of Statistics*, 29(5): 1189–1232, 2001.

Harold N. Gabow. Data structures for weighted matching and nearest common ancestors with linking. In *Proceedings of the 1st Annual ACM-SIAM Symposium on Discrete Algorithms (SODA)*, pages 434–443, 1990.

Bruce L. Golden and William R. Stewart. Empirical analysis of heuristics. In E. L. Lawler, J. K. Lenstra, A. H. G. Rinnooy Kan, and D. B. Shmoys, editors, *The Traveling Salesman Problem: A Guided Tour of Combinatorial Optimization*, pages 207–249. Wiley, 1985.

J. C. Gower. Some distance properties of latent root and vector methods used in multivariate analysis. *Biometrika*, 53 (3–4):325–338, 1966. doi: 10.1093/biomet/53.3-4.325.

Charles R. Harris, K. Jarrod Millman, Stéfan J. van der Walt, Ralf Gommers, Pauli Virtanen, David Cournapeau, Eric Wieser, Julian Taylor, Sebastian Berg, Nathaniel J. Smith, Robert Kern, Matti Picus, Stephan Hoyer, Marten H. van Kerkwijk, Matthew Brett, Allan Haldane, Jaime Fernández del Río, Mark Wiebe, Pearu Peterson, Pierre Gérard-Marchant, Kevin Sheppard, Tyler Reddy, Warren Weckesser, Hameer Abbasi, Christoph Gohlke, and Travis E. Oliphant. Array programming with NumPy. *Nature*, 585:357–362, 2020. doi: 10.1038/s41586-020-2649-2.

Michael Held and Richard M. Karp. A dynamic programming approach to sequencing problems. *Journal of the Society for Industrial and Applied Mathematics*, 10(1):196–210, 1962. doi: 10.1137/0110015.

Michael Held and Richard M. Karp. The traveling-salesman problem and minimum spanning trees. *Operations Research*, 18(6):1138–1162, 1970. doi: 10.1287/opre.18.6.1138.

Michael Held and Richard M. Karp. The traveling-salesman problem and minimum spanning trees: Part II. *Mathematical Programming*, 1(1):6–25, 1971. doi: 10.1007/BF01584070.

Keld Helsgaun. An effective implementation of the Lin-Kernighan traveling salesman heuristic. *European Journal of Operational Research*, 126(1):106–130, 2000.

Keld Helsgaun. An extension of the Lin-Kernighan-Helsgaun TSP solver for constrained traveling salesman and vehicle routing problems. Technical report, Roskilde University, 2017.

David Hilbert. Ueber die stetige Abbildung einer Linie auf ein Flächenstück. *Mathematische Annalen*, 38(3):459–460, 1891. doi: 10.1007/BF01199431.

Chuck Holland, Jack Levis, Ranganath Nuggehalli, Bob Santilli, and Jeff Winters. UPS optimizes delivery routes. *Interfaces*, 47(1):8–23, 2017. doi: 10.1287/inte.2016.0875. Winner of the 2016 Franz Edelman Award.

David S. Johnson, Lyle A. McGeoch, and Edward E. Rothberg. Asymptotic experimental analysis for the Held–Karp traveling salesman bound. In *Proceedings of the 7th Annual ACM-SIAM Symposium on Discrete Algorithms (SODA)*, pages 341–350, 1996.

I.T. Jolliffe. *Principal Component Analysis*. Springer, 2002.

Anna R. Karlin, Nathan Klein, and Shayan Oveis Gharan. A (slightly) improved approximation algorithm for metric TSP. In *Proceedings of the 53rd Annual ACM Symposium on Theory of Computing*, pages 32–45, 2021. doi: 10.1145/3406325.3451009.

Richard M. Karp. Reducibility among combinatorial problems. In Raymond E. Miller and James W. Thatcher, editors, *Complexity of Computer Computations*, pages 85–103. Plenum Press, 1972. doi: 10.1007/978-1-4684-2001-2_9.

Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. LightGBM: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems 30*, pages 3146–3154, 2017.

Shuhan Kou, Bruce L. Golden, and Stefan Poikonen. Optimal TSP tour length estimation using standard deviation as a predictor. *Computers & Operations Research*, 148:105993, 2022. doi: 10.1016/j.cor.2022.105993.

Shuhan Kou, Bruce L. Golden, and Stefan Poikonen. Optimal TSP tour length estimation using Sammon maps. *Optimization Letters*, 17(1):89–105, 2023. doi: 10.1007/s11590-022-01937-y.

Shuhan Kou, Bruce L. Golden, and Stefan Poikonen. Estimating optimal objective values for the TSP, VRP, and other combinatorial problems using randomization. *International Transactions in Operational Research*, 31(5):3443–3458, 2024. doi: 10.1111/itor.13260.

Joseph B. Kruskal. On the shortest spanning subtree of a graph and the traveling salesman problem. *Proceedings of the American Mathematical Society*, 7(1):48–50, 1956. doi: 10.1090/S0002-9939-1956-0078686-7.

Ohseok Kwon, Bruce L. Golden, and Edward A. Wasil. Estimating the length of the optimal TSP tour: An empirical study using regression and neural networks. *Computers & Operations Research*, 22(10):1039–1046, 1995. doi: 10.1016/0305-0548(94)00093-N.

Siu Kwan Lam, Antoine Pitrou, and Stanley Seibert. Numba: A LLVM-based Python JIT compiler. In *Proceedings of the Second Workshop on the LLVM Compiler Infrastructure in HPC*, pages 1–6, 2015. doi: 10.1145/2833157.2833162.

Stuart Lloyd. Least squares quantization in PCM. *IEEE Transactions on Information Theory*, 28(2):129–137, 1982.

Scott M. Lundberg and Su-In Lee. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems 30*, pages 4765–4774, 2017.

Radu Mariescu-Istodor and Pasi Fränti. Solving the large-scale TSP problem in 1 h: Santa Claus Challenge 2020. *Frontiers in Robotics and AI*, 8:689908, 2021. doi: 10.3389/frobt.2021.689908.

Xuanhao Pan, Yan Jin, Yuandong Ding, Mingxiao Feng, Li Zhao, Lei Song, and Jiang Bian. H-TSP: Hierarchically solving the large-scale traveling salesman problem. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 37, pages 9345–9353, Washington, DC, USA, 2023. AAAI Press. doi: 10.1609/aaai.v37i8.26120.

Christos H. Papadimitriou. The Euclidean traveling salesman problem is NP-complete. *Theoretical Computer Science*, 4(3):237–244, 1977. doi: 10.1016/0304-3975(77)90012-3.

Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, Peter Prettenhofer, Ron Weiss, Vincent Dubourg, et al. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.

Allon G. Percus and Olivier C. Martin. Finite size and dimensional dependence in the Euclidean traveling salesman problem. *Physical Review Letters*, 76(8):1188–1191, 1996. doi: 10.1103/PhysRevLett.76.1188.

Boris T. Polyak. Minimization of unsmooth functionals. *USSR Computational Mathematics and Mathematical Physics*, 9(3):14–29, 1969. doi: 10.1016/0041-5553(69)90061-5.

Robert Clay Prim. Shortest connection networks and some generalizations. *Bell System Technical Journal*, 36(6): 1389–1401, 1957.

Gerhard Reinelt. TSPLIB—a traveling salesman problem library. *ORSA Journal on Computing*, 3(4):376–384, 1991. doi: 10.1287/ijoc.3.4.376.

Daniel J. Rosenkrantz, Richard E. Stearns, and Philip M. Lewis, II. An analysis of several heuristics for the traveling salesman problem. *SIAM Journal on Computing*, 6(3):563–581, 1977. doi: 10.1137/0206041.

Anatoliy I. Serdyukov. On some extremal tours in graphs (in Russian). *Upravlyaemye Sistemy*, 17:76–79, 1978.

Michael Ian Shamos and Dan Hoey. Closest-point problems. In *16th Annual Symposium on Foundations of Computer Science (FOCS 1975)*, pages 151–162. IEEE, 1975. doi: 10.1109/SFCS.1975.8.

Kate Smith-Miles, Jano I. van Hemert, and Xin Yu Lim. Understanding TSP difficulty by learning from evolved instances. In *Learning and Intelligent Optimization*, volume 6073 of *Lecture Notes in Computer Science*, pages 266–280. Springer, 2010. doi: 10.1007/978-3-642-13800-3_29.

J. Michael Steele. Growth rates of Euclidean minimal spanning trees with power weighted edges. *The Annals of Probability*, 16(4):1767–1787, 1988. doi: 10.1214/aop/1176991596.

Warren S. Torgerson. Multidimensional scaling: I. Theory and method. *Psychometrika*, 17(4):401–419, 1952. doi: 10.1007/BF02288916.

Taha Varol, Okan Örsan Özener, and Erinç Albey. Neural network estimators for optimal tour lengths of traveling salesperson problem instances with arbitrary node distributions. *Transportation Science*, 58(1):45–66, 2024. doi: 10.1287/trsc.2022.0015.

Pauli Virtanen, Ralf Gommers, Travis E. Oliphant, Matt Haberland, Tyler Reddy, David Cournapeau, Evgeni Burovski, Pearu Peterson, Warren Weckesser, Jonathan Bright, et al. SciPy 1.0: fundamental algorithms for scientific computing in Python. *Nature Methods*, 17:261–272, 2020. doi: 10.1038/s41592-019-0686-2.

Ton Volgenant and Roy Jonker. A branch and bound algorithm for the symmetric traveling salesman problem based on the 1-tree relaxation. *European Journal of Operational Research*, 9(1):83–89, 1982. doi: 10.1016/0377-2217(82)90015-7.

Yuan Yuan and Yuxing Peng. RaceTrack: An approximation algorithm for the mobile sink routing problem. In *Ad-Hoc, Mobile and Wireless Networks*, volume 6288 of *Lecture Notes in Computer Science*, pages 135–148. Springer, 2010. doi: 10.1007/978-3-642-14785-2_11.

# A Code and Data Availability

Our implementation, trained model weights, and benchmark scripts are available at `https://github.com/sfeldmanMIG25/Area-and-Distribution-Free-Estimator-for-TSP`. Synthetic instances can be regenerated using the provided scripts with fixed random seeds. The classical estimators and the calibrated constant-ratio baselines are released as separate modules, and the fitted constant-ratio table is frozen as a released artifact rather than refitted at inference. The eight per-benchmark results tables are generated from the scored benchmark outputs and written into the manuscript mechanically rather than typed; a verification pass re-derives all 1,638 table cells from those outputs and reports any that disagrees with the typeset value, and a second pass does the same for the 182 cells of the cost/accuracy ladders of Section 5. The remaining tables, and scalars quoted in the running prose such as the timing split and the quarantine and bound-check counts, are transcribed by hand from the same artifacts and are not covered by either check. The label validation of Appendix E, the refits of Section 4.5, the MDS diagnostics of Appendix Q and the figures of this paper are each released as a standalone reproduction script, and the versions of the Python analysis dependencies are pinned in the repository. For direct audit of target clipping, the two released out-of-interval rows are `N1000_D15_G100_pchergphhtgncoe_59` (training, raw $\alpha = 0.974823$) and `N5_D2_G100_oc_33` (validation, raw $\alpha = 2.059518$).

The GART 2.0 pipeline uses SciPy [Virtanen et al., 2020] for Delaunay triangulation [Delaunay, 1934] and MST construction, Numba [Lam et al., 2015] for centroid-distance computation, NumPy [Harris et al., 2020] symmetric eigendecomposition for classical MDS, LightGBM [Ke et al., 2017] for regression, and scikit-learn [Pedregosa et al., 2011] for metric computation. Estimator timings were collected on one Intel i7 13th Gen machine with 64 GB RAM and Windows 11. Concorde 03.12.19 and LKH-3 3.0.13 generated the stored reference tour costs; the canonical table does not retain solver certificates.

# B Training Data Generation Details

We constructed the training dataset by generating synthetic TSP instances across four distribution classes, assigning the generator independently per axis.

## B.1 Training Corpus

The uniform class draws coordinates from $\mathcal{U}(0, G)$ where $G$ is the grid size. The normal class draws from $\mathcal{N}(G/2, G/6)$ truncated to $[0, G]$. The clustered class places $k \in [3, 15]$ cluster centers uniformly and draws points from $\mathcal{N}(\mu_k, \sigma_k)$ around each center. The correlated class uses autoregressive structure across axes, $x_{j+1} = \rho \cdot x_j + \epsilon$ with $j$ indexing the axis, producing axis-aligned anisotropy.

Node counts span $n \in [5, 1{,}000]$ with training-visible dimensions $d \in \{2, 3, 4, 5, 6, 7, 8, 9, 10, 15, 20, 25, 30, 35, 40, 45, 50\}$, plus a held-out bucket at $d = 100$ (5,904 test instances). The pipeline ran Concorde and LKH-3 and retained the shorter tour cost.

The 106,272 solved instances are partitioned into training (69,768, 65.65%), validation (19,584, 18.43%), and test (16,920, 15.92%) using a stratified sample on $(d, n, \text{grid size})$. The stratification rule ranks instances within each stratum by a seeded uniform and assigns roughly the first tenth to test and the next fifth to validation, with the remainder to training; $d = 100$ instances bypass this rule and are locked to the test partition, which lifts the corpus-level test share to the 15.92% above. Table 11 lists the per-$(d, n)$ training-instance count; grid size $G \in \{100, 1{,}000, 10{,}000\}$ is sampled uniformly within each cell.

### B.2 2D Benchmark Generator Design

Four classes (isotropic, biased, geometric, clustered) follow conventions used in the TSP-instance-generation literature, including the variants reported by Çavdar and Sokol [2015] for the distribution-free baseline. The fifth class, Line Noise, samples $y = mx + b + \epsilon$ with small $\epsilon$ and then clips out-of-range coordinates to the grid boundary rather than resampling them. Because the slope reaches $\pm 5$ while $x$ is drawn across the full grid width, most points fall outside the box and are pinned to a face. The realized geometry is therefore a union of two to four thin segments with dense collinear masses along the boundary, not a single noisy line: at $n \geq 200$ a median 59.9% of points lie exactly on a face, and $\alpha$ tracks that fraction closely (Spearman 0.86, rising monotonically from 1.180 to 1.431 across its quartiles) while showing no relationship to the transverse noise width. We describe the class as realized rather than as intended because its difficulty comes from the segment-union structure. The `grid` member of the Geometric Struct. class draws a jittered square lattice, which no product of the four independent per-axis draws of Section 3.2 can produce, and the training corpus contains no lattice instance.

### B.3 Multidimensional Benchmark Grid

The full $(d, n)$ count grid of the 16,920-instance multidimensional benchmark:

Table 10: Multidimensional synthetic benchmark grid: instance count per $(d, n)$ bucket (16,920 instances total).

| $d \setminus n$ | $[5, 10]$ | $[11, 50]$ | $[51, 100]$ | $[101, 200]$ | $[201, 500]$ | $[501, 1000]$ | Total |
|---|---|---|---|---|---|---|---|
| $d = 2$ | 162 | 108 | 135 | 27 | 81 | 135 | **648** |
| $d =$ 3–5 | 486 | 324 | 405 | 81 | 243 | 405 | **1,944** |
| $d =$ 6–10 | 810 | 540 | 675 | 135 | 405 | 675 | **3,240** |
| $d =$ 15–25 | 486 | 324 | 405 | 81 | 243 | 405 | **1,944** |
| $d =$ 30–50 | 810 | 540 | 675 | 135 | 405 | 675 | **3,240** |
| $d = 100$ | 1,476 | 984 | 1,230 | 246 | 738 | 1,230 | **5,904** |
| **Total** | **4,230** | **2,820** | **3,525** | **705** | **2,115** | **3,525** | **16,920** |

Table 11: Training-set instance count per $(d, n)$ cell (69,768 total); grid size $G$ is sampled uniformly from $\{100,\ 1{,}000,\ 10{,}000\}$.

| $d \setminus n$ | $[5, 10]$ | $[11, 50]$ | $[51, 100]$ | $[101, 200]$ | $[201, 500]$ | $[501, 1000]$ | Total |
|---|---|---|---|---|---|---|---|
| $d = 2$ | 1,026 | 684 | 855 | 171 | 513 | 855 | **4,104** |
| $d =$ 3–5 | 3,078 | 2,052 | 2,565 | 513 | 1,539 | 2,565 | **12,312** |
| $d =$ 6–10 | 5,130 | 3,420 | 4,275 | 855 | 2,565 | 4,275 | **20,520** |
| $d =$ 15–25 | 3,078 | 2,052 | 2,565 | 513 | 1,539 | 2,565 | **12,312** |
| $d =$ 30–50 | 5,130 | 3,420 | 4,275 | 855 | 2,565 | 4,275 | **20,520** |
| **Total** | **17,442** | **11,628** | **14,535** | **2,907** | **8,721** | **14,535** | **69,768** |

## C Detailed Feature Specifications

Table 24 (Appendix O) lists the 11 geometric and centroid features, Tables 12 and 13 the 19 MST-derived ones, and Table 14 the single constructive ratio, 31 in all (Section 3.1). Listed complexities assume the MST has already been built; MST construction itself dominates the asymptotic cost and is analyzed in Section 3.

### C.1 MST Edge Distribution Statistics

The first category collected here captures the local uniformity of the point set through the statistical moments of the MST edge weights. Unlike coordinate-based variance, edge weights are invariant to rotation and translation. High skewness or kurtosis can indicate a mixture of scales, such as dense clusters separated by voids; these moments are included as candidate predictors of the $\alpha$ ratio rather than as a tested causal mechanism.

Table 12: MST edge distribution statistics (10 features).

| Feature Name | Description & Rationale | Complexity | Origin / Citation |
|---|---|---|---|
| **MST Edge Max** | Weight of the longest edge in the MST. | $O(n)$ | Standard |
| **MST Quantiles** | 10%, 25%, 50%, 75%, 90% percentiles. Provides a non-parametric distribution profile. | $O(n \log n)$ | **Proposed**; via MST [Prim, 1957] |
| **MST Edge Mean** | Average edge length in the MST. | $O(n)$ | Smith-Miles et al. [2010] |
| **MST Edge Std** | Standard deviation of edge lengths in the MST. | $O(n)$ | Smith-Miles et al. [2010] |
| **MST Edge Skewness** | Asymmetry of edge distribution. High positive skew implies many short edges and few long bridges. | $O(n)$ | Smith-Miles et al. [2010] |
| **MST Edge Kurtosis** | Tailedness of distribution. Indicates the presence of extreme outliers. | $O(n)$ | Smith-Miles et al. [2010] |

### C.2 Topological Structure and Clustering Proxies

The second category contains features designed to detect higher-order structures—such as clusters, filaments, and hubs—without performing expensive clustering algorithms. We introduce specific ratios (Dominance, Gap, Normalized Diameter) as $O(n)$ candidate proxies for these geometric properties. The Leaf Ratio is motivated by the hypothesis that low-leaf trees are more common on linear manifolds than in isotropic clouds; this interpretation is not evaluated independently here.

Table 13: Topological and clustering feature specifications (9 features).

| Feature Name | Description & Rationale | Complexity | Origin / Citation |
|---|---|---|---|
| **MST Dominance Ratio** | Sum of largest $\sqrt{n}$ edges / Total Length. Captures weight of inter-cluster bridges. | $O(n)$ | **Proposed**; via MST [Prim, 1957] |
| **MST Gap Ratio** | Max edge / Median edge. Measures magnitude of sparsity jumps. | $O(1)$ | **Proposed**; via MST [Prim, 1957] |
| **MST Leaf Ratio** | Fraction of nodes with degree $k = 1$. Topology proxy for dimension. | $O(n)$ | Smith-Miles et al. [2010] |
| **MST Degree (Mean, Std)** | Branching factor statistics. | $O(n)$ | Smith-Miles et al. [2010] |
| **MST Degree Max** | Maximum node degree. Identifies "hub" nodes common in High-D spaces. | $O(n)$ | Smith-Miles et al. [2010] |
| **MST Diameter** | Sum of edge weights on the longest path between any two leaves, computed via double-BFS. | $O(n)$ | Graph Theory |
| **MST Normalized Diameter** | Diameter / Total Length. Scale-invariant shape proxy ($= 1.0$ for path-manifolds exactly). | $O(1)$ | **Proposed**; via double-BFS |
| **Large Edge Count** | Count of edges $> \mu + \sigma$. Secondary cluster indicator. | $O(n)$ | **Proposed**; via MST [Prim, 1957] |

### C.3 Constructive Ratio

The third and last category holds a single feature. It is the only input GART 2.0 adds to the 30-feature block it inherits, it ranks second by SHAP magnitude (Appendix D), and it is the feature the non-Euclidean coverage gate of Section 6 tests.

Table 14: Constructive ratio specification (1 feature).

| Feature Name | Description & Rationale | Complexity | Origin / Citation |
|---|---|---|---|
| **Greedy-to-MST Ratio** | Length of an exact greedy nearest-neighbor tour divided by $L_{\mathrm{MST}}$. A greedy tour is feasible, so the ratio is a constructed upper bound on the target $\alpha$. Values outside $[1.035, 2.209]$, the range spanned by the full 106,272-row corpus, trigger the coverage gate; the training split alone spans $[1.0465, 2.1295]$. | $O(n^2d)$ dense for $n \leq 3000$; $O(nmd)$ with $m = 16$ candidate neighbors above it | **Proposed**; greedy nearest-neighbor construction |

# D Feature Importance: SHAP Values

To characterize the 31-feature set, we report the SHAP values [Lundberg and Lee, 2017] against the released booster on a 5,000-row sample of the 16,920-row held-out test split. Table 15 ranks all 31 features by mean absolute SHAP contribution.

Section 3.4 gives the per-feature shares. By group, the nineteen MST-derived features carry 50.7% of total mean absolute SHAP magnitude, the eleven geometric features 25.5%, and the single constructive ratio 23.9%. SHAP magnitude describes model reliance, not a causal effect or proof that a feature group improves generalization.

We used SHAP analysis as a diagnostic during feature reduction, yielding the 31-feature set described in Section 3.1; because this selection used validation feedback, it should not be interpreted as an independent confirmatory test.

Table 15: All 31 features ranked by mean absolute SHAP share, against the released booster on a 5,000-row test sample.

| Rank | Feature | Mean \|SHAP\| | Share (%) |
|---|---|---|---|
| 1 | `mst_dominance_ratio` | 0.2866 | 26.52 |
| 2 | `greedy_nn_over_mst` | 0.2578 | 23.86 |
| 3 | `n_customers` | 0.0922 | 8.53 |
| 4 | `mst_diameter_normalized` | 0.0886 | 8.20 |
| 5 | `dimension` | 0.0605 | 5.60 |
| 6 | `log_bounding_hypervolume` | 0.0513 | 4.75 |
| 7 | `mst_degree_mean` | 0.0414 | 3.84 |
| 8 | `bounding_hypervolume` | 0.0261 | 2.41 |
| 9 | `mst_gap_ratio` | 0.0195 | 1.81 |
| 10 | `centroid_dist_std` | 0.0193 | 1.79 |
| 11 | `large_edge_count` | 0.0183 | 1.70 |
| 12 | `mst_leaf_ratio` | 0.0179 | 1.66 |
| 13 | `mst_diameter` | 0.0154 | 1.42 |
| 14 | `mst_degree_std` | 0.0119 | 1.10 |
| 15 | `mst_edge_skew` | 0.0069 | 0.64 |
| 16 | `aspect_ratio` | 0.0068 | 0.63 |
| 17 | `mst_degree_max` | 0.0063 | 0.59 |
| 18 | `centroid_dist_mean` | 0.0059 | 0.55 |
| 19 | `mst_edge_std` | 0.0051 | 0.47 |
| 20 | `centroid_dist_iqr` | 0.0046 | 0.42 |
| 21 | `mst_edge_q25` | 0.0045 | 0.42 |
| 22 | `mst_edge_max` | 0.0045 | 0.42 |
| 23 | `mst_edge_kurtosis` | 0.0044 | 0.41 |
| 24 | `mst_edge_q75` | 0.0039 | 0.36 |
| 25 | `node_density` | 0.0035 | 0.33 |
| 26 | `mst_edge_q90` | 0.0035 | 0.33 |
| 27 | `mst_edge_mean` | 0.0029 | 0.27 |
| 28 | `mst_edge_q50` | 0.0029 | 0.27 |
| 29 | `mst_edge_q10` | 0.0027 | 0.25 |
| 30 | `centroid_dist_max` | 0.0026 | 0.24 |
| 31 | `log_node_density` | 0.0026 | 0.24 |

# E Label Validation and Certification

A benchmark is only as sound as its labels, so every label is screened before any table is built. The first screen is consistency. We recompute the length of every stored tour on its own released coordinates, over all $106{,}272$ instances; a stored cost that its own tour cannot reproduce to within rounding describes no released point set, and no bound is tight enough to reconstruct the optimum from the coordinates alone. The 184 instances ($0.173\%$) that fail are *quarantined* rather than scored: they carry an empty reference cost and the status `quarantined_label_unrecoverable` in every released results CSV, an aggregate that ignores the status column yields NaN and never a plausible number, and the scoring harness raises on any attempt to score one. 74 of the 184 fall in the multidimensional test partition, which is therefore scored on the remaining $16{,}846$ instances throughout.

A certified lower bound is also an acceptance test. If $B$ is a proven lower bound on an instance's optimum and $L$ is its stored label, then $B \le \mathrm{OPT} \le L$ must hold whenever $L$ is an upper bound on the optimum, and $B > L$ refutes the label outright, with no tour entering the test. The released labels are constructed to face that test with no slack. Every reference cost is recomputed in float64 from the released coordinates: at $n \le 10$ the label is the exact Held–Karp dynamic-programming optimum, a certificate; above that size the label is the float64 length of the retained tour, an upper bound that is exact wherever that tour is optimal; and where a 1-tree bound closes on the tour the label is certified after the fact, 2,182 instances. The 2D benchmark's labels are the better of the stored tour and a float-metric 2-opt descent from it, because a solver that minimizes a rounded metric can select a tour the float64 metric can beat. TSPLIB is scored on its published optima, with one documented exception: `linhp318` declares a fixed-edge Hamiltonian-path problem on `lin318`'s coordinates, so its published 41,345 is a path optimum no tour on those coordinates can attain; every estimator and the bound are scored there against the tour optimum on those coordinates, 42,029, which is `lin318`'s published value; a scan of the full instance set found no second fixed-edge declaration and no second duplicated geometry.

Across the 145,013 bound checks the released artifacts supply there are zero violations of $B \leq L$, the worst relative excess being $5.8 \times 10^{-15}$. Two coverage limits bound the test. The 184 quarantined instances above carry no recoverable label and are outside it. And seven TSPLIB instances carry no bound, because the test runs in TSPLIB's integer metric there and the dense perturbed matrix does not fit in memory, five of them inside the scored EUC_2D set; nothing in this section is claimed about those instances.

## F Benchmark-Model Implementation Notes

This appendix records what each scored classical estimator receives and why three further estimators are surveyed in the literature review and scored nowhere.

**BHH.** For points drawn i.i.d. from a density $f$ of bounded support, $L_n/n^{(d-1)/d} \to \beta_d \int f(x)^{(d-1)/d}\,dx$ [Beardwood et al., 1959]; for $f$ uniform on a region of measure $\mathcal{V}$ the integral is $\mathcal{V}^{1/d}$, giving $\beta_d n^{(d-1)/d}\mathcal{V}^{1/d}$. Here $\mathcal{V}$ is the measure of the sampling region, and the convex hull of the realized sample is a different, downward-biased statistic that converges to it only as $n$ grows. Our synthetic generators draw on $[0, G]^d$, so $\mathcal{V} = G^d$ is known exactly, and we report that form on the matched domain. We use $\beta_2 = 0.7124 \pm 0.0002$ [Johnson et al., 1996] and $\beta_3 = 0.6979 \pm 0.0002$ [Percus and Martin, 1996]. For $d \geq 4$ no comparably pinned estimate exists and we substitute the large-$d$ asymptotic $\sqrt{d/(2\pi e)}$ [Bertsimas and van Ryzin, 1990], a form that source proves for the minimum-spanning-tree and minimum-matching constants and only conjectures for the TSP constant; that substitution, and the fact that $n \leq 1000$ with $d \geq 4$ is far from the asymptotic regime, both bound how much the $d \geq 4$ rows can be read as a test of the theorem.

**Not scored: Daganzo, Chien, and Kwon–Golden–Wasil.** We survey all three in Section 1.2 and score none of them, for provenance rather than performance. Their equations reach the modern record through the literature review in Figliozzi [2009], not through the articles themselves: Daganzo [1984a], which is the Daganzo paper that review carries, Chien [1992] and Kwon et al. [1995] are paywalled, a direct check of each DOI returned no open-access location, and we could not obtain them. Nor is the secondary record self-consistent. For Chien, Çavdar and Sokol [2015] give a Daganzo-form coefficient of 0.69 and Choi [2021] give 0.88, against the 0.67 of the $2.1\bar{r} + 0.67\sqrt{nR}$ form that review carries; for Kwon–Golden–Wasil, the published renderings disagree on whether the bracket carries $n$ or $(n+1)$ and on the precision of its coefficients. Choosing between those renderings requires the primaries, which is exactly what we lack, so we print no number for any of the three. They stay in the survey because they are part of the record this work sits in, and omitting them would misdescribe the field. The scored pair, BHH and Çavdar–Sokol, are the two whose primary documents we hold.

**Çavdar–Sokol.** This estimator is implemented directly from a primary document we hold: Çavdar [2014], Chapter 4, the same work as Çavdar and Sokol [2015] and by the same author. The area term is the area of the rectangle covering the nodes, and $\bar{c}_j$ and $\mathrm{cstdev}_j$ are the mean and standard deviation of the absolute distances to that rectangle's midpoint line on axis $j$, not to the sample mean. The divisor $\bar{c}_x\bar{c}_y$ sits inside the radical; placing it outside leaves the second term dimensionless. We apply two further elements of the published method. The first is the frame: the estimator is defined on an axis-aligned rectangle and is not rotation invariant, so the implementation rotates each instance into its minimum-area enclosing rectangle by scanning convex-hull edges [Freeman and Shapira, 1975], which is what the source prescribes. The second is the finite-$n$ correction $0.9325e^{0.00005298n} - 0.2972e^{-0.01452n}$, Eq. (21) of the dissertation, fitted over $n \in \{100, 125, \ldots, 975\}$ at $R^2 = 0.9867$; the raw estimate is divided by it. Choi [2021] records that the uncorrected estimator underestimates for $n < 1000$, which is the direction the correction moves it. That correction is bounded to the range it was fitted over, and the bound binds on this benchmark: 1,320 of the 2,580 2D instances have $n < 100$, so for slightly over half the benchmark the ratio is evaluated at the $n = 100$ endpoint rather than extrapolated below it. The authors' own figures show the ratio continuing to fall toward $0.4$ as $n$ approaches 10, but publish no fitted form there, so extrapolating would substitute our curve for theirs. We hold the upper end fixed for the same reason: above $n = 975$ no correction is applied, because Eq. (21) grows without bound ($E/T = 1.22$ at $n = 5000$) and would contradict the training fit it exists to repair. That leaves a $1.8\%$ step at the upper boundary, where the fitted ratio reads 0.982. Two properties of the source, not of our evaluation, remain: the regression targets Lin–Kernighan tour lengths rather than optima, and every training graph carried a node at each corner of its rectangle, which pins the graph area to the covering rectangle in a way our instances do not.

**Constant-ratio references.** The $\alpha = 1$ row returns $L_{\mathrm{MST}}$. The asymptotic row uses $\beta_{\mathrm{TSP}}/\beta_{\mathrm{MST}}$ for uniform planar points. The numerator is well established at $0.7124$; the denominator is not. We quote $0.6331$ following Cortina-Borja and Robinson [2000]. That figure is a simulation estimate and not a theorem, and published estimates of the planar Euclidean MST constant span a range wider than its printed precision, so the ratio $1.1253$ should be read as approximate. The calibrated rows take the mean of $\alpha$ within each dimension, and within each dimension–size cell, over the training split only; the fitted values are frozen and released rather than refitted at inference. At $d = 100$, which no training row covers, the table extrapolates from $d = 50$.

**Custom Hilbert sort.** We independently min–max normalize every axis to a 16-bit grid, sort by a $d$-dimensional Hilbert index [Hilbert, 1891], visit nodes in that order, and close the tour. This is a custom generalization inspired by the planar space-filling-curve heuristic of Bartholdi and Platzman [1982]; its planar performance guarantee does not transfer to our anisotropically normalized multidimensional ordering. It constructs a tour rather than estimating one, so its error is bounded below by the gap between a space-filling-curve tour and the optimum.

**Sources.** Every constant this paper scores is read from a primary document. The BHH constants come from Johnson et al. [1996] and Percus and Martin [1996]; the Çavdar–Sokol form and both of its constant sets are verified directly against Çavdar [2014], which is open access. No scored constant is transcribed from a secondary source, and where the only available rendering was secondary we declined to score the estimator rather than publish the number.

**Hybrid non-Euclidean builder.** The hybrid builder of Section 6 computes every feature the released model consumes, and the downstream feature lookup raises on a missing feature rather than defaulting it. Bounding hypervolume is stored as $\exp(\min(\log \mathrm{hv},\, 690))$, so the raw feature stays finite at high embedding dimension while its log-scale companions carry the magnitude past the cap.

**Convex-hull cap.** Where an estimator receives the convex-hull plug-in we compute the hull only for $d \leq 3$ and substitute the coordinate bounding box above it, recording which was used. QHull's output size grows as $n^{\lfloor d/2 \rfloor}$, the same blow-up cited in Section 1 as a reason not to build estimators on hull geometry; at $d = 8$ with $n = 1000$ a single hull dominates the entire benchmark's runtime.

## G Classical Estimators: Full Benchmark and Matched Domain

Table 16 reports the two classical estimators twice. The upper panels apply them to the complete 2D and TSPLIB EUC_2D benchmarks, with the convex hull of each instance standing in for BHH's region measure, which is the only option on TSPLIB and the fairer option on the degenerate 2D classes. The lower panel restricts to the 210 i.i.d. uniform

2D instances, where the sampling region $[0, G]^2$ is exact and BHH receives it. Çavdar–Sokol receives the same input in both, the rectangle covering the nodes, so only its instance set changes between the panels.

Table 16: Classical estimators on the full benchmarks and on their matched domains; a large MSPE relative to MAPE identifies a systematic offset.

| Domain | Estimator | $N$ | SDPE (%) [95% CI] | MAPE (%) | \|Median\| (%) | MSPE (%) |
|---|---|---|---|---|---|---|
| Full 2D benchmark | BHH | 2,580 | 32.00 [30.00, 33.97] | 23.80 | 15.53 | -13.71 |
| | Çavdar–Sokol | 2,580 | 34.61 [31.96, 37.12] | 18.24 | 6.63 | -0.96 |
| | **GART 2.0** | 2,580 | 4.68 [4.42, 4.92] | 2.89 | 1.49 | -0.10 |
| | Calibrated MST ratio $\hat{\rho}(d, n)$ | 2,580 | 9.38 [8.96, 9.78] | 5.76 | 2.31 | -0.75 |
| TSPLIB EUC_2D | BHH | 78 | 40.75 [26.25, 53.61] | 25.14 | 13.41 | 13.66 |
| | Çavdar–Sokol | 78 | 36.34 [24.38, 48.39] | 23.87 | 11.32 | 20.07 |
| | **GART 2.0** | 78 | 2.92 [2.40, 3.38] | 2.53 | 1.87 | 2.08 |
| | Calibrated MST ratio $\hat{\rho}(d, n)$ | 78 | 4.48 [3.59, 5.32] | 3.74 | 3.18 | 1.18 |
| 2D `random` (uniform on $[0, G]^2$) matched domain | **GART 2.0** | 210 | 2.07 [1.65, 2.48] | 1.31 | 0.75 | -0.01 |
| | GART 1.0 | 210 | 4.96 [4.53, 5.35] | 5.75 | 6.05 | 4.69 |
| | BHH (sampling region) | 210 | 7.76 [6.39, 8.98] | 8.91 | 6.70 | -8.65 |
| | Çavdar–Sokol | 210 | 14.28 [11.63, 16.53] | 8.16 | 2.84 | -5.97 |
| | $L_{\mathrm{MST}}$ ($\alpha = 1$) | 210 | 7.70 [6.39, 8.82] | 15.60 | 12.52 | -15.60 |

## H Per-Class Detail for the 2D Diverse Benchmark

Table 17 disaggregates the 2D benchmark by the five stress classes of Table 2, with the Geometric Struct. class split into its `grid` generator and its other two. Two of the six rows carry geometry the training corpus does not contain, Line Noise and `grid`. Reported inside the 630-instance class aggregate, `grid`'s signed error is what carries that aggregate to $+2.74$ MSPE while the two represented generators sit at $+0.58$; splitting the row is what makes the second unrepresented generator visible.

Table 17: 2D benchmark by generator class; `grid` is its own row because it has no training counterpart. †: not represented in training.

| Class | Estimator | SDPE (%) [95% CI] | MAPE (%) | \|Median\| (%) | MSPE (%) | $R^2_\alpha$ | Time (ms) |
|---|---|---|---|---|---|---|---|
| Isotropic Count=840 | **GART 2.0** | **2.23** [2.02, 2.44] | 1.47 | 0.91 | 0.13 | 0.954 | 108 |
| | Calibrated MST ratio $\hat{\rho}(d, n)$ | 5.11 [4.65, 5.53] | 3.31 | 1.89 | 0.38 | 0.744 | 25.4 |
| | GART 1.0 | 5.18 [4.94, 5.43] | 5.38 | 5.29 | 3.82 | 0.701 | 49.5 |
| | $L_{\mathrm{MST}}$ ($\alpha = 1$) | 7.79 [7.26, 8.29] | 17.15 | 14.51 | -17.15 | -2.451 | 24.9 |
| | Asymptotic MST ratio | 8.77 [8.05, 9.37] | 7.12 | 3.81 | -6.77 | -0.455 | 22.1 |
| | Custom Hilbert sort | 11.50 [10.84, 12.16] | 30.78 | 35.07 | 30.78 | — | 0.71 |
| Biased Count=840 | **GART 2.0** | **2.51** [2.31, 2.69] | 1.79 | 1.24 | 0.30 | 0.959 | 114 |
| | Calibrated MST ratio $\hat{\rho}(d, n)$ | 5.24 [4.79, 5.66] | 3.21 | 1.54 | -1.26 | 0.757 | 24.3 |
| | GART 1.0 | 7.97 [7.58, 8.34] | 7.93 | 6.19 | 6.70 | 0.487 | 49.7 |
| | $L_{\mathrm{MST}}$ ($\alpha = 1$) | 8.97 [8.36, 9.53] | 18.36 | 14.75 | -18.36 | -2.012 | 23.6 |
| | Asymptotic MST ratio | 10.09 [9.38, 10.69] | 8.22 | 4.08 | -8.13 | -0.475 | 21.5 |
| | Custom Hilbert sort | 13.14 [12.49, 13.78] | 32.04 | 35.69 | 32.04 | — | 0.719 |

*Table 17 continued from the previous page*

| Class | Estimator | SDPE (%) [95% CI] | MAPE (%) | \|Median\| (%) | MSPE (%) | $R^2_\alpha$ | Time (ms) |
|---|---|---|---|---|---|---|---|
| | **GART 2.0** | **1.92** [1.66, 2.19] | 7.07 | 7.64 | 7.07 | 0.090 | 103 |
| | GART 1.0 | 4.97 [4.51, 5.43] | 11.90 | 9.77 | 11.90 | -1.871 | 45.1 |
| Geometric: Grid† | $L_{\mathrm{MST}}$ ($\alpha = 1$) | 6.01 [4.73, 7.17] | 4.54 | 2.35 | -4.54 | -0.429 | 27.1 |
| Count=210 | Calibrated MST ratio $\hat{\rho}(d, n)$ | 6.42 [5.38, 7.37] | 15.99 | 14.82 | 15.99 | -4.371 | 26.6 |
| | Asymptotic MST ratio | 6.76 [5.23, 8.04] | 9.43 | 10.08 | 7.41 | -0.817 | 24.6 |
| | Custom Hilbert sort | 6.86 [6.16, 7.52] | 16.14 | 15.81 | 16.14 | — | 0.929 |
| | **GART 2.0** | **2.60** [2.29, 2.93] | 1.96 | 1.48 | 0.58 | 0.938 | 117 |
| | Calibrated MST ratio $\hat{\rho}(d, n)$ | 5.06 [4.37, 5.71] | 2.99 | 1.14 | 0.53 | 0.736 | 24.7 |
| Geometric: Boundary, X-Central | GART 1.0 | 5.33 [4.94, 5.72] | 5.50 | 5.28 | 3.81 | 0.700 | 50.8 |
| Count=420 | $L_{\mathrm{MST}}$ ($\alpha = 1$) | 7.89 [6.97, 8.68] | 17.01 | 13.76 | -17.01 | -2.299 | 24.6 |
| | Asymptotic MST ratio | 8.87 [7.93, 9.75] | 6.69 | 2.95 | -6.62 | -0.419 | 22.5 |
| | Custom Hilbert sort | 11.61 [10.66, 12.56] | 31.37 | 35.84 | 31.37 | — | 0.823 |
| | **GART 2.0** | **2.75** [2.06, 3.36] | 2.24 | 1.76 | -1.06 | 0.828 | 148 |
| | Calibrated MST ratio $\hat{\rho}(d, n)$ | 5.36 [4.31, 6.23] | 7.94 | 7.01 | -7.94 | -0.950 | 33.8 |
| Clustered | $L_{\mathrm{MST}}$ ($\alpha = 1$) | 5.65 [4.69, 6.46] | 20.46 | 19.41 | -20.46 | -7.816 | 30.7 |
| Count=60 | Asymptotic MST ratio | 6.36 [5.29, 7.29] | 10.50 | 9.31 | -10.50 | -2.157 | 31.8 |
| | GART 1.0 | 6.71 [5.46, 7.86] | 5.41 | 4.06 | -2.53 | -0.071 | 69.2 |
| | Custom Hilbert sort | 9.43 [7.18, 11.20] | 40.97 | 43.87 | 40.97 | — | 7.14 |
| | **GART 2.0** | **6.09** [5.67, 6.47] | 10.82 | 11.04 | -10.82 | 0.294 | 106 |
| | Calibrated MST ratio $\hat{\rho}(d, n)$ | 8.64 [8.04, 9.18] | 20.45 | 20.17 | -20.45 | -1.364 | 22.8 |
| Line Noise† | $L_{\mathrm{MST}}$ ($\alpha = 1$) | 10.67 [10.01, 11.26] | 34.13 | 38.24 | -34.13 | -5.250 | 24.2 |
| Count=210 | Asymptotic MST ratio | 12.01 [11.26, 12.64] | 25.88 | 30.51 | -25.88 | -3.154 | 23.3 |
| | GART 1.0 | 14.90 [13.48, 16.16] | 12.03 | 10.50 | 2.17 | 0.262 | 47.1 |
| | Custom Hilbert sort | 24.64 [23.00, 26.22] | 38.32 | 42.86 | 38.32 | — | 0.662 |
| | **GART 2.0** | **4.68** [4.42, 4.92] | 2.89 | 1.49 | -0.10 | 0.867 | 113 |
| | GART 1.0 | 7.90 [7.60, 8.16] | 7.30 | 6.27 | 5.13 | 0.637 | 50.1 |
| **Total** | Calibrated MST ratio $\hat{\rho}(d, n)$ | 9.38 [8.96, 9.78] | 5.76 | 2.31 | -0.75 | 0.462 | 25 |
| Count=2,580 | $L_{\mathrm{MST}}$ ($\alpha = 1$) | 10.27 [9.91, 10.61] | 17.95 | 14.61 | -17.95 | -1.621 | 24.9 |
| | Asymptotic MST ratio | 11.55 [11.15, 11.91] | 9.20 | 5.34 | -7.68 | -0.378 | 22.6 |
| | Custom Hilbert sort | 14.19 [13.75, 14.60] | 30.95 | 34.97 | 30.95 | — | 0.726 |

# I Rank Agreement

Table 18 reports global rank correlation and close-pair ordering accuracy with $L_{\mathrm{MST}}$ as the control. Close-pair accuracy is the fraction of instance pairs whose reference costs differ by less than the stated threshold that the estimator orders concordantly with the reference. Pairs whose reference costs are exactly equal carry no reference ordering and are excluded. Every benchmark is enumerated exhaustively, so the statistic is exact and no random seed enters it: $\binom{2580}{2} = 3{,}326{,}910$ pairs on 2D, $\binom{78}{2} = 3{,}003$ on TSPLIB EUC_2D and $\binom{16846}{2} = 141{,}885{,}435$ on the multidimensional set, the 74 quarantined instances of Appendix E having no reference cost to order. Of the multidimensional pairs 1,647,719 qualify at the 5% threshold and 3,262,149 at 10%; the corresponding 2D counts are 74,826 and 152,306, and the TSPLIB EUC_2D counts 54 and 105. The 142 million multidimensional pairs are never held at once: because the relative gap to a larger partner increases with that partner's cost, sorting the reference costs makes each instance's qualifying partners a contiguous run whose end binary search locates, so only the qualifying pairs are built. That enumeration was verified to agree pair-for-pair with a direct scan of all 141,885,435 pairs.

Table 18: Rank agreement with reference tour cost; the $\alpha = 1$ row is the ordering every constant multiple of $L_{\mathrm{MST}}$ induces.

| Benchmark | Estimator | $N$ | Spearman $\rho$ | Kendall $\tau$ | Close pair $< 5\%$ (%) | Close pair $< 10\%$ (%) |
|---|---|---|---|---|---|---|
| 2D | **GART 2.0** | 2,580 | 0.9994 | 0.9827 | 74.48 | 83.56 |
| | GART 1.0 | 2,580 | 0.9985 | 0.9713 | 69.14 | 76.71 |
| | Calibrated MST ratio $\hat{\rho}(d, n)$ | 2,580 | 0.9979 | 0.9664 | 68.86 | 76.07 |
| | Asymptotic MST ratio | 2,580 | 0.9961 | 0.9615 | 71.04 | 77.76 |
| | $L_{\mathrm{MST}}$ ($\alpha = 1$) | 2,580 | 0.9961 | 0.9615 | 71.04 | 77.76 |
| Multidimensional | **GART 2.0** | 16,846 | 1.0000 | 0.9982 | 92.41 | 96.11 |
| | Calibrated MST ratio $\hat{\rho}(d, n)$ | 16,846 | 0.9999 | 0.9948 | 81.72 | 89.04 |
| | $L_{\mathrm{MST}}$ ($\alpha = 1$) | 16,846 | 0.9996 | 0.9856 | 69.05 | 75.86 |
| | Custom Hilbert sort | 16,846 | 0.9989 | 0.9726 | 58.53 | 64.31 |
| TSPLIB EUC_2D | **GART 2.0** | 78 | 0.9993 | 0.9867 | 72.22 | 80.95 |
| | GART 1.0 | 78 | 0.9982 | 0.9773 | 70.37 | 72.38 |
| | Asymptotic MST ratio | 78 | 0.9986 | 0.9793 | 61.11 | 73.33 |
| | $L_{\mathrm{MST}}$ ($\alpha = 1$) | 78 | 0.9986 | 0.9793 | 61.11 | 73.33 |

## J Paired Comparisons Against GART 2.0

We compare aggregate margins on the instances where both estimators return a finite prediction. We report the mean paired difference in absolute percent error, GART 2.0 minus the baseline, with a 95% paired bootstrap interval over 1,000 resamples of instances, and a Wilcoxon signed-rank $p$-value. Negative values favor GART 2.0. Table 19 reports the comparisons the body cites; the complete set is released with the replication materials.

Table 19: Paired difference in absolute percent error, GART 2.0 minus baseline, in percentage points; negative favors GART 2.0.

| Benchmark | Baseline | $N$ | Mean diff. (pp) [95% CI] | Wilcoxon $p$ |
|---|---|---|---|---|
| TSPLIB EUC_2D | Asymptotic MST ratio | 78 | $-1.00$ [$-1.77$, $-0.22$] | 0.0049 |
| TSPLIB EUC_2D | $L_{\mathrm{MST}}$ ($\alpha = 1$) | 78 | $-8.84$ [$-9.99$, $-7.63$] | $1.1 \times 10^{-13}$ |
| 2D | Asymptotic MST ratio | 2,580 | $-6.31$ [$-6.68$, $-5.99$] | $2.5 \times 10^{-289}$ |
| 2D | Calibrated MST ratio $\hat{\rho}(d, n)$ | 2,580 | $-2.87$ [$-3.07$, $-2.69$] | $1.3 \times 10^{-159}$ |
| 2D | $L_{\mathrm{MST}}$ ($\alpha = 1$) | 2,580 | $-15.06$ [$-15.44$, $-14.71$] | $< 10^{-300}$ |
| Multidimensional | Calibrated MST ratio $\hat{\rho}(d, n)$ | 16,846 | $-1.20$ [$-1.23$, $-1.16$] | $< 10^{-300}$ |
| Multidimensional | $L_{\mathrm{MST}}$ ($\alpha = 1$) | 16,846 | $-9.09$ [$-9.19$, $-8.98$] | $< 10^{-300}$ |
| 2D uniform | BHH (sampling region) | 210 | $-7.60$ [$-8.56$, $-6.71$] | $2.0 \times 10^{-35}$ |
| 2D uniform | Çavdar–Sokol | 210 | $-6.85$ [$-8.57$, $-5.30$] | $7.2 \times 10^{-27}$ |
| 2D uniform | $L_{\mathrm{MST}}$ ($\alpha = 1$) | 210 | $-14.29$ [$-15.35$, $-13.38$] | $3.3 \times 10^{-36}$ |

## K Per-Dimension Detail for the Multidimensional Benchmark

Table 20 reports the multidimensional benchmark broken down by dimension $d$.

Table 20: Multidimensional benchmark by dimension; same 16,846 scored instances as Table 21. Within each dimension rows are ordered by SDPE.

| Dimension | Estimator | SDPE (%) [95% CI] | MAPE (%) | \|Median\| (%) | MSPE (%) | $R^2_\alpha$ | Time (ms) |
|---|---|---|---|---|---|---|---|
| $d = 2$ Count=647 | **GART 2.0** | **2.37** [2.13, 2.60] | 1.57 | 0.91 | 0.28 | 0.966 | 67.9 |
| | Calibrated MST ratio $\hat{\rho}(d, n)$ | 6.19 [5.67, 6.71] | 4.05 | 2.06 | 0.53 | 0.745 | 38.6 |
| | $L_{\mathrm{MST}}$ ($\alpha = 1$) | 9.78 [9.21, 10.35] | 19.14 | 15.18 | -19.14 | -1.988 | 50.3 |
| | Custom Hilbert sort | 15.75 [14.80, 16.87] | 29.74 | 35.22 | 29.74 | — | 1.47 |
| | BHH (sampling region) | 30.92 [25.44, 37.70] | 18.93 | 11.63 | 5.11 | — | 0.094 |
| | *Reference-tour generation* | — | — | — | — | — | 102 ms |
| $d = 3$–5 Count=1,944 | **GART 2.0** | **1.71** [1.60, 1.82] | 1.07 | 0.59 | 0.03 | 0.966 | 76 |
| | Calibrated MST ratio $\hat{\rho}(d, n)$ | 4.05 [3.81, 4.29] | 2.53 | 1.29 | 0.13 | 0.793 | 41 |
| | $L_{\mathrm{MST}}$ ($\alpha = 1$) | 7.67 [7.39, 7.93] | 14.96 | 11.73 | -14.96 | -2.285 | 45 |
| | Custom Hilbert sort | 15.72 [15.24, 16.20] | 35.70 | 40.14 | 35.70 | — | 1.62 |
| | BHH (sampling region) | 24.14 [22.93, 25.27] | 26.81 | 27.99 | -19.14 | — | 0.107 |
| | *Reference-tour generation* | — | — | — | — | — | 109 ms |
| $d = 6$–10 Count=3,238 | **GART 2.0** | **1.01** [0.96, 1.06] | 0.64 | 0.36 | -0.01 | 0.982 | 74.4 |
| | Calibrated MST ratio $\hat{\rho}(d, n)$ | 2.98 [2.84, 3.11] | 1.79 | 0.80 | 0.16 | 0.830 | 28.7 |
| | $L_{\mathrm{MST}}$ ($\alpha = 1$) | 6.46 [6.26, 6.64] | 11.92 | 8.96 | -11.92 | -2.304 | 28.5 |
| | BHH (sampling region) | 10.28 [9.99, 10.59] | 27.58 | 27.88 | -27.43 | — | 0.117 |
| | Custom Hilbert sort | 13.82 [13.53, 14.11] | 32.52 | 35.92 | 32.52 | — | 10.7 |
| | *Reference-tour generation* | — | — | — | — | — | 122 ms |
| $d = 15$–25 Count=1,942 | **GART 2.0** | **0.68** [0.63, 0.73] | 0.42 | 0.23 | -0.01 | 0.990 | 90.7 |
| | Calibrated MST ratio $\hat{\rho}(d, n)$ | 2.46 [2.31, 2.61] | 1.40 | 0.51 | 0.07 | 0.855 | 30.8 |
| | $L_{\mathrm{MST}}$ ($\alpha = 1$) | 5.98 [5.76, 6.20] | 9.40 | 6.41 | -9.40 | -1.813 | 32.2 |
| | BHH (sampling region) | 6.17 [5.96, 6.41] | 26.98 | 26.59 | -26.97 | — | 0.151 |
| | Custom Hilbert sort | 11.71 [11.43, 11.98] | 24.58 | 26.33 | 24.58 | — | 40.6 |
| | *Reference-tour generation* | — | — | — | — | — | 123 ms |
| $d = 30$–50 Count=3,228 | **GART 2.0** | **0.48** [0.45, 0.50] | 0.29 | 0.15 | -0.03 | 0.995 | 110 |
| | Calibrated MST ratio $\hat{\rho}(d, n)$ | 2.31 [2.20, 2.41] | 1.29 | 0.40 | 0.03 | 0.866 | 32.8 |
| | BHH (sampling region) | 4.34 [4.22, 4.45] | 28.17 | 27.55 | -28.17 | — | 0.216 |
| | $L_{\mathrm{MST}}$ ($\alpha = 1$) | 5.94 [5.77, 6.10] | 8.02 | 4.96 | -8.02 | -1.405 | 33.2 |
| | Custom Hilbert sort | 7.99 [7.84, 8.14] | 17.07 | 18.12 | 17.07 | — | 54.8 |
| | *Reference-tour generation* | — | — | — | — | — | 136 ms |
| $d = 100$ Count=5,847 | **GART 2.0** | **0.44** [0.43, 0.45] | 0.59 | 0.63 | 0.51 | 0.991 | 162 |
| | Calibrated MST ratio $\hat{\rho}(d, n)$ | 2.21 [2.14, 2.28] | 1.78 | 1.07 | 1.03 | 0.856 | 35.4 |
| | BHH (sampling region) | 2.76 [2.71, 2.81] | 29.92 | 29.39 | -29.92 | — | 0.767 |
| | Custom Hilbert sort | 4.60 [4.54, 4.66] | 10.35 | 11.03 | 10.35 | — | 97.9 |
| | $L_{\mathrm{MST}}$ ($\alpha = 1$) | 5.93 [5.82, 6.06] | 6.73 | 3.60 | -6.73 | -1.043 | 37.6 |
| | *Reference-tour generation* | — | — | — | — | — | 120 ms |

*Table 20 continued from the previous page*

| Dimension | Estimator | SDPE (%) [95% CI] | MAPE (%) | \|Median\| (%) | MSPE (%) | $R_\alpha^2$ | Time (ms) |
|---|---|---|---|---|---|---|---|
| **Total** Count=16,846 | **GART 2.0** | **0.99** [0.95, 1.02] | 0.62 | 0.42 | 0.18 | 0.984 | 121 |
| | Calibrated MST ratio $\hat{\rho}(d, n)$ | 2.94 [2.87, 3.02] | 1.82 | 0.98 | 0.44 | 0.858 | 33.8 |
| | $L_{\text{MST}}$ ($\alpha = 1$) | 7.26 [7.16, 7.36] | 9.71 | 7.28 | -9.71 | -1.245 | 36.4 |
| | BHH (sampling region) | 13.57 [12.85, 14.40] | 28.01 | 28.56 | -26.18 | — | 0.248 |
| | Custom Hilbert sort | 14.40 [14.26, 14.53] | 21.21 | 16.23 | 21.21 | — | 51.5 |
| | *Reference-tour generation* | — | — | — | — | — | 122 ms |

## L Per-Size Detail for the Multidimensional Benchmark

Table 21 reports the multidimensional benchmark broken down by instance-size bucket.

Table 21: Multidimensional benchmark by instance-size bucket. Within each bucket rows are ordered by SDPE.

| Size bucket | Estimator | SDPE (%) [95% CI] | MAPE (%) | \|Median\| (%) | MSPE (%) | $R_\alpha^2$ | Time (ms) |
|---|---|---|---|---|---|---|---|
| $[5, 10]$ Count=4,229 | **GART 2.0** | **1.56** [1.47, 1.64] | 0.99 | 0.61 | 0.08 | 0.960 | 53.2 |
| | Calibrated MST ratio $\hat{\rho}(d, n)$ | 5.42 [5.29, 5.55] | 4.42 | 4.07 | 0.68 | 0.534 | 3.73 |
| | $L_{\text{MST}}$ ($\alpha = 1$) | 6.05 [5.86, 6.24] | 19.78 | 18.68 | -19.78 | -5.633 | 3.96 |
| | Custom Hilbert sort | 7.49 [7.15, 7.82] | 8.15 | 5.90 | 8.15 | — | 0.672 |
| | BHH (sampling region) | 11.23 [10.28, 12.10] | 33.49 | 33.68 | -32.59 | — | 0.139 |
| | *Reference-tour generation* | — | — | — | — | — | 1 ms |
| $[11, 50]$ Count=2,820 | **GART 2.0** | **1.03** [0.97, 1.10] | 0.65 | 0.41 | 0.10 | 0.946 | 78.4 |
| | Calibrated MST ratio $\hat{\rho}(d, n)$ | 2.16 [2.05, 2.26] | 1.66 | 1.37 | 0.39 | 0.771 | 17.7 |
| | $L_{\text{MST}}$ ($\alpha = 1$) | 4.14 [3.98, 4.32] | 8.61 | 7.46 | -8.61 | -3.235 | 18.1 |
| | Custom Hilbert sort | 12.41 [11.94, 12.85] | 19.59 | 15.81 | 19.59 | — | 15.9 |
| | BHH (sampling region) | 14.43 [12.24, 16.94] | 28.50 | 29.34 | -26.34 | — | 0.152 |
| | *Reference-tour generation* | — | — | — | — | — | 108 ms |
| $[51, 100]$ Count=3,520 | **GART 2.0** | **0.69** [0.66, 0.72] | 0.53 | 0.42 | 0.22 | 0.964 | 108 |
| | Calibrated MST ratio $\hat{\rho}(d, n)$ | 1.21 [1.13, 1.30] | 0.92 | 0.79 | 0.35 | 0.885 | 32.2 |
| | $L_{\text{MST}}$ ($\alpha = 1$) | 3.50 [3.39, 3.61] | 6.55 | 5.28 | -6.55 | -2.860 | 34.2 |
| | Custom Hilbert sort | 13.55 [13.18, 13.92] | 24.13 | 20.19 | 24.13 | — | 40 |
| | BHH (sampling region) | 15.69 [13.62, 18.66] | 26.49 | 27.69 | -23.59 | — | 0.184 |
| | *Reference-tour generation* | — | — | — | — | — | 76 ms |
| $[101, 200]$ Count=705 | **GART 2.0** | **0.49** [0.45, 0.53] | 0.41 | 0.35 | 0.19 | 0.973 | 145 |
| | Calibrated MST ratio $\hat{\rho}(d, n)$ | 0.89 [0.79, 0.99] | 0.73 | 0.68 | 0.35 | 0.906 | 56.9 |
| | $L_{\text{MST}}$ ($\alpha = 1$) | 2.93 [2.77, 3.10] | 5.57 | 4.45 | -5.57 | -3.115 | 57.8 |
| | BHH (sampling region) | 11.36 [9.62, 12.92] | 25.24 | 27.21 | -23.65 | — | 0.383 |
| | Custom Hilbert sort | 13.12 [12.62, 13.65] | 26.84 | 23.82 | 26.84 | — | 73 |
| | *Reference-tour generation* | — | — | — | — | — | 224 ms |

*Table 21 continued from the previous page*

| Size bucket | Estimator | SDPE (%) [95% CI] | MAPE (%) | \|Median\| (%) | MSPE (%) | $R^2_\alpha$ | Time (ms) |
|---|---|---|---|---|---|---|---|
| $[201, 500]$ Count=2,107 | **GART 2.0** | **0.46** [0.45, 0.47] | 0.41 | 0.29 | 0.27 | 0.969 | 194 |
| | Calibrated MST ratio $\hat{\rho}(d, n)$ | 0.74 [0.71, 0.78] | 0.65 | 0.63 | 0.35 | 0.923 | 95.9 |
| | $L_{\mathrm{MST}}$ $(\alpha = 1)$ | 2.76 [2.68, 2.84] | 5.25 | 4.34 | -5.25 | -3.170 | 99.3 |
| | BHH (sampling region) | 11.70 [10.58, 12.73] | 25.31 | 27.11 | -23.60 | — | 1.27 |
| | Custom Hilbert sort | 12.63 [12.40, 12.84] | 28.23 | 26.34 | 28.23 | — | 144 |
| | *Reference-tour generation* | — | — | — | — | — | 886 ms |
| $[501, 1000]$ Count=3,465 | **GART 2.0** | **0.45** [0.44, 0.45] | 0.40 | 0.24 | 0.28 | 0.966 | 460 |
| | Calibrated MST ratio $\hat{\rho}(d, n)$ | 0.69 [0.66, 0.71] | 0.60 | 0.58 | 0.33 | 0.925 | 273 |
| | $L_{\mathrm{MST}}$ $(\alpha = 1)$ | 2.61 [2.55, 2.67] | 5.06 | 4.21 | -5.06 | -3.321 | 326 |
| | BHH (sampling region) | 11.83 [11.03, 12.69] | 24.70 | 26.63 | -22.93 | — | 109 |
| | Custom Hilbert sort | 12.70 [12.55, 12.87] | 30.08 | 29.02 | 30.08 | — | 795 |
| | *Reference-tour generation* | — | — | — | — | — | 3,799 ms |
| **Total** Count=16,846 | **GART 2.0** | **0.99** [0.95, 1.02] | 0.62 | 0.42 | 0.18 | 0.984 | 121 |
| | Calibrated MST ratio $\hat{\rho}(d, n)$ | 2.94 [2.87, 3.02] | 1.82 | 0.98 | 0.44 | 0.858 | 33.8 |
| | $L_{\mathrm{MST}}$ $(\alpha = 1)$ | 7.26 [7.16, 7.36] | 9.71 | 7.28 | -9.71 | -1.245 | 36.4 |
| | BHH (sampling region) | 13.57 [12.85, 14.40] | 28.01 | 28.56 | -26.18 | — | 0.248 |
| | Custom Hilbert sort | 14.40 [14.26, 14.53] | 21.21 | 16.23 | 21.21 | — | 51.5 |
| | *Reference-tour generation* | — | — | — | — | — | 122 ms |

## M Per-Size Detail for the 2D Diverse Benchmark

Table 22 reports the 2D diverse benchmark broken down by instance-size bucket.

Table 22: 2D diverse synthetic benchmark by instance-size bucket. Within each bucket rows are ordered by SDPE.

| Size bucket | Estimator | SDPE (%) [95% CI] | MAPE (%) | \|Median\| (%) | MSPE (%) | $R^2_\alpha$ | Time (ms) |
|---|---|---|---|---|---|---|---|
| $[5, 10]$ Count=360 | **GART 2.0** | **4.28** [3.92, 4.67] | 3.21 | 2.54 | -0.00 | 0.907 | 58.7 |
| | GART 1.0 | 6.93 [6.27, 7.52] | 5.27 | 4.08 | 0.85 | 0.784 | 31.1 |
| | Custom Hilbert sort | 8.84 [8.11, 9.50] | 8.34 | 5.77 | 8.34 | — | 0.178 |
| | $L_{\mathrm{MST}}$ $(\alpha = 1)$ | 9.29 [8.63, 9.91] | 33.91 | 34.08 | -33.91 | -6.433 | 10.3 |
| | Asymptotic MST ratio | 10.46 [9.73, 11.13] | 25.64 | 25.83 | -25.63 | -3.806 | 10.5 |
| | Calibrated MST ratio $\hat{\rho}(d, n)$ | 14.22 [13.24, 15.16] | 11.69 | 10.19 | 1.14 | -0.003 | 10.6 |
| | *Reference-tour generation* | — | — | — | — | — | 82 ms |
| $[11, 50]$ Count=480 | **GART 2.0** | **5.83** [5.31, 6.35] | 4.07 | 2.66 | -0.81 | 0.708 | 82.6 |
| | GART 1.0 | 6.42 [5.85, 6.95] | 4.56 | 2.96 | 0.67 | 0.715 | 37.3 |
| | $L_{\mathrm{MST}}$ $(\alpha = 1)$ | 8.98 [8.25, 9.68] | 20.70 | 19.65 | -20.70 | -2.731 | 14.6 |
| | Asymptotic MST ratio | 10.11 [9.27, 10.90] | 11.99 | 9.59 | -10.76 | -0.833 | 13.5 |
| | Custom Hilbert sort | 10.37 [9.73, 11.07] | 27.17 | 26.96 | 27.17 | — | 0.346 |
| | Calibrated MST ratio $\hat{\rho}(d, n)$ | 11.17 [10.22, 12.09] | 7.53 | 4.11 | -1.42 | -0.047 | 14.5 |
| | *Reference-tour generation* | — | — | — | — | — | 85 ms |

*Table 22 continued from the previous page*

| Size bucket | Estimator | SDPE (%) [95% CI] | MAPE (%) | \|Median\| (%) | MSPE (%) | $R^2_\alpha$ | Time (ms) |
|---|---|---|---|---|---|---|---|
| $[51, 100]$ Count=630 | **GART 2.0** | **5.58** [4.99, 6.14] | 3.41 | 1.86 | -0.36 | 0.636 | 89 |
| | GART 1.0 | 7.38 [6.79, 8.01] | 5.39 | 3.52 | 2.33 | 0.494 | 42.9 |
| | $L_{\mathrm{MST}}$ ($\alpha = 1$) | 7.99 [7.24, 8.65] | 16.93 | 15.69 | -16.93 | -2.370 | 20.7 |
| | Asymptotic MST ratio | 8.99 [8.14, 9.79] | 8.09 | 5.98 | -6.53 | -0.425 | 19.5 |
| | Calibrated MST ratio $\hat{\rho}(d, n)$ | 9.46 [8.57, 10.30] | 5.72 | 2.65 | -1.67 | -0.056 | 20.6 |
| | Custom Hilbert sort | 9.82 [9.11, 10.53] | 34.21 | 34.06 | 34.21 | — | 0.605 |
| | *Reference-tour generation* | — | — | — | — | — | 248 ms |
| $[101, 500]$ Count=500 | **GART 2.0** | **4.25** [3.69, 4.75] | 2.39 | 1.09 | 0.18 | 0.593 | 177 |
| | GART 1.0 | 5.81 [5.22, 6.42] | 9.35 | 8.48 | 8.93 | -0.876 | 74.8 |
| | $L_{\mathrm{MST}}$ ($\alpha = 1$) | 5.88 [5.19, 6.51] | 13.08 | 12.52 | -13.08 | -3.061 | 37.5 |
| | Asymptotic MST ratio | 6.61 [5.79, 7.38] | 4.09 | 2.01 | -2.19 | -0.121 | 34 |
| | Calibrated MST ratio $\hat{\rho}(d, n)$ | 6.69 [5.92, 7.44] | 3.67 | 1.33 | -0.91 | -0.028 | 37.6 |
| | Custom Hilbert sort | 10.07 [8.92, 11.20] | 38.03 | 37.89 | 38.03 | — | 14.6 |
| | *Reference-tour generation* | — | — | — | — | — | 2,965 ms |
| $[501, 1000]$ Count=610 | **GART 2.0** | **2.75** [2.46, 3.04] | 1.64 | 0.79 | 0.45 | 0.650 | 277 |
| | $L_{\mathrm{MST}}$ ($\alpha = 1$) | 4.24 [3.82, 4.64] | 11.43 | 11.46 | -11.43 | -5.552 | 61.7 |
| | Asymptotic MST ratio | 4.77 [4.31, 5.21] | 2.64 | 1.04 | -0.33 | -0.013 | 54.7 |
| | Calibrated MST ratio $\hat{\rho}(d, n)$ | 4.78 [4.33, 5.21] | 2.63 | 1.03 | -0.25 | -0.009 | 57.6 |
| | GART 1.0 | 6.27 [5.67, 6.86] | 10.96 | 9.84 | 10.96 | -5.600 | 108 |
| | Custom Hilbert sort | 11.07 [9.96, 12.17] | 38.08 | 38.05 | 38.08 | — | 50.9 |
| | *Reference-tour generation* | — | — | — | — | — | 9,032 ms |
| **Total** Count=2,580 | **GART 2.0** | **4.68** [4.42, 4.92] | 2.89 | 1.49 | -0.10 | 0.867 | 113 |
| | GART 1.0 | 7.90 [7.60, 8.16] | 7.30 | 6.27 | 5.13 | 0.637 | 50.1 |
| | Calibrated MST ratio $\hat{\rho}(d, n)$ | 9.38 [8.96, 9.78] | 5.76 | 2.31 | -0.75 | 0.462 | 25 |
| | $L_{\mathrm{MST}}$ ($\alpha = 1$) | 10.27 [9.91, 10.61] | 17.95 | 14.61 | -17.95 | -1.621 | 24.9 |
| | Asymptotic MST ratio | 11.55 [11.15, 11.91] | 9.20 | 5.34 | -7.68 | -0.378 | 22.6 |
| | Custom Hilbert sort | 14.19 [13.75, 14.60] | 30.95 | 34.97 | 30.95 | — | 0.726 |
| | *Reference-tour generation* | — | — | — | — | — | 459 ms |

## N Per-Size Detail for the TSPLIB95 EUC_2D Benchmark

Table 23 reports the TSPLIB95 EUC_2D benchmark broken down by instance-size bucket.

Table 23: TSPLIB95 EUC_2D benchmark. Time is median wall clock; within each bucket rows are ordered by SDPE.

| $n$ range | Estimator | SDPE (%) [95% CI] | MAPE (%) | \|Median\| (%) | MSPE (%) | $R^2_\alpha$ | Time (ms) |
|---|---|---|---|---|---|---|---|
| $n \in [51, 150]$ Count=23 | **GART 2.0** | **2.63** [1.34, 3.58] | 2.01 | 1.22 | 1.47 | 0.580 | 3.64 |
| | $L_{\mathrm{MST}}$ ($\alpha = 1$) | 3.77 [2.51, 4.50] | 13.43 | 12.67 | -13.43 | -9.539 | 1.68 |
| | Asymptotic MST ratio | 4.24 [2.82, 5.07] | 3.55 | 2.13 | -2.58 | -0.395 | 1.16 |
| | Calibrated MST ratio $\hat{\rho}(d, n)$ | 4.68 [3.07, 5.68] | 4.09 | 3.93 | 1.04 | -0.176 | 1.14 |
| | GART 1.0 | 5.64 [3.58, 7.19] | 5.44 | 3.92 | 4.07 | -1.307 | 4.05 |
| | Custom Hilbert sort | 11.46 [8.20, 13.91] | 39.91 | 38.92 | 39.91 | — | 0.656 |
| $n \in [151, 400]$ Count=16 | **GART 2.0** | **2.86** [1.95, 3.51] | 2.38 | 2.50 | 1.51 | 0.706 | 4.51 |
| | $L_{\mathrm{MST}}$ ($\alpha = 1$) | 4.82 [2.13, 6.65] | 12.38 | 11.56 | -12.38 | -4.801 | 2.33 |
| | Calibrated MST ratio $\hat{\rho}(d, n)$ | 5.37 [2.49, 7.39] | 3.73 | 2.43 | 0.01 | 0.049 | 1.91 |
| | Asymptotic MST ratio | 5.42 [2.39, 7.48] | 3.39 | 1.27 | -1.41 | -0.088 | 1.84 |
| | GART 1.0 | 5.53 [3.93, 6.54] | 5.92 | 4.46 | 4.93 | -0.497 | 5.03 |
| | Custom Hilbert sort | 7.35 [3.56, 10.39] | 44.24 | 42.42 | 44.24 | — | 1.44 |
| $n > 400$ Count=39 | **GART 2.0** | **3.05** [2.32, 3.62] | 2.90 | 1.98 | 2.68 | -0.071 | 20.5 |
| | $L_{\mathrm{MST}}$ ($\alpha = 1$) | 3.58 [2.84, 4.17] | 9.74 | 9.39 | -9.74 | -6.448 | 9.55 |
| | Calibrated MST ratio $\hat{\rho}(d, n)$ | 3.95 [3.13, 4.60] | 3.54 | 3.24 | 1.75 | -0.122 | 9.52 |
| | Asymptotic MST ratio | 4.03 [3.20, 4.70] | 3.57 | 3.15 | 1.57 | -0.132 | 9.27 |
| | GART 1.0 | 7.64 [6.07, 8.92] | 11.23 | 9.66 | 11.18 | -10.726 | 18.4 |
| | Custom Hilbert sort | 12.55 [9.60, 15.71] | 48.66 | 47.03 | 48.66 | — | 9.14 |
| **Total** Count=78 | **GART 2.0** | **2.92** [2.40, 3.38] | 2.53 | 1.87 | 2.08 | 0.484 | 6.12 |
| | $L_{\mathrm{MST}}$ ($\alpha = 1$) | 4.21 [3.30, 4.99] | 11.37 | 11.56 | -11.37 | -5.591 | 3.08 |
| | Calibrated MST ratio $\hat{\rho}(d, n)$ | 4.48 [3.59, 5.32] | 3.74 | 3.18 | 1.18 | 0.087 | 2.62 |
| | Asymptotic MST ratio | 4.74 [3.72, 5.62] | 3.53 | 2.67 | -0.27 | -0.010 | 2.57 |
| | GART 1.0 | 7.45 [6.24, 8.46] | 8.43 | 6.51 | 7.80 | -3.579 | 6.12 |
| | Custom Hilbert sort | 11.85 [9.86, 13.59] | 45.17 | 44.60 | 45.17 | — | 2.64 |

## O Geometric and Centroid Feature Specifications

Table 24 lists the 11 geometric and centroid features that complement the MST-derived features in Tables 12, 13 and the constructive ratio of Table 14.

## P Trained Hyperparameters and Booster Statistics

Table 25 reports the hyperparameters and the resulting booster statistics for the final GART 2.0 model. The hyperparameters are inherited, frozen and not tuned for this model. The search we rejected was Optuna's tree-structured Parzen estimator (TPE), 200 trials, 60 of them completed and 140 pruned, against in-distribution validation. The tuned booster reaches 0.6112% on the multidimensional test split against 0.6226% for the same booster on the frozen hyperparameters, and 2.6364% against 2.4725% on TSPLIB EUC_2D; both are unconstrained and both are on the logit target, so the monotone constraint of the released model does not enter the difference, and scoring the tuned booster against the released model instead would confound the search with that constraint.

Table 24: Geometric and centroid feature specifications (11 features).

| Feature Name | Description & Rationale | Complexity | Origin / Citation |
|---|---|---|---|
| **Dimension** ($d$) | The number of spatial coordinates defining each point. | $O(1)$ | Standard |
| **Number of Nodes** ($n$) | The total count of nodes in the instance; released feature name `n_customers`. | $O(1)$ | Standard |
| **Aspect Ratio** | Ratio of the largest dimension range to the smallest. Detects subspace distortion. | $O(n \cdot d)$ | **Proposed** |
| **Bounding Hypervolume** | Product of coordinate ranges: $\prod_{j=1}^{d}(\max_j - \min_j)$. Approximates the sampling-region measure $\mathcal{V}$ for BHH bounds; stored capped at $e^{690}$, beyond which the log twin carries the scale. | $O(n \cdot d)$ | Adapted from Beardwood et al. [1959] |
| **Log Bounding Hypervolume** | Logarithm of hypervolume: $\sum_{j=1}^{d} \ln(\max_j - \min_j)$. Log-space stable for $d \geq 100$. | $O(n \cdot d)$ | **Proposed** |
| **Node Density** | Ratio of $n$ to Bounding Hypervolume. Proxy for point intensity $\delta = n/\mathcal{V}$. | $O(1)$ | **Proposed**; cf. Beardwood et al. [1959] |
| **Log Node Density** | Logarithm of node density: $\ln(n) - \sum \ln(\max_d - \min_d)$. Log-space stable. | $O(1)$ | **Proposed** |
| **Centroid Dist (Mean, Std)** | Average and deviation of distances to the centroid. Measures central tendency. | $O(n \cdot d)$ | Adapted from Çavdar and Sokol [2015] |
| **Centroid Dist Max** | Radius of the enclosing sphere. | $O(n \cdot d)$ | Adapted from Çavdar and Sokol [2015] |
| **Centroid Dist IQR** | Interquartile range of centroid distances. Robust dispersion metric. | $O(n \cdot d)$ | **Proposed**; extends Çavdar and Sokol [2015] |

Table 25: GART 2.0 hyperparameters and trained-model statistics; the hyperparameter block is inherited and frozen.

| **Group** | **Parameter** | **Value** |
|---|---|---|
| *Hyperparameters (inherited, frozen, not tuned)* | | |
| *Tree structure* | num leaves | 148 |
| | min child samples | 23 |
| | feature fraction | 0.548 |
| | bagging fraction / freq | 0.609 / 6 |
| *Optim. & reg.* | learning rate | 0.0260 |
| | L1 regularization | $7.76\times10^{-2}$ |
| | L2 regularization | $1.19\times10^{-5}$ |
| *Training configuration* | | |
| *Training setup* | boosting type | gbdt |
| | objective | regression_l2 (squared error) |
| | early-stopping patience | 100 |
| | random seed | 42 |
| *Trained-booster statistics* | | |
| *Booster* | trees (after early stop) | 1,118 |
| | avg leaves / tree | 147.80 |
| | max leaves / tree | 148 |
| | avg tree depth | 11.83 |
| | max tree depth | 40 |

Two other learners were fitted on the same feature pipeline before the gradient-boosted ensemble was frozen, and neither displaced it. A residual feed-forward network predicts the same $\alpha$ from the same MST-and-geometry descriptors, and on the 2D diverse benchmark it costs 137.5 to 146.2 ms per instance against GART 2.0's 103.3 to 147.8 ms, slower on five of the six generator classes of Table 17 and slower in aggregate. It does not buy accuracy with that time so much as move it: the network reads 2.98% MAPE overall against 2.89% and loses on five of the six classes, but it wins the near-collinear Line Noise class at 9.14% against 10.82% and finishes marginally ahead on aggregate dispersion, 4.61% SDPE against 4.68%. An ordinary least-squares fit is the cheapest of the three at 61.9 to 92.1 ms and pays for it in the tail, reaching 17.9% MAPE on the clustered class against 2.2%, eight times the released model's error, at $R^2_\alpha = -9.53$. Neither is a control on the released feature vector and we do not report them as one: the network was fitted on 30 columns and the linear model on 28 against the released block's 31, and neither was fitted on the released feature table. What they bound is the model-class question rather than settling it, and nothing in them gave a reason to trade the boosted ensemble away. Their per-instance predictions are released with every other row we score.

## Q MDS Embedding for Non-Euclidean Instances

For TSPLIB ATT, GEO (geographic coordinates with great-circle distances), and EXPLICIT (distance matrix only) instances, we apply classical MDS to obtain an approximate positive-eigenvalue Euclidean representation. CEIL_2D uses native coordinates. The embedding dimension $\kappa$ is chosen by the rule of Section 6; this retained-mass rule is a dimension-selection heuristic, not a distance-preservation guarantee.

Given an $n \times n$ distance matrix $D$, classical MDS computes the eigendecomposition of the doubly-centered matrix $B = -\frac{1}{2} J D^{(2)} J$ where $J = I - \frac{1}{n}\mathbf{1}\mathbf{1}^T$ and $D^{(2)}$ is the element-wise squared distance matrix. The embedding coordinates are $X = V_\kappa \Lambda_\kappa^{1/2}$ where $V_\kappa$ and $\Lambda_\kappa$ are the top $\kappa$ eigenvectors and eigenvalues.

We embed into $\kappa$ dimensions, selecting the smallest $\kappa$ that retains $\geq$99.9% of positive eigenvalue mass (capped at $\kappa = 100$). For GEO instances, the selected dimension ranges from 2 to 61; for screened EXPLICIT instances, from 10 to 100. The MST features (edge weights, degrees, diameter) are computed from the original distance matrix, while geometric features (centroid statistics, bounding volume, aspect ratio) use the approximate MDS coordinates. The dimension $\kappa$ is passed to the model as the `dimension` feature.

We recomputed these diagnostics directly from the released TSPLIB files. Normalized distance stress is $\sqrt{\sum_{i<j}(D_{ij} - \hat{D}_{ij})^2 / \sum_{i<j} D_{ij}^2}$, where $\hat{D}_{ij}$ is the Euclidean distance induced by the retained MDS coordinates; correlation is Pearson correlation between the same distance vectors. Table 26 reports medians and ranges. CEIL_2D stress measures only integer-ceiling distortion on a fixed-seed sample of 100,000–1,000,000 pairs; other rows use all pairs. Five capped instances do not reach the 99.9% target: `att532`, `pa561`, `si175`, `si535`, and `si1032`.

Table 26: Embedding diagnostics on the 23 screened instances; medians with [minimum, maximum]. "Below target": eigenvalue mass under 99.9% from the $\kappa = 100$ cap.

| Distance type | $N$ | $\kappa$ range | Normalized stress | Distance correlation | Below target |
|---|---|---|---|---|---|
| CEIL_2D | 4 | 2 | 0.0000 [0.0000, 0.0000] | 1.00000 [1.00000, 1.00000] | 0 |
| ATT | 2 | 6–100 | 0.0028 [0.0007, 0.0048] | 0.99999 [0.99998, 1.00000] | 1 |
| GEO | 10 | 2–61 | 0.0207 [0.0005, 0.2363] | 0.99951 [0.96944, 1.00000] | 0 |
| EXPLICIT screened | 7 | 10–100 | 0.0198 [0.0041, 0.1208] | 0.99954 [0.99598, 0.99999] | 4 |

The screened application set excludes ten EXPLICIT matrices with structural triangle-inequality violations. Seven EXPLICIT matrices remain, including three with mild rounding-level violations. GART 2.0 declines one of them, `si1032`, and obtains 3.01% MAPE on the six it accepts. Distortion varies substantially within type (maximum stress 0.2363 for GEO and 0.1208 for screened EXPLICIT), so we do not attribute between-type accuracy differences to the distance label alone.